Lattice Yang–Mills theory at strong coupling IV: the mass gap is $4\log(1/\beta) + O(1)$

A sequel to the foundations monograph of the project "Solving Yang–Mills and the Mass Gap"

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Abstract

For lattice Yang–Mills theory with Wilson's action in dimension $d \ge 3$ and $0 < \beta < \beta_0(d)$ (the threshold of the cluster expansion of [F]), we prove that the mass gap of the infinite-volume theory of Part II, and the finite-volume gaps of [F] on spatial lattices with all sides ≥ 4, are at least $4\log(1/q(\beta))$, with $q(\beta) = e^3 K(e^{4\beta} - 1)$ and $K = 4(2d-3)$. As $\beta \downarrow 0$ this is $4\log(1/\beta) - O(1)$. The leading coefficient 4 is the one predicted by the strong-coupling expansion (the lightest glueball is a tube with four plaquettes per unit of time, with predicted mass $-4\log u(\beta) + o(1)$, $u(\beta) \sim \beta/4$ for $SU(2)$), whereas the Dobrushin bounds of [F] and Part II give only $\log(1/\beta) - O(1)$. The proof combines three ingredients: duplicate variables, by which every term of the expansion of a covariance that does not connect the two observables cancels by the swap symmetry; the loop expansion and the ratio bound of Part III for the duplicated system; and a geometric lemma: a surface without free edges has at least four temporal plaquettes in each time layer it crosses, because these plaquettes form a graph of minimum degree two on the spatial lattice, whose girth is four (sides ≥ 4 are needed: with a side of length 3 the girth is three). In $d = 2$ the girth is the spatial size, which gives an infinite mass gap in infinite volume and a finite-volume gap growing linearly with the spatial size, in agreement with the exact solution. As in [F], II and III nothing non-abelian is used, so the result belongs to the strong-coupling endpoint of the problem. We also prove the matching upper bound: for an irreducible non-trivial ρ of dimension N, $m \le 4\log(1/u(\beta)) + D(\beta)$ with $u = c_\rho/(Nc_0)$ and an explicit $D(\beta) = O(\beta)$, on the whole range $0 < \beta < \beta_0(d)$ when $N \le 4$ ($N \le 5$ if $\rho \cong \bar{\rho}$), with $D < 0.32$ when $N \le 3$ (numerically $D < 0.05$ for $U(1)$, $SU(2)$, $SU(3)$ in $d = 3, 4$), and for every ρ when β is small. The proof uses a positivity property of the transfer matrix that isolates the minimal tube (the four sides of a unit cube between two plaquettes), an exact evaluation of that tube by character orthogonality, a lemma that every other term of the resulting expansion has at least five further plaquettes, and the spectral inequality $\mathrm{Cov} \le e^{-m}\mathrm{Var}$ of the ground-state chain of Part II. Together, for these ρ: $4\log(1/q(\beta)) \le m(\mu_\beta) \le 4\log(1/u(\beta)) + D(\beta)$ for all $\beta < \beta_0(d)$, so $m(\mu_\beta) = 4\log(1/\beta) + O(1)$, which is what principle 3 ("the strong-coupling test") of the research program requires. The additive constants of the two bounds still differ by about 35 for $SU(2)$ in $d = 4$. A transfer-matrix bound that weighs the temporal plaquettes of all but one time layer by the operator norm $u_* = \max_{r \ne 0} u_r$ of their kernel ($u_* = u$ for $\dim \rho \le 49$) improves the lower bound to $4\log(1/u(\beta)) - 4\log(eK) - O(\beta)$, so that for these ρ the two bounds differ by at most $4\log(eK) + O(\beta)$ (about 16.0 for $d = 4$ on the whole range). Counting the connecting surfaces by their fronts (the temporal plaquettes of one time layer, which can change only where a spatial plaquette or a support

touches them) removes the remaining entropy and gives the sharp constant on the whole range: $m(\mu_\beta) \geq 4\log(1/u(\beta)) - \log(1+\delta(\beta)) - O(\beta^5)$ with an explicit $\delta(\beta) = O(\beta)$, so that $m(\mu_\beta) = 4\log(1/u(\beta)) + O(\beta)$. Grouping everything that meets at one time slice into a junction, the condition that no link of a connecting surface is free leaves a single junction of order β through which a front of four plaquettes can pass, the cap; this gives the first-order constant of the method, $\delta(\beta) = 2\beta(1+O(\beta))$ when ρ has no trivial summand, i.e. $m(\mu_\beta) \geq 4\log(1/u(\beta)) - 2\beta + O(\beta^2)$. Evaluating the cap, and two sup-norm bounds of the upper bound, by characters gives the first-order term exactly: $m(\mu_\beta) = 4\log(1/u(\beta)) - \kappa(\beta) + O(\beta^2)$ with $\kappa(\beta) = \tau_0^{-1}\int(\operatorname{Re}\chi_\rho)^2 f_\beta\, dU = \frac{\beta}{N\tau_0}\int(\operatorname{Re}\chi_\rho)^3\, dU + O(\beta^2)$, which vanishes to first order for $U(1)$ and for $SU(N)$, $N \neq 3$, and is $\beta/6 + O(\beta^2)$ for $SU(3)$, where three fundamental fluxes can meet at a link. For $SU(2)$ the two bounds give $-1.3\cdot 10^{-4} \leq m(\mu_\beta) - 4\log(1/u(\beta)) \leq 4.8\cdot 10^{-4}$ for all $\beta < \beta_0(4)$ in $d = 4$, and $-2.5\cdot 10^{-4} \leq m(\mu_\beta) - 4\log(1/u(\beta)) \leq 2.4\cdot 10^{-3}$ for all $\beta < \beta_0(3)$ in $d = 3$ (a numerical evaluation of the explicit constants). The asymptotics itself, indeed the full expansion of the glueball mass, is classical (Schor; O'Carroll et al.); what is new here is a self-contained proof with explicit constants on the whole range, for general compact G and the infinite-volume theory of Part II.

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1 Introduction and main result

The foundations monograph [F] proves that the finite-volume mass gap is bounded below uniformly in the volume for small β, by Dobrushin's method; Part II [II] constructs the infinite-volume theory, identifies its transfer operator with a reversible Markov chain of time slices, and proves the same kind of bound, $m \geq \log(1/(6(d-1)\tanh\beta))$, for the infinite-volume mass gap $m(\mu_\beta)$. These bounds behave like $\log(1/\beta)$ as $\beta \downarrow 0$. The strong-coupling expansion predicts $m(\beta) = -4\log u(\beta) + O(u)$, where $u(\beta) = c_\rho(\beta)/(Nc_0(\beta))$ ($\approx \beta/4$ for $SU(2)$, $\approx \beta/(2N^2)$ for $N \geq 3$): the lightest state in the gauge-invariant sector propagates as a tube of plaquettes with four plaquettes per unit of Euclidean time. Principle 3 of the research program (`docs/03-research-program.md`, §3.1) asks every argument in the strong-coupling regime to reproduce this behaviour. Here we prove a lower bound with the correct leading coefficient for all $\beta < \beta_0(d)$ (Theorem 1.2) and the matching upper bound, on the

same range for representations of small dimension and for every representation when β is small (Theorem 1.3); together, $m(\mu_\beta) = 4\log(1/\beta) + O(1)$.

Literature. The strong-coupling glueball spectrum was studied rigorously in the 1980s. According to the account of Faria da Veiga and O'Carroll [FO25, Sections II and V], Schor [Sch83, Sch84a, Sch84b] showed, for $SU(2)$ in three dimensions at fixed lattice spacing and small β, by the hyperplane decoupling method applied to the truncated plaquette–plaquette correlation, that the low-lying energy-momentum spectrum contains a single isolated one-glueball dispersion curve, whose mass is $-4\log\beta + r(\beta)$ with r analytic. By their abstracts, O'Carroll, Braga and Schor [OBS85] ($2+1$ dimensions) and O'Carroll and Barbosa [OB85] ($3+1$ dimensions) studied the masses in the subspace generated by the time-zero plaquette fields (in [OB85]: at least one and at most four, with convergent expansions), and O'Carroll [OC85] the excited masses near $-6\log\beta$ in $2+1$ dimensions. We have read [FO25] and the abstracts of [OB85, OC85], not the original papers (marked "verify"), and nothing below relies on them. These results are much sharper than ours: they determine the glueball mass to all orders in β, whereas the bounds proved here determine it up to $O(\beta^2)$ (Corollary 9.6). The title and the first lines of the abstract of [OBS85] mention a mass splitting for complex characters; for $SU(3)$ we find a first-order splitting of the C-even and C-odd plaquette states (Remark 9.7(iii)). What this article adds for the project is a short self-contained proof, with explicit constants and the explicit range $\beta < \beta_0(d)$, for general compact G and for the infinite-volume theory of [II], together with a covariance bound for all bounded local observables.

Convention 1.1 (Labels). As in [F, Convention 'Labels']: every Theorem, Proposition, Lemma and Corollary below has a complete proof here or in [F, II, III]; Remarks are commentary. We use the notation of [III]: $K := 4(2d-3)$, $\epsilon(\beta) = e^{2\beta} - 1$, $x(\beta) = e^2 K\epsilon(\beta)$, $\theta(\beta) = x/(1-x)$, and $\beta_0(d) = \frac{1}{2}\log(1 + \frac{1}{e^2K(K+2)})$, the threshold below which [F, Theorem 'Convergence of the strong-coupling cluster expansion' (`thm:conv`)] holds.

Theorem 1.2 (Main theorem). *Let G, ρ be as in [F, Convention 'Group'] with G metrisable, $d \geq 3$, $0 < \beta < \beta_0(d)$, and*

$$q(\beta) := e^3 K(e^{4\beta} - 1).$$

Then $q(\beta) < \frac{e(2+\epsilon_0)}{K+2} < 0.389$, where $\epsilon_0 = \frac{1}{e^2K(K+2)}$, and:

(i) for every periodic spatial lattice Λ_s with all sides $L_i \geq 4$, the finite-volume mass gap of [F, Definition 'Hamiltonian and mass gap' (`def:H`)] satisfies $m_{\Lambda_s}(\beta) \geq 4\log(1/q(\beta))$;

(ii) the mass gap of the infinite-volume theory [II, Definition 'Mass gap of an infinite-volume lattice theory' (`iv:def:gap`)] satisfies $m(\mu_\beta) \geq 4\log(1/q(\beta))$;

(iii) for bounded measurable A, B on $G^{E(\mathbb{Z}^d)}$ depending only on finite link sets S_A, S_B, all of whose endpoints have time coordinates in $[a_1, a_2]$, resp. $[b_1, b_2]$, with $a_2 < b_1$,

$$|\mathrm{Cov}_{\mu_\beta}(A, B)| \leq \frac{2\|A\|_\infty \|B\|_\infty}{1 - q(\beta)}\, c(\beta)^{|S_A|+|S_B|}\, q(\beta)^{4(b_1-a_2)},$$

where $c(\beta) := \left(e^{2\theta(\beta)}(1 + \frac{1}{K})\right)^{2(d-1)}$.

As $\beta \downarrow 0$, $4\log(1/q(\beta)) = 4\log(1/\beta) - 4\log(4e^3K) + O(\beta)$.

Theorem 1.3 (Upper bound; the gap is $4\log(1/\beta) + O(1)$). *Let G, ρ be as in Theorem 1.2 (G metrisable), $d \geq 3$, and let ρ be irreducible and non-trivial, of dimension N; put $\iota := 1$ if $\rho \cong \bar{\rho}$ and $\iota := 0$ otherwise, and $u(\beta) := c_\rho(\beta)/(Nc_0(\beta))$. Suppose $N \leq 3$ (for instance $U(1)$, $SU(2)$, $SU(3)$). There is an explicit function $D(\beta)$ (Theorem 5.6) with $0 < D(\beta) < 0.32$ and $D(\beta) = O(\beta)$ such that for every $0 < \beta < \beta_0(d)$*

$$4\log\frac{1}{q(\beta)} \;\leq\; m(\mu_\beta) \;\leq\; 4\log\frac{1}{u(\beta)} + D(\beta);$$

the upper bound also holds for $m_{\Lambda_s}(\beta)$ on every periodic spatial lattice with all sides ≥ 3, and both bounds on every periodic spatial lattice with all sides ≥ 4. Hence

$$-4\log(4e^3K) - O(\beta) \;\leq\; m(\mu_\beta) - 4\log\frac{1}{\beta} \;\leq\; 4\log\frac{2N^2}{1+\iota} + O(\beta) \qquad (0<\beta<\beta_0(d)).$$

Moreover, for every $0<\beta<\beta_0(d)$, $m(\mu_\beta) \geq 4\log(1/u(\beta)) - 8(K+1)\theta_5(\beta) - \log(1+\delta_\infty(\beta))$ with $\theta_5 = O(\beta^5)$ and an explicit $\delta_\infty(\beta) = O(\beta)$, and the same holds for $m_{\Lambda_s}(\beta)$ when all spatial sides are ≥ 4, with a slightly larger δ (Theorem 7.8, Lemma 7.9, Corollary 7.11). So $m(\mu_\beta) - 4\log(1/u(\beta))$ lies in an interval of length at most $8(K+1)\theta_5 + \log(1+\delta_\infty) + D(\beta)$, and $m(\mu_\beta) = 4\log(1/u(\beta)) + O(\beta)$ as $\beta \downarrow 0$. Grouping by junctions (Theorem 8.11, Corollary 8.13) improves δ_∞ to $\delta_J = 2\beta(1+O(\beta))$: $m(\mu_\beta) \geq 4\log(1/u(\beta)) - 2\beta + O(\beta^2)$. Evaluating the cap and the upper bound by characters (Theorems 9.4 and 9.5, Corollary 9.6) gives the first-order term, $m(\mu_\beta) = 4\log(1/u(\beta)) - \kappa(\beta) + O(\beta^2)$ with $\kappa(\beta) = \tau_0^{-1}\int(\operatorname{Re}\chi_\rho)^2 f_\beta\, dU$, which is $O(\beta^2)$ for $U(1)$ and $SU(2)$ and $\beta/6 + O(\beta^2)$ for $SU(3)$; for $SU(2)$ the interval has length at most $6.1\cdot 10^{-4}$ ($d=4$) and $2.7\cdot 10^{-3}$ ($d=3$) on the whole range (Remark 9.7; Numerical). The upper bound holds, with a larger explicit bound on D, whenever $N \leq 4$, or $N=5$ and $\rho \cong \bar\rho$. Without any restriction on N, the upper bound holds for all sufficiently small β, so $m(\mu_\beta) = 4\log(1/\beta) + O(1)$ as $\beta \downarrow 0$.

Here $\mathrm{Cov}(A,B) := \mu(AB) - \mu(A)\mu(B)$. Statement (iii) is the infinite-volume form of the covariance bound of Theorem 4.1; (i) follows from Theorem 4.1 by [F, Corollary 'Gap from Euclidean correlations' (`cor:gapfromcorr`)], and (ii) from (iii) by [II, Theorem 'OS reconstruction of the lattice theory' (`iv:thm:OS`)](iii). Theorem 1.3 is Corollary 5.8, together with Theorem 5.6(ii) and Lemma 5.7 for the upper bound on spatial lattices with sides ≥ 3, Corollary 7.11 (with Theorem 7.8 and Lemma 7.9) for the sharper lower bound and the asymptotics, and the last sentence of Theorem 5.6 for arbitrary N.

Remark 1.4 (Comparison). For $d=4$: the bound of Part II, $\log(1/(18\tanh\beta))$, is $\log(1/\beta) - 2.89 + O(\beta^2)$; the bound here is $4\log(1/\beta) - 29.5 + O(\beta)$. In the common range of validity $\beta < \beta_0(4) \approx 1.54\cdot 10^{-4}$ the two are comparable near $\beta_0(4)$ (5.890 and 5.591 at $\beta_0(4)$), and the present one is larger for $\beta < 1.39\cdot 10^{-4}$, by an additive amount that tends to infinity as $\beta \to 0$ (the ratio of the two bounds increases to 4; it is 1.92 at $\beta = 10^{-5}$). These values were computed with `python3`. The point of the theorem is the coefficient 4, not the numerical strength at fixed β; the additive constant -29.5 is far from the predicted $+4\log 4$ for $SU(2)$ (from $u \approx \beta/4$).

Strategy. (1) Let U and U' be two independent copies of the field, each with law μ. Then $2\,\mathrm{Cov}_\mu(A,B)$ is the integral of $(A-A')(B-B')$ against $\mu\otimes\mu$. The duplicated Wilson weight is $\prod_p(1+f_\beta(U_p))(1+f_\beta(U'_p)) = \prod_p(1+g_p)$ with $|g_p| \leq e^{4\beta}-1$, so the loop expansion of [III] applies to the duplicated system, with partition functions $\Xi(\cdot)^2$. (2) In this expansion every set of plaquettes that does not contain a component connecting the supports of A and B gives a product of two integrals, each odd under the exchange $U \leftrightarrow U'$, hence zero (Lemma 2.5). (3) A connecting component has no free edges away from the supports, so in every time layer between the supports it contains at least four temporal plaquettes (Lemma 3.2), hence at least $4\times$(number of layers) plaquettes (Corollary 3.4). (4) The sum over such sets is controlled by the ratio bound and the rooted counting of [III]. (5) The upper bound (Section 5) uses positivity instead of cancellation. Splitting the one-link operator of the transfer matrix into its trivial and non-trivial parts writes the covariance of two plaquettes at times 0 and 1 as a sum of non-negative terms; one of them is the expectation of a local function in which the four sides of the unit cube between the plaquettes carry non-trivial characters. Its loop expansion is evaluated exactly on the cube by character orthogonality, every other term has at least five further plaquettes, and the spectral inequality $\mathrm{Cov} \leq e^{-m}\,\mathrm{Var}$ gives $m \leq 4\log(1/u) + O(\beta)$,

on the whole range $\beta < \beta_0(d)$ when $\dim \rho \le 4$ (≤ 5 if $\rho \cong \bar{\rho}$) and for every ρ when β is small. (6) Returning to the lower bound (Section 6): in the transfer-matrix representation of the terms of the duplicated expansion, a temporal plaquette is an operator between two time slices of norm $u_* = \max_{r \neq 0} u_r$, far below the sup norm $e^{4\beta} - 1$ used in (4); this gives $m \ge 4\log(1/u) - 4\log(eK) - O(\beta)$ (when $u_* = u$, e.g. $\dim \rho \le 49$). (7) Finally (Section 7) the connecting sets are encoded by their fronts: a front changes only where a marker (a spatial plaquette or a support) touches it, so the sum over connecting sets becomes a sum over paths of segments and clusters with decorations, in which an unchanged front of k edges costs a^k per layer, $a = e^{2(K+1)\theta_5} u_*$, and nothing in entropy. A generating function in which only the segments of one path carry the weight $z = 1/(a^4(1+\delta'))$ then gives $m \ge 4\log(1/u_*) - O(\beta)$ ($= 4\log(1/u) - O(\beta)$ when $u_* = u$). Counting at vertices, with the exact numbers of fronts with four and five edges, a sharper sup norm of g_p and a ratio bound that sees only closed polymers (which have at least five plaquettes), makes the constant effective on the whole range. (8) Grouping all markers and fronts that meet at a slice into junctions (Section 8) and splitting the weight of each segment between its two ends, the local form of the no-free-link condition shows that the only junction of order $b = e^{2(K+1)\theta_5} \|g_p\|_\infty$ (about 2β when ρ has no trivial summand) through which a four-edge front can pass is the cap, a plaquette closing the front at a slice where it continues; this gives $\delta = b(1 + O(\beta))$, the first-order constant of the method. (9) Finally (Section 9) the transfer bound is applied junction by junction, with the gauge projections at the vertices of each junction; the cap then becomes an operator on the class functions of the holonomy of its plaquette, whose norm relative to the tube is $\kappa(\beta) + O(\beta^2)$, $\kappa(\beta) = \tau_0^{-1} \int (\operatorname{Re} \chi_\rho)^2 f_\beta \, dU$, and the same number replaces two sup-norm bounds of the upper bound; so $m(\mu_\beta) = 4\log(1/u(\beta)) - \kappa(\beta) + O(\beta^2)$.

2 The duplicated system

Throughout this section and the next, $\Lambda = \prod_{\lambda=0}^{d-1} \mathbb{Z}/L_\lambda\mathbb{Z}$ is a periodic lattice with all $L_\lambda \ge 3$, with the conventions of [F, III]: $\mathcal{P} = \mathcal{P}(\Lambda)$, $E(\Lambda)$, $E(p)$, $E(X)$, adjacency and connectedness of sets of plaquettes, $f_\beta = w_\beta/c_0(\beta) - 1$, and for $\mathcal{Q} \subseteq \mathcal{P}$ the restricted partition function $\Xi(\mathcal{Q}) = \int \prod_{p \in \mathcal{Q}} (1 + f_\beta(U_p)) \, dU$ of [III, Definition 'Restricted partition functions, loop polymers' (`al:def:Xi`)], so that $\Xi(\mathcal{P}) = Z_\Lambda(\beta)/c_0(\beta)^{|\mathcal{P}|}$.

Definition 2.1 (Duplicated system)**.** On $\Omega_\Lambda \times \Omega_\Lambda$ with the product Haar measure $dU\, dU'$ put, for $p \in \mathcal{P}$,

$$g_p(U, U') := \big(1 + f_\beta(U_p)\big)\big(1 + f_\beta(U'_p)\big) - 1.$$

For $X \subseteq \mathcal{P}$ and bounded measurable F on $\Omega_\Lambda \times \Omega_\Lambda$ let $\Phi_F(X) := \iint F \prod_{p \in X} g_p \, dU \, dU'$.

Lemma 2.2 (Properties of the duplicated system)**.** *(a)* $\|g_p\|_\infty \le \epsilon(2\beta) = e^{4\beta} - 1$.

(b) For $\mathcal{Q} \subseteq \mathcal{P}$, $\iint \prod_{p \in \mathcal{Q}} (1 + g_p) \, dU \, dU' = \Xi(\mathcal{Q})^2$; and $\mu_{\Lambda,\beta} \otimes \mu_{\Lambda,\beta}$ has density $\prod_{p \in \mathcal{P}} (1 + g_p)/\Xi(\mathcal{P})^2$ with respect to $dU\, dU'$.

(c) Let $X \subseteq \mathcal{P}$, let ℓ be a link lying in exactly one plaquette of X, and let H be bounded measurable and independent of (U_ℓ, U'_ℓ). Then $\iint H \prod_{p \in X} g_p \, dU \, dU' = 0$.

(d) If ρ has no trivial summand (for instance if ρ is irreducible and non-trivial), then $\|g_p\|_\infty \le \gamma_2(\beta) := e^{2\beta} - 1 + \beta^2 e^\beta$.

Proof. (a) By [III, Lemma `al:lem:facts`](a),(b), $e^{-\beta} \le w_\beta \le e^\beta$ and $1 \le c_0 \le e^\beta$, so $1 + f_\beta = w_\beta/c_0 \in [e^{-2\beta}, e^\beta]$ and $(1 + f_\beta(U_p))(1 + f_\beta(U'_p)) \in [e^{-4\beta}, e^{2\beta}]$ and $g_p \in [e^{-4\beta} - 1, e^{2\beta} - 1]$; both $1 - e^{-4\beta}$ and $e^{2\beta} - 1$ are at most $e^{4\beta} - 1$. (b) The integrand is $\prod_{p \in \mathcal{Q}} (1 + f_\beta(U_p)) \cdot \prod_{p \in \mathcal{Q}} (1 + f_\beta(U'_p))$; integrate over U and U' separately. For the density: $W_\beta(U) W_\beta(U') = c_0^{2|\mathcal{P}|} \prod_p (1 + g_p)$ and

$Z_\Lambda^2 = c_0^{2|\mathcal{P}|}\Xi(\mathcal{P})^2$. (c) Let p be the plaquette of X containing ℓ. Then $U_p = AU_\ell^{\pm 1}B$ and $U_p' = A'U_\ell'^{\pm 1}B'$ with A, B, A', B' independent of (U_ℓ, U_ℓ'), and no other factor of the integrand depends on (U_ℓ, U_ℓ'). By Fubini it suffices that $\iint g_p\, dU_\ell\, dU_\ell' = \int(1 + f_\beta(AU^{\pm 1}B))\, dU \cdot \int(1 + f_\beta(A'U^{\pm 1}B'))\, dU - 1 = 1 \cdot 1 - 1 = 0$, by [III, Lemma `al:lem:facts`](c). (d) Put $y := \frac{\beta}{N}\operatorname{Re}\operatorname{Tr}\rho(U) \in [-\beta, \beta]$. As in the proof of [III, Lemma `al:lem:facts`](b), $\int \operatorname{Tr}\rho(U)\, dU$ is the multiplicity of the trivial representation in ρ, here 0, so $\int y\, dU = 0$; and $e^y \le 1 + y + \frac{1}{2}y^2 e^{|y|}$. Hence $1 \le c_0 \le 1 + \frac{1}{2}\beta^2 e^\beta$, so $1 + f_\beta = w_\beta/c_0 \in [e^{-\beta}(1 - \frac{1}{2}\beta^2 e^\beta), e^\beta]$, $(1 + f_\beta(U_p))(1 + f_\beta(U_p')) \in [e^{-2\beta}(1 - \beta^2 e^\beta), e^{2\beta}]$, and $g_p \in [-(1 - e^{-2\beta}) - \beta^2 e^{-\beta}, e^{2\beta} - 1]$; finally $1 - e^{-2\beta} \le e^{2\beta} - 1$. □

Definition 2.3 (Loop polymers for a set of links)**.** For $S \subseteq E(\Lambda)$ let $\mathcal{P}_S := \{p : E(p) \cap S \neq \emptyset\}$, let $\mathcal{X}_S$ be the family of sets $X \subseteq \mathcal{P}$ each of whose connected components contains a plaquette of $\mathcal{P}_S$ (including $X = \emptyset$), and for $X \subseteq \mathcal{P}$ let $\overline{X} := \{p : E(p) \cap (E(X) \cup S) \neq \emptyset\}$. Put $J_S := 2(d-1)|S|$. Then $|\mathcal{P}_S| \le J_S$ and $|\overline{X}| \le (K+1)|X| + J_S$ (each link lies in $2(d-1)$ plaquettes, and each plaquette is adjacent to at most K plaquettes, [III, Lemma `al:lem:facts`](d)).

Lemma 2.4 (Duplicated loop expansion)**.** *Let $S \subseteq E(\Lambda)$ and let F be bounded, measurable, and dependent only on the variables (U_ℓ, U_ℓ'), $\ell \in S$. Then*

$$\int F\, d(\mu_{\Lambda,\beta} \otimes \mu_{\Lambda,\beta}) = \sum_{X \in \mathcal{X}_S} \Phi_F(X) \Big(\frac{\Xi(\mathcal{P} \setminus \overline{X})}{\Xi(\mathcal{P})}\Big)^2.$$

Proof. This is the proof of [III, Lemma 'Loop expansion' (`al:lem:loopexp`)] with f_β replaced by g, single link variables by pairs (U_ℓ, U_ℓ'), dU by $dU\, dU'$, and E_C by S. By Lemma 2.2(b), $\Xi(\mathcal{P})^2 \int F\, d(\mu \otimes \mu) = \sum_{T \subseteq \mathcal{P}} \iint F \prod_{p \in T} g_p\, dU\, dU'$. For $T \subseteq \mathcal{P}$ let $X(T)$ be the union of the components of T meeting $\mathcal{P}_S$ and $Y(T) := T \setminus X(T)$. The proof of the claim in [III, Lemma `al:lem:loopexp`], which uses only the combinatorics of components, of $\mathcal{P}_S$ (there $\mathcal{P}_C$) and of $\overline{X}$, shows that $T \mapsto (X(T), Y(T))$ is a bijection onto the pairs with $X \in \mathcal{X}_S$ and $Y \subseteq \mathcal{P} \setminus \overline{X}$. For such a pair, $F \prod_{p \in X} g_p$ depends only on the variables of the links in $S \cup E(X)$ and $\prod_{p \in Y} g_p$ only on those in $E(Y)$, a disjoint set; so the integral factorises, and $\sum_{Y \subseteq \mathcal{P} \setminus \overline{X}} \iint \prod_{p \in Y} g_p = \iint \prod_{p \in \mathcal{P} \setminus \overline{X}}(1 + g_p) = \Xi(\mathcal{P} \setminus \overline{X})^2$ by Lemma 2.2(b). Divide by $\Xi(\mathcal{P})^2$. □

Lemma 2.5 (Cancellation)**.** *Let A, B be bounded measurable functions on Ω_Λ depending only on disjoint link sets S_A, S_B, put $S := S_A \cup S_B$ and $F(U, U') := (A(U) - A(U'))(B(U) - B(U'))$. Then:*

(i) $\int F\, d(\mu \otimes \mu) = 2\operatorname{Cov}_\mu(A, B)$ *for every probability measure μ on Ω_Λ, and $\|F\|_\infty \le 4\|A\|_\infty \|B\|_\infty$;*

(ii) if $X \in \mathcal{X}_S$ and no connected component of X contains both a plaquette of $\mathcal{P}_{S_A}$ and a plaquette of $\mathcal{P}_{S_B}$, then $\Phi_F(X) = 0$;

(iii) if some link of $E(X) \setminus S$ lies in exactly one plaquette of X, then $\Phi_F(X) = 0$.

Proof. (i) Expand: $\int (A - A')(B - B')\, d(\mu \otimes \mu) = 2\mu(AB) - 2\mu(A)\mu(B)$. (ii) Let X_1 be the union of the components of X that contain a plaquette of $\mathcal{P}_{S_A}$, and $X_2 := X \setminus X_1$; by hypothesis no component of X_1 meets $\mathcal{P}_{S_B}$, and every component of X_2 meets $\mathcal{P}_{S_B}$ but not $\mathcal{P}_{S_A}$. A component containing a link of S_B contains a plaquette of $\mathcal{P}_{S_B}$; hence $E(X_1) \cap S_B = \emptyset$, and likewise $E(X_2) \cap S_A = \emptyset$; and $E(X_1) \cap E(X_2) = \emptyset$, since distinct components share no link. So $S_A \cup E(X_1)$ and $S_B \cup E(X_2)$ are disjoint, and since $dU\, dU'$ is a product measure,

$$\Phi_F(X) = \iint (A - A') \prod_{p \in X_1} g_p \cdot \iint (B - B') \prod_{p \in X_2} g_p.$$

The exchange $(U, U') \mapsto (U', U)$ preserves $dU\, dU'$ and every g_p (which is symmetric in U_p, U_p'), and changes the sign of $A - A'$; so the first factor equals its negative and vanishes. (iii) Apply Lemma 2.2(c) with $H = F \prod_{p' \in X \setminus \{p\}} g_{p'}$, p the plaquette containing the link. □

3 Geometry: four plaquettes per time layer

Time coordinates are taken in $\mathbb{Z}/L_0\mathbb{Z}$. A link $(x, 0)$ is *temporal of layer* x_0 (it joins times x_0 and x_0+1); a plaquette $(x, 0i)$, $1 \le i \le d-1$, is *temporal of layer* x_0. For $Y \subseteq \mathcal{P}$ and a layer s, Y_s denotes the set of temporal plaquettes of layer s in Y, and $T(p)$ the set of time coordinates of the corners of p (one time for a spatial plaquette, two consecutive times for a temporal one). Let $\Lambda_s := \prod_{i=1}^{d-1} \mathbb{Z}/L_i\mathbb{Z}$ be the spatial torus and g the *girth* of its nearest-neighbour graph (the length of a shortest cycle).

Lemma 3.1 (Girth)**.** *Let all $L_i \ge 3$. If $d \ge 3$, then $g = 4$ when all $L_i \ge 4$, and $g = 3$ when some $L_i = 3$. If $d = 2$, then $g = L_1$.*

Proof. Since all $L_i \ge 3$ the graph is simple. A cycle is a closed walk without backtracking, i.e. a sum of unit vectors $\pm e_i$ equal to 0 in Λ_s with no two consecutive steps opposite. For $d = 2$ the graph is the cycle $\mathbb{Z}/L_1\mathbb{Z}$ itself. For $d \ge 3$ a unit square is a cycle of length 4; a cycle of length 3 needs $\pm e_i \pm e_j \pm e_k = 0$ with no two opposite consecutive steps, which forces $i = j = k$ and $3e_i = 0$, i.e. $L_i = 3$; and conversely $L_i = 3$ gives the cycle $0, e_i, 2e_i$. □

Lemma 3.2 (Plaquettes per layer)**.** *Let all $L_i \ge 3$ ($1 \le i \le d-1$). Let $S \subseteq E(\Lambda)$ and $X \subseteq \mathcal{P}$ be such that every link of $E(X) \setminus S$ lies in at least two plaquettes of X. Let s be a layer none of whose temporal links belongs to S, and let Y be a connected component of X with $Y_s \ne \emptyset$. Then $|Y_s| \ge g$. In particular $|Y_s| \ge 4$ if $d \ge 3$ and all $L_i \ge 4$.*

Proof. The temporal links of layer s are $\ell_{\vec{x}} := ((s, \vec{x}), 0)$, $\vec{x} \in \Lambda_s$, and the plaquettes containing $\ell_{\vec{x}}$ are the temporal plaquettes $((s, \vec{x}), 0i)$ and $((s, \vec{x} - e_i), 0i)$ of layer s. Let Γ be the graph on Λ_s whose edges are the plaquettes $((s, \vec{x}), 0i) \in Y_s$, each joining $\vec{x}$ and $\vec{x} + e_i$ (the positions of its two temporal links). Since $L_i \ge 3$, distinct plaquettes give distinct edges and there are no loops, so Γ is a subgraph of the nearest-neighbour graph of Λ_s. If $\vec{x}$ is an endpoint of an edge, then $\ell_{\vec{x}} \in E(Y) \setminus S$, so $\ell_{\vec{x}}$ lies in at least two plaquettes of X; these are temporal plaquettes of layer s sharing the link $\ell_{\vec{x}}$ with a plaquette of Y, hence they belong to the component Y, i.e. to Y_s. So every non-isolated vertex of Γ has degree at least 2, and since $Y_s \ne \emptyset$, Γ contains a cycle (follow edges without immediately backtracking until a vertex repeats), which has at least g edges; they are distinct plaquettes of Y_s. The last statement is Lemma 3.1. □

Lemma 3.3 (Crossing layers)**.** *Let $Y \subseteq \mathcal{P}$ be connected and let $s \ne s'$ be layers with $Y_s = Y_{s'} = \emptyset$. Then either $T(p) \subseteq I_1$ for all $p \in Y$, or $T(p) \subseteq I_2$ for all $p \in Y$, where $I_1 := \{s+1, \dots, s'\}$ and $I_2 := \{s'+1, \dots, s\}$ are the two cyclic intervals of $\mathbb{Z}/L_0\mathbb{Z}$ determined by s, s'.*

Proof. I_1 and I_2 partition $\mathbb{Z}/L_0\mathbb{Z}$. A spatial plaquette has $T(p) = \{r\}$, contained in one of them. A temporal plaquette of layer $r \notin \{s, s'\}$ has $T(p) = \{r, r+1\}$, and $r, r+1$ lie in the same interval because the only consecutive pairs split between I_1 and I_2 are $\{s, s+1\}$ and $\{s', s'+1\}$. If $p, p' \in Y$ are adjacent, the time coordinates of the endpoints of a common link lie in $T(p) \cap T(p')$, so p, p' lie in the same class. By connectedness all plaquettes of Y lie in the same class. □

Corollary 3.4 (Size of connecting components)**.** *Let all $L_\lambda \ge 3$. Let $S_A, S_B \subseteq E(\Lambda)$ and integers $a_1 \le a_2 < b_1 \le b_2$ with $b_2 - a_1 < L_0$ be such that the endpoints of the links of S_A have time coordinates in $\{a_1, \dots, a_2\}$ and those of S_B in $\{b_1, \dots, b_2\}$ (modulo L_0). Put*

$$\tau_{\text{in}} := b_1 - a_2, \qquad \tau_{\text{out}} := L_0 - (b_2 - a_1), \qquad \tau := \min(\tau_{\text{in}}, \tau_{\text{out}}) \ \ (\ge 1).$$

Let $X \subseteq \mathcal{P}$ be such that every link of $E(X) \setminus (S_A \cup S_B)$ lies in at least two plaquettes of X, and let Y be a component of X containing a plaquette of $\mathcal{P}_{S_A}$ and a plaquette of $\mathcal{P}_{S_B}$. Then $|Y| \geq g\,\tau$. More precisely, with the inner layers $\{a_2, \dots, b_1 - 1\}$ and the outer layers $\{b_2, \dots, a_1 - 1 + L_0\}$, Y contains at least g temporal plaquettes in every inner layer, or at least g temporal plaquettes in every outer layer.

Proof. Call the layers $s \in \{a_2, \dots, b_1 - 1\}$ inner (τ_{in} of them) and the layers $s' \in \{b_2, \dots, a_1 - 1 + L_0\}$ outer (τ_{out} of them); as integers, $s < s' < s + L_0$ for every inner s and outer s', so inner and outer layers are distinct. *These layers are free.* A temporal link of an inner layer s joins s and $s+1$, where $s+1 \in \{a_2 + 1, \dots, b_1\}$ is not in $\{a_1, \dots, a_2\}$ and $s \in \{a_2, \dots, b_1 - 1\}$ is not in $\{b_1, \dots, b_2\}$ modulo L_0 (because $b_2 - a_1 < L_0$); so it belongs neither to S_A nor to S_B. The same argument applies to outer layers. *Times of the plaquettes of Y at the supports.* Let $p \in Y \cap \mathcal{P}_{S_A}$, containing a link $\ell \in S_A$. If p is spatial, $T(p) = \{r\}$ with $r \in \{a_1, \dots, a_2\}$; if p is temporal of layer r, then ℓ is either its temporal link of layer r or a spatial link at time r or $r+1$, so $r \in \{a_1 - 1, \dots, a_2\}$. Now let s be inner and s' outer with $Y_s = Y_{s'} = \emptyset$, and I_1, I_2 as in Lemma 3.3. Then $T(p) \subseteq I_2 = \{s' + 1, \dots, s\}$: the times $a_1, \dots, a_2$ lie in I_2 because $s' + 1 \leq a_1 + L_0$ and $s \geq a_2$; the time $a_2 + 1$ occurs in $T(p)$ only if p is temporal of layer a_2, which is excluded when $s = a_2$ (as $Y_s = \emptyset$) and lies in I_2 when $s \geq a_2 + 1$; similarly the time $a_1 - 1$ occurs only if p is temporal of layer $a_1 - 1$, which is excluded when $s' \equiv a_1 - 1$ and lies in I_2 otherwise. In the same way $T(p) \subseteq I_1 = \{s + 1, \dots, s'\}$ for every $p \in Y \cap \mathcal{P}_{S_B}$. *Conclusion.* If $Y_s = Y_{s'} = \emptyset$ for some inner s and some outer s', Lemma 3.3 puts all of Y into one of the disjoint classes I_1, I_2, which contradicts the previous paragraph, since Y meets both $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$. Hence $Y_s \neq \emptyset$ for all inner layers, or $Y_{s'} \neq \emptyset$ for all outer layers. In either case Lemma 3.2 (with $S = S_A \cup S_B$) gives at least g plaquettes of Y in each of these layers, and plaquettes of different layers are different. □

4 The covariance bound and the mass gap

Theorem 4.1 (Covariance bound in finite volume)**.** *Let $d \geq 2$, $0 < \beta < \beta_0(d)$, and let Λ be a periodic lattice with all $L_\lambda \geq 3$; let g be the girth of Lemma 3.1. Let A, B be bounded measurable functions on Ω_Λ depending only on disjoint link sets S_A, S_B as in Corollary 3.4, with $\tau = \min(\tau_{\text{in}}, \tau_{\text{out}})$. Then, with $q = q(\beta) < 1$ (Lemma 4.2) and $c = c(\beta)$ as in Theorem 1.2,*

$$|\mathrm{Cov}_{\mu_{\Lambda,\beta}}(A, B)| \leq \frac{2\|A\|_\infty \|B\|_\infty}{1 - q}\, c^{|S_A| + |S_B|}\, q^{g\tau}.$$

Proof. If S_A or S_B is empty, A or B is constant and the covariance vanishes; so let both be non-empty. Let $S := S_A \cup S_B$ and F as in Lemma 2.5. By Lemmas 2.4 and 2.5, $2\,\mathrm{Cov}(A, B) = \sum_X \Phi_F(X)(\Xi(\mathcal{P} \setminus \overline{X})/\Xi(\mathcal{P}))^2$, where only sets $X \in \mathcal{X}_S$ contribute in which every link of $E(X) \setminus S$ lies in at least two plaquettes of X and some component meets both $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$; by Corollary 3.4 these have $|X| \geq n_0 := g\tau$. For each of them: $|\Phi_F(X)| \leq 4\|A\|_\infty \|B\|_\infty\, \epsilon(2\beta)^{|X|}$ by Lemmas 2.5(i) and 2.2(a); and by [III, Proposition 'Removing plaquettes' (`al:prop:remove`)] and Definition 2.3,

$$\Big(\frac{\Xi(\mathcal{P} \setminus \overline{X})}{\Xi(\mathcal{P})}\Big)^2 \leq e^{2\theta|\overline{X}|} \leq e^{2\theta(K+1)|X|} e^{2\theta J_S} \leq e^{2|X|} e^{2\theta J_S},$$

using $(K+1)\theta < 1$ [III, Lemma `al:lem:constants`]. The number of $X \in \mathcal{X}_S$ with $|X| = n$ is at most $\binom{J_S + Kn}{n}$: this is [III, Corollary 'Counting loop polymers' (`al:cor:countloop`)] with E_C replaced by S (its proof uses only $|\mathcal{P}_S| \leq J_S$ and the adjacency bound K; it requires $J_S \geq 1$, i.e. $S \neq \emptyset$). With $\binom{m}{n} t^n \leq (1 + t)^m$ at $t = 1/K$, $\binom{J_S + Kn}{n} \leq (1 + \frac{1}{K})^{J_S} (1 + \frac{1}{K})^{Kn} K^n \leq$

$(1+\frac{1}{K})^{J_S}(eK)^n$. Altogether

$$2|\operatorname{Cov}(A,B)| \le 4\|A\|_\infty\|B\|_\infty\, e^{2\theta J_S}\Big(1+\frac{1}{K}\Big)^{J_S}\sum_{n\ge n_0}\big(eK\cdot e^2\epsilon(2\beta)\big)^n = \frac{4\|A\|_\infty\|B\|_\infty\, c^{|S|}}{1-q}\,q^{n_0},$$

since $e^3K\epsilon(2\beta)=q$ and $(e^{2\theta}(1+\frac{1}{K}))^{J_S}=c^{|S|}$. □

Lemma 4.2 (Constants)**.** *For $d\ge 2$ and $0<\beta<\beta_0(d)$: $q(\beta)<\frac{e(2+\epsilon_0)}{K+2}$, which is <0.909 for $d=2$ and <0.389 for $d\ge 3$; and $4\log\frac{1}{q(\beta)} = 4\log\frac{1}{\beta} - 4\log(4e^3K) + O(\beta)$ as $\beta\downarrow 0$. Moreover $\beta_0(d)<\beta_*(d)=\operatorname{artanh}\frac{1}{6(d-1)}$, the threshold of [II].*

Proof. With $\epsilon = e^{2\beta}-1<\epsilon_0$ (which is the condition $\beta<\beta_0(d)$), $e^{4\beta}-1=\epsilon(2+\epsilon)<\epsilon_0(2+\epsilon_0)$ and $e^3K\epsilon_0=\frac{e}{K+2}$; for $d=2$, $K=4$ and $\frac{e(2+\epsilon_0)}{6}<0.909$; for $d\ge 3$, $K\ge 12$ and $\frac{e(2+\epsilon_0)}{14}<0.389$. The asymptotics follow from $e^{4\beta}-1=4\beta+O(\beta^2)$. The last claim is [II, Corollary 'Area law in infinite volume' (`iv:cor:arealaw`)] (first line of its proof). □

Proof of Theorem 1.2. The bound on q is Lemma 4.2. (iii) Since $\beta<\beta_*(d)$, the infinite-volume Gibbs measure μ_β of [II, Theorem 'Uniqueness and convergence' (`iv:thm:unique`)] exists and is the limit of the periodised measures $\tilde\mu_{\Lambda_k,\beta}$ on every bounded measurable local function, for every sequence of periodic lattices with all sides tending to infinity. Take such Λ_k with spatial sides ≥ 4. For k large, the reduction map is injective on the links of $S_A\cup S_B$, the functions $A\circ\iota$, $B\circ\iota$ on Ω_{Λ_k} (in the notation of [II]) depend on the images of S_A, S_B, which satisfy the hypotheses of Theorem 4.1 with the same a_i,b_i, $g=4$, $\tau_{\text{in}}=b_1-a_2$ and $\tau_{\text{out}}\to\infty$. Theorem 4.1 bounds $\operatorname{Cov}_{\tilde\mu_{\Lambda_k,\beta}}(A,B)$ uniformly in k, and the left side converges to $\operatorname{Cov}_{\mu_\beta}(A,B)$.

(ii) Let $F\in\mathcal{A}_+$ in the notation of [II, Definition 'Reflection, shifts, half-space algebras' (`iv:def:reflection`)], depending only on a finite set of links whose endpoints have time coordinates in $[0,n_F]$. Then ΘF depends only on links with endpoint times in $[-n_F,0]$ and S_tF on links with endpoint times in $[t,t+n_F]$, with the same number of links. For $t\ge 1$ the two sets are disjoint and (iii) gives, by θ- and translation invariance of μ_β (so that $\mu_\beta(\Theta F)\mu_\beta(S_tF)=|\mu_\beta(F)|^2$),

$$\big|\mu_\beta(\Theta F\cdot S_tF)-|\mu_\beta(F)|^2\big| = \big|\operatorname{Cov}_{\mu_\beta}(\Theta F,S_tF)\big| \le C_F\,q^{4t}\qquad (t\ge 1),$$

with C_F independent of t (here $b_1-a_2=t$); for $t=0$ the left side is at most $2\|F\|_\infty^2$. So the hypothesis of [II, Theorem `iv:thm:OS`](iii) holds with $m=4\log(1/q)$ and the constant $\max(C_F,2\|F\|_\infty^2)$, and $m(\mu_\beta)\ge 4\log(1/q)$.

(i) Let A be a bounded gauge-invariant function on $\Omega_s=G^{E(\Lambda_s)}$ and $\Lambda=(\mathbb{Z}/L_0\mathbb{Z})\times\Lambda_s$ with $L_0\ge 3$. The functions $\overline{A}(t)$ and $A(0)$ depend on the spatial links of the slices t and 0: in Corollary 3.4 take $a_1=a_2=0$, $b_1=b_2=t$; for $1\le t\le L_0/2$, $\tau=\min(t,L_0-t)=t$, and $g=4$ since all $L_i\ge 4$. Theorem 4.1 gives $|\langle\overline{A}(t)A(0)\rangle-\langle\overline{A}(t)\rangle\langle A(0)\rangle|\le C_Aq^{4t}$ with C_A depending on $\|A\|_\infty$ and $|E(\Lambda_s)|$ only, not on L_0; for $t=0$ the left side is at most $2\|A\|_\infty^2$. [F, Corollary `cor:gapfromcorr`] (whose proof lets $L_0\to\infty$ at fixed t, so that the bound is needed only for large L_0) gives $m_{\Lambda_s}(\beta)\ge 4\log(1/q)$. □

5 The matching upper bound

Throughout this section $d\ge 3$, G is metrisable (as in Theorem 1.2), and ρ is irreducible and non-trivial (automatic for an irreducible faithful ρ when $G\ne\{1\}$); $N=d_\rho$, $\iota:=1$ if $\rho\cong\bar\rho$ and $\iota:=0$ otherwise, $\tau_0:=\frac{1+\iota}{2}$, and

$$u_r(\beta):=\frac{c_r(\beta)}{d_r\,c_0(\beta)}\quad (r\text{ irreducible}),\qquad u(\beta):=u_\rho(\beta)=\frac{c_\rho(\beta)}{Nc_0(\beta)},$$

$$\varphi_\beta := 1 - \frac{c_0(\beta)}{w_\beta} = \frac{f_\beta}{1+f_\beta}.$$

Since w_β is real and invariant under inversion, $c_r = \int \overline{\chi_r}\, w_\beta\, dU = \int \chi_r\, w_\beta\, dU = c_{\bar r}$ is real (and $c_r > 0$ by [F, Lemma 'Positivity of the character coefficients' (`lem:cr`)]); so $u_{\bar r} = u_r > 0$, and $f_\beta = \sum_{r\neq 0} d_r u_r \chi_r$, absolutely and uniformly convergent ([F, Corollary `cor:uniform`]). Fix two spatial directions, labelled $1, 2$. On a periodic lattice Λ (all $L_\lambda \geq 3$) let $p_0 := (0, 12)$ be the spatial plaquette at the origin and time 0, $p_1 := (e_0, 12)$ its translate at time 1, Q the unit cube with bottom p_0 and top p_1, $\mathcal{Q}$ the set of its six faces, $X_0 \subseteq \mathcal{Q}$ its four sides (the temporal plaquettes $(y, 0i)$ of layer 0 with $(y, i) \in E(p_0)$), $S_Q := E(\mathcal{Q})$ its twelve edges and $S_A := E(p_0)$; and

$$A := \operatorname{Re}\chi_\rho(U_{p_0}), \qquad A_1 := \operatorname{Re}\chi_\rho(U_{p_1}).$$

The same notation is used on $\mathbb{Z}^d$.

The proof has three steps. (a) A positivity property of the transfer matrix (Lemma 5.2) bounds $\operatorname{Cov}(A, A_1)$ from below by the expectation of one function of the twelve edges of Q, in which the four sides carry the factor φ_β, i.e. are forced to carry non-trivial characters. (b) The loop expansion of [III], applied to this local expectation, gives an exact integral over the cube, evaluated by character orthogonality (Lemma 5.1), plus terms with at least five further plaquettes (Lemma 5.4). (c) The spectral inequality $\operatorname{Cov} \leq e^{-m}\operatorname{Var}$. Unlike the duplicated expansion of Section 2, nothing here compares the leading term with the other surfaces connecting p_0 and p_1: the remainder is relative to the minimal tube itself. This is why the result holds on the whole range $\beta < \beta_0(d)$.

Lemma 5.1 (Character computations)**.** *For every $\beta > 0$:*

(i) $\int A\, dU = 0$ *and* $\int A^2\, dU = \tau_0 = \frac{1+\iota}{2}$.

(ii) Let a_0, a_1 be real continuous class functions on G, invariant under $U \mapsto U^{-1}$, with $\sum_r |\hat a_{k,r}| d_r < \infty$, where $\hat a_{k,r} := \int a_k \overline{\chi_r}\, dU$. Then $\hat a_{k,r} = \hat a_{k,\bar r} \in \mathbb{R}$ and

$$\int a_0(U_{p_0})\, a_1(U_{p_1}) \prod_{s\in X_0} f_\beta(U_s)\, dU = \sum_{r\neq 0} \hat a_{0,r}\, \hat a_{1,r}\, u_r^4.$$

In particular $\int A A_1 \prod_{s\in X_0} f_\beta(U_s)\, dU = \tau_0 u^4$.

(iii) $\frac{(1+\iota)\beta e^{-\beta}}{2N^2} \leq u(\beta) \leq \frac{(1+\iota)\beta + \beta^2 e^\beta}{2N^2}$.

(iv) $\int |\varphi_\beta|\, dU \leq e^{3\beta}\beta\sqrt{\tau_0}/N \leq e^{4\beta} N\, u(\beta)/\sqrt{\tau_0}$.

Proof. (i) $\int \chi_\rho = 0$ since ρ is irreducible and non-trivial. With $\operatorname{Re}\chi = \frac12(\chi + \bar\chi)$, $\int (\operatorname{Re}\chi_\rho)^2 = \frac14(\int \chi_\rho^2 + 2\int |\chi_\rho|^2 + \int \overline{\chi_\rho}^2)$. By Schur orthogonality [F, Cited Theorem 'Haar measure, Peter–Weyl' (`thm:pw`)](ii), $\int |\chi_\rho|^2 = 1$ and, since $\chi_{\bar\rho} = \overline{\chi_\rho}$, $\int \chi_\rho^2 = \int \chi_\rho \overline{\chi_{\bar\rho}} = \iota$; likewise $\int \overline{\chi_\rho}^2 = \iota$.

(ii) Substituting $U \mapsto U^{-1}$ (Haar measure is inversion invariant) and using $a_k(U^{-1}) = a_k(U)$, $\chi_r(U^{-1}) = \overline{\chi_r(U)}$: $\hat a_{k,r} = \int a_k \chi_r\, dU$, which is $\overline{\hat a_{k,r}}$ (a_k is real) and equals $\int a_k \overline{\chi_{\bar r}} = \hat a_{k,\bar r}$. Orient the six faces of Q as its boundary. The proof of [F, Proposition 'Cube activity' (`prop:cube`)] establishes, in its line "Altogether $\int \prod_p \chi_{r_p}(U_p)\, dU = \delta_{\text{all } r_p = r}\, d_r^{-4}$" (whose derivation does not use $r_p \neq 0$): for irreducible representations attached to the six faces, $\int \prod_{\text{faces}} \chi_{r_{\text{face}}}(U_{\text{face}})\, dU$ equals d_r^{-4} if all six are equivalent to the same r, and 0 otherwise. Each boundary-oriented face holonomy is conjugate to U_p or U_p^{-1}, and a_0, a_1, f_β are class functions invariant under inversion, so they are the same functions of the boundary-oriented holonomies. Insert $a_k = \sum_r \hat a_{k,r}\chi_r$ for the bottom and the top and $f_\beta = \sum_{r\neq 0} d_r u_r \chi_r$ for the four sides. The resulting six-fold series converges absolutely, because $|\int \prod \chi_{r_{\text{face}}}| \leq \prod d_{r_{\text{face}}}$, $\sum_r |\hat a_{k,r}| d_r < \infty$ and $\sum_r d_r^2 u_r = \sum_r d_r c_r / c_0 = e^\beta/c_0$ ([F, eq. `eq:trace`]); so it may be integrated term by term. Only the terms in which all six faces carry the same r survive, and

$r \neq 0$ because f_β has no trivial component; such a term contributes $\hat{a}_{0,r}\hat{a}_{1,r}(d_r u_r)^4 d_r^{-4}$. For $a_0 = a_1 = \operatorname{Re}\chi_\rho$: if $\iota = 1$, χ_ρ is real, $\hat{a}_{k,\rho} = 1$, and the sum is u^4; if $\iota = 0$, $\hat{a}_{k,\rho} = \hat{a}_{k,\bar{\rho}} = \frac{1}{2}$ and the sum is $2 \cdot \frac{1}{4}u^4$. In both cases it is $\tau_0 u^4$.

(iii) By the proof of [F, Lemma `lem:cr`], $c_\rho = \sum_{n,m\geq 0} \frac{t^{n+m}}{n!\,m!} m_\rho(n,m)$ with $t = \beta/(2N)$ and $m_\rho(n,m)$ the multiplicity of ρ in $\rho^{\otimes n} \otimes \bar{\rho}^{\otimes m}$. Here $m_\rho(0,0) = 0$, $m_\rho(1,0) = 1$, $m_\rho(0,1) = \iota$, and $m_\rho(n,m) \leq N^{n+m-1}$ (dimension count). Hence $(1+\iota)t \leq c_\rho \leq (1+\iota)t + N^{-1}(e^{2Nt} - 1 - 2Nt) \leq \frac{(1+\iota)\beta}{2N} + \frac{\beta^2 e^\beta}{2N}$, using $e^y - 1 - y \leq \frac{y^2}{2}e^y$ with $y = 2Nt = \beta$. Divide by Nc_0 and use $1 \leq c_0 \leq e^\beta$ ([III, Lemma `al:lem:facts`](b)).

(iv) As $1 + f_\beta = w_\beta/c_0 \geq e^{-2\beta}$ ([III, Lemma `al:lem:facts`](a),(b)), $|\varphi_\beta| \leq e^{2\beta}|f_\beta|$, and by the Cauchy–Schwarz inequality $\int |\varphi_\beta| \leq e^{2\beta}(\int f_\beta^2)^{1/2}$. Now $\int f_\beta^2 = c_0^{-2}\int (w_\beta - c_0)^2 \leq \int (w_\beta - 1)^2$, because $c_0 \geq 1$ and $c = c_0 = \int w_\beta$ minimises $\int (w_\beta - c)^2$. With $y := \frac{\beta}{N}\operatorname{Re}\chi_\rho$ we have $|y| \leq \beta$, $|e^y - 1| \leq |y|e^{|y|}$, and so $\int (w_\beta - 1)^2 \leq e^{2\beta}\int y^2 = e^{2\beta}\beta^2\tau_0/N^2$ by (i). This gives the first inequality; the second follows from $\beta \leq e^\beta N^2 u/\tau_0$, which is (iii). □

Lemma 5.2 (Positivity)**.** *Let $\Lambda = (\mathbb{Z}/L_0\mathbb{Z}) \times \Lambda_s$ be periodic with all sides ≥ 3, $\beta > 0$, B a real bounded measurable function on Ω_s, $b \in \mathbb{R}$, and $E_0 \subseteq E_s$ a set of spatial links; for $\ell \in E_s$ let s_ℓ be the temporal plaquette of layer 0 (between the slices 0 and 1) based on ℓ. Then, with $\Delta_B := (B(0) - b)(B(1) - b)$,*

$$\langle \Delta_B \rangle_{\Lambda,\beta} \;\geq\; \Big\langle \Delta_B \prod_{\ell \in E_0} \varphi_\beta(U_{s_\ell}) \Big\rangle_{\Lambda,\beta}.$$

In particular, if $b = \langle B(0)\rangle_{\Lambda,\beta}$, which equals $\langle B(1)\rangle_{\Lambda,\beta}$ by translation invariance, the left side is $\operatorname{Cov}_{\Lambda,\beta}(B(0), B(1))$.

Proof. We use the transfer matrix $\hat{T} = VK_1P_{\mathcal{G}}V$ of [F, Definition `def:T`] on $L^2(\Omega_s)$, where $P_{\mathcal{G}}$ is the projection onto the gauge-invariant subspace (denoted P in [F]). By [F, Lemma 'The one-link operator' (`lem:kop`)], the one-link operator is $k_\beta = c_0\Pi_0 + k'_\beta$ with $k'_\beta := \sum_{r\neq 0}(c_r/d_r)\Pi_r$; both summands are positive operators ($c_r > 0$), with kernels c_0 and $w_\beta(UV^{-1}) - c_0$. For $F \subseteq E_0$ let $K_1^{(F)}$ be the tensor product over $\ell \in E_s$ of k'_β ($\ell \in F$), $c_0\Pi_0$ ($\ell \in E_0 \setminus F$) and k_β ($\ell \notin E_0$). Expanding $\bigotimes_{\ell\in E_0}(c_0\Pi_0 + k'_\beta)$ gives $K_1 = \sum_{F\subseteq E_0} K_1^{(F)}$. Each $K_1^{(F)}$ is positive, as a tensor product of positive operators, and its kernel is a product over the links of class functions of $U_\ell U_\ell'^{-1}$; so, as in the proof of [F, Lemma `lem:K`](i), it commutes with every R_g and hence with $P_{\mathcal{G}}$. Put $\hat{T}_F := VK_1^{(F)}P_{\mathcal{G}}V = P_{\mathcal{G}}(VK_1^{(F)}V)P_{\mathcal{G}}$. Then $\hat{T}_F \geq 0$ and $\hat{T} = \sum_F \hat{T}_F$, so $0 \leq \hat{T}_F \leq \hat{T}$ and $\hat{T}_F$ is trace class.

Let $\tilde{B}$ be multiplication by $B - b$. By [F, Theorem 'Transfer matrix formalism' (`thm:tm`)],

$$\langle \Delta_B \rangle = \frac{\operatorname{Tr}(\hat{T}^{L_0-1}\tilde{B}\hat{T}\tilde{B})}{\operatorname{Tr}\hat{T}^{L_0}} = \sum_{F\subseteq E_0} \frac{\operatorname{Tr}(\hat{T}^{L_0-1}\tilde{B}\hat{T}_F\tilde{B})}{\operatorname{Tr}\hat{T}^{L_0}}.$$

Every term is non-negative: by cyclicity of the trace ($\hat{T}_F$ is trace class, the other factors are bounded), $\operatorname{Tr}(\hat{T}^{L_0-1}\tilde{B}\hat{T}_F\tilde{B}) = \operatorname{Tr}(\hat{T}_F^{1/2}\,\tilde{B}\hat{T}^{L_0-1}\tilde{B}\,\hat{T}_F^{1/2})$, the trace of a positive operator. Finally, the term $F = E_0$ is the right side of the claim. Indeed, the proof of [F, Theorem `thm:tm`] applies verbatim when, in layer 0, the weights $w_\beta(U_{s_\ell})$, $\ell \in E_0$, are replaced by $w_\beta(U_{s_\ell})\varphi_\beta(U_{s_\ell}) = w_\beta(U_{s_\ell}) - c_0$, again a class function invariant under inversion: integrating out the temporal links of layer 0 then gives, by the computation in the proof of [F, Lemma `lem:K`](ii), the kernel of $K_1^{(E_0)}P_{\mathcal{G}}$ instead of that of $K = K_1P_{\mathcal{G}}$, i.e. $\hat{T}_{E_0}$ instead of $\hat{T}$ between the slices 0 and 1. (Sides ≥ 2 would suffice here.) □

Remark 5.3. Lemma 5.2 is a form of reflection positivity for the reflection through the plane $x_0 = \frac{1}{2}$, which bisects links ([F, Remark 'Other reflections'](ii)), applied to each piece of a

decomposition of the weights of the crossing plaquettes into positive-definite parts. It holds for every $\beta > 0$. Its point is that it discards, with the right sign, every contribution to the covariance except the one in which the four sides of Q carry non-trivial characters.

Lemma 5.4 (Five plaquettes)**.** *Let Λ be periodic with all sides ≥ 3, or $\Lambda = \mathbb{Z}^d$. Let $\mathcal{L} \subseteq E(\Lambda)$, and let $Y \neq \emptyset$ be a finite set of plaquettes such that every plaquette of Y contains at most one link of $\mathcal{L}$ and every link of $E(Y) \setminus \mathcal{L}$ lies in at least two plaquettes of Y. Then $|Y| \geq 5$.*

Proof. Two facts. (a) Distinct plaquettes share at most one link. (b) For a link m and a vertex $v \notin m$, at most one plaquette contains both. For (a): two links of a plaquette either meet at a corner, and two perpendicular links at a common vertex lie in exactly one plaquette; or they are opposite, (z, μ) and $(z + e_\nu, \mu)$, and then the plaquette is $(z, \mu\nu)$, determined because $z + e_\nu \neq z - e_\nu$ when $L_\nu \geq 3$. For (b): with $m = (z, \mu)$, the plaquettes containing m are $(z, \mu\nu)$ and $(z - e_\nu, \mu\nu)$, $\nu \neq \mu$; their vertices off m are $z \pm e_\nu$, $z + e_\mu \pm e_\nu$, and these $4(d-1)$ points are distinct when all sides are ≥ 3, so v determines the plaquette.

Let $q \in Y$. It has at least three links outside $\mathcal{L}$, each of which lies in a further plaquette of Y, and by (a) these further plaquettes are distinct; so $|Y| \geq 4$. Suppose $|Y| = 4$. Applying this to every plaquette of Y: each has exactly three links outside $\mathcal{L}$ (four would need four further plaquettes) and exactly one in $\mathcal{L}$, and it shares each of its three outside links with a different one of the other three plaquettes; by (a), any two plaquettes of Y share exactly one link. Label the links of q cyclically $m_{\mathcal{L}}, \ell_1, \ell_2, \ell_3$ with $m_{\mathcal{L}} \in \mathcal{L}$ (so ℓ_1 and ℓ_3 are opposite), let q_i be the plaquette of $Y \setminus \{q\}$ containing ℓ_i, and let $v := \ell_1 \cap \ell_2$, $v' := \ell_2 \cap \ell_3$. The plaquettes q_1, q_2 both contain v and share a link m; by (b), $v \in m$, since otherwise $q_1 = q_2$. Likewise q_2, q_3 share a link $m' \ni v'$. As q_1 shares only ℓ_1 with q, $m \neq \ell_2$, and likewise $m' \neq \ell_2$. So q_2 contains ℓ_2 and a further link at each endpoint of ℓ_2: it is spanned by ℓ_2 and a unit vector $e \perp \ell_2$, with m the link from v in direction e and m' the link from v' in direction e. The vector e is not parallel to ℓ_1: if $m = \ell_1$, then $q_2 \ni \ell_1$, but q_2 shares only ℓ_2 with q; and if m and ℓ_1 point in opposite directions from v, they are collinear and no plaquette contains both, whereas q_1 does. Hence e is perpendicular to ℓ_1 and ℓ_2, q_1 is the plaquette spanned at v by ℓ_1 and e, and q_3 the plaquette spanned at v' by ℓ_3 and e. Since $\ell_3 = \ell_1 + (v' - v)$, $q_3 = q_1 + (v' - v)$ is the translate of q_1 by a unit vector perpendicular to its plane, so q_1 and q_3 have no common vertex and share no link: a contradiction. Hence $|Y| \geq 5$. □

The bound is sharp: the five faces other than p_0 of a unit cube containing p_0 satisfy the hypotheses with $\mathcal{L} = E(p_0)$.

Lemma 5.5 (Two-point bounds in finite volume)**.** *Let $0 < \beta < \beta_0(d)$ and let Λ be periodic with all sides ≥ 3. Put*

$$q_* := eK\, e^{(K+1)\theta}\epsilon \;\leq\; x < 1,$$

$$\mathcal{T}_Q := (1+\epsilon)^6 e^{(24d-42)\theta}\Big(1 + \frac{1}{K}\Big)^{24(d-2)} \frac{q_*^5}{1-q_*}, \qquad \mathcal{T}_p := (1+\epsilon) e^{(8d-11)\theta}\Big(1 + \frac{1}{K}\Big)^{K} \frac{q_*^5}{1-q_*},$$

$$\bar{u} := \frac{(1+\iota)\beta + \beta^2 e^{\beta}}{2N^2}, \qquad a_* := N\bar{u}(1+\mathcal{T}_p) + \sqrt{\tau_0}\,\mathcal{T}_p, \qquad \eta := \frac{\epsilon}{\sqrt{\tau_0}} + \frac{a_* N \bar{u}}{\tau_0},$$

$$L(\beta) := \frac{(1-\eta)^2(1-\mathcal{T}_Q)}{1+\epsilon^6} - \Big(1 + \frac{a_*^2}{\tau_0}\Big) e^{16\beta}\, \frac{N^4}{\tau_0^2}\, \mathcal{T}_Q,$$

$$V(\beta) := (1+\epsilon)(1+\mathcal{T}_p) + \mathcal{T}_p,$$

with ϵ, x, θ, K as in the Convention. Then $a := \langle A \rangle_{\Lambda,\beta}$ satisfies $|a| \leq a_*$ and $\mathrm{Var}_{\Lambda,\beta}(A) \leq \tau_0\, V(\beta)$; and if $\eta \leq 1$,

$$\mathrm{Cov}_{\Lambda,\beta}(A, A_1) \;\geq\; \tau_0\, u(\beta)^4\, L(\beta).$$

Proof. The expansion. For $S \subseteq E(\Lambda)$ and a bounded measurable H depending only on the links of S, the proof of [III, Lemma 'Loop expansion' (`al:lem:loopexp`)] applies verbatim (with E_C replaced by S, as in Definition 2.3) and gives

$$\langle H\rangle_{\Lambda,\beta} = \sum_{X\in\mathcal{X}_S} \Phi_H(X)\, r_X, \qquad \Phi_H(X) := \int H \prod_{p\in X} f_\beta(U_p)\, dU, \qquad r_X := \frac{\Xi(\mathcal{P}\setminus\overline{X})}{\Xi(\mathcal{P})}.$$

If a link of $E(X)\setminus S$ lies in exactly one plaquette of X, then $\Phi_H(X)=0$ ([III, Lemma `al:lem:facts`](c)). Always $|\Phi_H(X)| \le \epsilon^{|X|}\int |H|\, dU$, by $\|f_\beta\|_\infty \le \epsilon$.

Let $(S,\mathcal{C})$ be $(S_Q,\mathcal{Q})$ or $(S_A,\{p_0\})$. Every plaquette not in $\mathcal{C}$ contains at most one link of S: two links of S in one plaquette either meet at a corner of Q (resp. p_0), and are then two perpendicular edges which span a face of Q (resp. p_0 itself), or are opposite links of the plaquette, i.e. parallel edges whose base points differ by a unit vector, which (sides ≥ 3) are edges of a common face. Each link lies in $2(d-1)$ plaquettes ([III, Lemma `al:lem:facts`](d)), and each edge of Q (resp. p_0) in two faces of Q (resp. in p_0); hence $|\mathcal{P}_{S_Q}| = 6+12(2d-4) = 24d-42$, $|\mathcal{P}_{S_Q}\setminus\mathcal{Q}| = 24(d-2)$, $|\mathcal{P}_{S_A}| = 1+4(2d-3) = 8d-11$ and $|\mathcal{P}_{S_A}\setminus\{p_0\}| = K$.

For $X\subseteq\mathcal{C}$ we have $E(X)\subseteq S$, hence $\overline{X}=\mathcal{P}_S$ and $r_X = r_\emptyset =: R_S$; and $\sum_{X\subseteq\mathcal{C}}\Phi_H(X) = \int H\prod_{p\in\mathcal{C}}(1+f_\beta(U_p))\,dU =: M_S(H)$. Let $X\in\mathcal{X}_S$ with $X\not\subseteq\mathcal{C}$ and $\Phi_H(X)\neq 0$, and put $Y := X\setminus\mathcal{C}$. A link of $E(Y)\setminus S$ lies in at least two plaquettes of X, none of which is in $\mathcal{C}$ (whose plaquettes have all their links in S); so Lemma 5.4 applies with $\mathcal{L}=S$ and gives $|Y|\ge 5$. Every component of Y contains a plaquette of $\mathcal{P}_S\setminus\mathcal{C}$: it lies in a component of X, which contains a plaquette of $\mathcal{P}_S$; if the component of Y is not itself a component of X, one of its plaquettes shares a link with a plaquette of $X\cap\mathcal{C}$, a link of S. Hence, by [III, Lemma 'Rooted counting' (`al:lem:rooted`)] applied to the graph on $(\mathcal{P}\setminus\mathcal{C})\cup\{o\}$ whose root o is joined to the J' plaquettes of $\mathcal{P}_S\setminus\mathcal{C}$, there are at most $\binom{J'+Kn}{n} \le (1+\frac{1}{K})^{J'}(eK)^n$ possible Y with $|Y|=n$. Moreover $\overline{X} = \mathcal{P}_S\cup\{p : E(p)\cap E(Y)\neq\emptyset\}$, so $|\overline{X}| \le |\mathcal{P}_S| + (K+1)|Y|$ and, by [III, Proposition 'Removing plaquettes' (`al:prop:remove`)], $r_X \le e^{\theta|\mathcal{P}_S|+\theta(K+1)|Y|}$. Summing over $X\cap\mathcal{C}$ (at most $(1+\epsilon)^{|\mathcal{C}|}$ in total weight) and over Y,

$$\big|\langle H\rangle_{\Lambda,\beta} - R_S\, M_S(H)\big| \;\le\; \int |H|\,dU\cdot(1+\epsilon)^{|\mathcal{C}|}e^{\theta|\mathcal{P}_S|}\Big(1+\frac{1}{K}\Big)^{J'}\sum_{n\ge 5} q_*^n, \tag{1}$$

which is $\int|H|\cdot\mathcal{T}_Q$ for $(S_Q,\mathcal{Q})$ and $\int|H|\cdot\mathcal{T}_p$ for $(S_A,\{p_0\})$. (Here $q_*\le x$ because $(K+1)\theta<1$, [III, Lemma `al:lem:constants`].)

The plaquette p_0. With $H=1$: $M_{S_A}(1) = \int(1+f_\beta) = 1$, so $|R_{S_A}-1|\le\mathcal{T}_p$. With $H=A$: $\int A(1+f_\beta(U_{p_0}))\,dU = \mathrm{Re}\int\chi_\rho w_\beta\,dU/c_0 = c_{\bar\rho}/c_0 = Nu$ (as $\int\chi_\rho\chi_r = \delta_{r\bar\rho}$), and $\int|A| \le (\int A^2)^{1/2} = \sqrt{\tau_0}$; so $|a|\le(1+\mathcal{T}_p)Nu+\sqrt{\tau_0}\,\mathcal{T}_p \le a_*$, by Lemma 5.1(iii). With $H=A^2$: $M_{S_A}(A^2)\le(1+\epsilon)\tau_0$, so $\mathrm{Var}(A)\le\langle A^2\rangle \le (1+\mathcal{T}_p)(1+\epsilon)\tau_0+\tau_0\mathcal{T}_p = \tau_0 V$.

Positivity. By Lemma 5.2 with $B=\mathrm{Re}\,\chi_\rho$ evaluated on p_0 (so that $B(0)=A$, $B(1)=A_1$), $b=a$ and $E_0=E(p_0)$,

$$\mathrm{Cov}(A,A_1) \;\ge\; \langle H_Q\rangle, \qquad H_Q := (A-a)(A_1-a)\prod_{s\in X_0}\varphi_\beta(U_s),$$

a function of the twelve edges of Q.

The cube. With $H=1$: $M_{S_Q}(1) = \int\prod_{p\in\mathcal{Q}}(1+f_\beta(U_p))\,dU = 1+\int\prod_{p\in\mathcal{Q}}f_\beta(U_p)\,dU =: M_1$. (For $\emptyset\neq X\subsetneq\mathcal{Q}$: the face-adjacency graph of the cube is connected, so some $p\in X$ and $p'\in\mathcal{Q}\setminus X$ share an edge of Q; that edge lies in exactly these two faces of Q, hence in exactly one plaquette of X, and $\int\prod_X f_\beta = 0$.) By Lemma 5.1(ii) with $a_0=a_1=f_\beta$, $M_1 = 1+\sum_{r\neq0} d_r^2 u_r^6 \in [1,1+\epsilon^6]$. So $R_{S_Q}M_1 \ge 1-\mathcal{T}_Q$ by (1). With $H=H_Q$: since $\varphi_\beta(1+f_\beta)=f_\beta$,

$$M_Q := M_{S_Q}(H_Q) = \int\tilde a(U_{p_0})\,\tilde a(U_{p_1})\prod_{s\in X_0} f_\beta(U_s)\,dU, \qquad \tilde a := (\mathrm{Re}\,\chi_\rho - a)(1+f_\beta).$$

The function $\tilde{a}$ is a real class function invariant under inversion, and its character expansion converges absolutely (that of f_β does, and multiplication by $\operatorname{Re}\chi_\rho$ maps χ_r to a combination of characters with total dimension-weighted coefficient $\le N d_r$). By Lemma 5.1(ii), $M_Q = \sum_{r\neq 0} \hat{a}_r^2 u_r^4 \ge \frac{2}{1+\iota}\hat{a}_\rho^2 u^4$ (the terms $r = \rho, \bar{\rho}$, equal), where

$$\hat{a}_\rho = \int \tilde{a}\,\overline{\chi_\rho}\,dU = \tau_0 + \int \operatorname{Re}\chi_\rho\, f_\beta\,\overline{\chi_\rho}\,dU - a\,Nu \;\ge\; \tau_0 - \epsilon\sqrt{\tau_0} - a_* N\bar{u} = \tau_0(1-\eta) \ge 0,$$

using $\int \operatorname{Re}\chi_\rho\overline{\chi_\rho} = \tau_0$, $\int(1+f_\beta)\overline{\chi_\rho} = Nu$ and $|\int \operatorname{Re}\chi_\rho f_\beta \overline{\chi_\rho}| \le \epsilon(\int(\operatorname{Re}\chi_\rho)^2)^{1/2}(\int|\chi_\rho|^2)^{1/2}$. Since $\frac{2}{1+\iota}\tau_0^2 = \tau_0$, $M_Q \ge \tau_0(1-\eta)^2u^4$.

Next, $\int|H_Q|\,dU \le (\tau_0 + a^2)(\int|\varphi_\beta|)^4$. Indeed, write b_ℓ, t_ℓ for the bottom and top edges of Q over $\ell \in E(p_0)$ and v_y for its temporal edges. The holonomy of the side over $\ell = (y,i)$ is conjugate to $U_{b_\ell}U_{v_{y+e_i}}U_{t_\ell}^{-1}U_{v_y}^{-1}$ or its inverse, so for fixed bottom and temporal edges the map from $(U_{t_\ell})_\ell$ to $W := (U_{s_\ell})_\ell$ is a bijection of G^4 preserving Haar measure. By the Cauchy–Schwarz inequality, $\int|H_Q| \le (\int(A-a)^2\prod_\ell|\varphi_\beta(W_\ell)|)^{1/2}(\int(A_1-a)^2\prod_\ell|\varphi_\beta(W_\ell)|)^{1/2}$. In the first integral, A depends only on the bottom edges and W is Haar distributed independently of them, so it equals $\int(A-a)^2\cdot(\int|\varphi_\beta|)^4 = (\tau_0+a^2)(\int|\varphi_\beta|)^4$ by Lemma 5.1(i); the second is the same, with bottom and top exchanged. By Lemma 5.1(iv), $\int|H_Q| \le (\tau_0 + a_*^2)e^{16\beta}N^4u^4/\tau_0^2$.

Finally, by (1), $M_Q \ge 0$ and $R_{S_Q} \ge (1-\mathcal{T}_Q)/M_1 \ge (1-\mathcal{T}_Q)/(1+\epsilon^6)$,

$$\langle H_Q\rangle \;\ge\; R_{S_Q}M_Q - \mathcal{T}_Q\int|H_Q| \;\ge\; \tau_0u^4\Big[\frac{(1-\eta)^2(1-\mathcal{T}_Q)}{1+\epsilon^6} - \Big(1+\frac{a_*^2}{\tau_0}\Big)e^{16\beta}\frac{N^4}{\tau_0^2}\mathcal{T}_Q\Big] = \tau_0u^4L(\beta).$$

□

Theorem 5.6 (Upper bound)**.** *Let $d \ge 3$, G metrisable, ρ irreducible and non-trivial, $0 < \beta < \beta_0(d)$, and with the notation of Lemma 5.5 suppose that $\eta(\beta) \le 1$ and $L(\beta) > 0$. Put $D(\beta) := \log(V(\beta)/L(\beta))$. Then*

(i) $m(\mu_\beta) \le 4\log\frac{1}{u(\beta)} + D(\beta)$;

(ii) $m_{\Lambda_s}(\beta) \le 4\log\frac{1}{u(\beta)} + D(\beta)$ for every periodic spatial lattice Λ_s with all sides ≥ 3.

As $\beta \downarrow 0$, $D(\beta) = 2(1+2/\sqrt{\tau_0})\beta + O(\beta^2)$; in particular the hypotheses hold for all sufficiently small β.

Proof. Write Γ_1 and Γ_0 for the covariance of A_1 and A and for the variance of A in the relevant theory. By Lemma 5.5, $\Gamma_1 \ge \tau_0u^4L > 0$ and $\Gamma_0 \le \tau_0 V$, so $\log(\Gamma_0/\Gamma_1) \le 4\log(1/u) + D(\beta)$, provided these bounds hold for the theory in question. It remains to show $m \le \log(\Gamma_0/\Gamma_1)$.

(ii) Let $\Lambda = (\mathbb{Z}/L_0\mathbb{Z})\times\Lambda_s$ with $L_0 \ge 3$, and A the corresponding gauge-invariant function on the spatial configurations, so that $A_1 = A(1)$ in the notation of [F, Theorem 'Transfer matrix formalism' (`thm:tm`)]. Since A is gauge invariant, [F, Theorem 'Ground-state expectations' (`thm:gs`)] applies: its first formula with $t = 1$ for the pair (A, A), and its second formula for A and for A^2, together with $\langle A(1)\rangle_\Lambda = \langle A(0)\rangle_\Lambda$ (translation invariance, [F, Theorem `thm:basic`](iv)), give, as $L_0 \to \infty$, $\operatorname{Cov}_\Lambda(A_1, A) \to \langle Q_\Omega\hat{A}\Omega, e^{-H}Q_\Omega\hat{A}\Omega\rangle$ and $\operatorname{Var}_\Lambda(A) \to \|Q_\Omega\hat{A}\Omega\|^2$, where Q_Ω is the projection orthogonal to the vacuum, as in the proof of [F, Theorem 'Gap ⇔ clustering in time' (`thm:gapclust`)](i); and $\langle Q_\Omega\hat{A}\Omega, e^{-H}Q_\Omega\hat{A}\Omega\rangle \le e^{-m_{\Lambda_s}(\beta)}\|Q_\Omega\hat{A}\Omega\|^2$. The bounds of Lemma 5.5 are uniform in L_0, so they hold for the limits Γ_1, Γ_0, and $e^{-m_{\Lambda_s}(\beta)} \ge \Gamma_1/\Gamma_0$.

(i) By [II, Theorem 'Uniqueness and convergence' (`iv:thm:unique`)](iii) (note $\beta < \beta_0(d) < \beta_*(d)$, Lemma 4.2), the covariance and the variance under μ_β are limits of those on periodic lattices with all sides tending to infinity, so the bounds of Lemma 5.5 hold for them. With ν and P as in [II, Corollary 'The time-slice chain' (`iv:cor:chain`)] and $\tilde{A} := A - \nu(A)$, a function of the time-zero slice, [II, Corollary `iv:cor:chain`](iv) gives $\Gamma_1 = \langle\tilde{A}, P\tilde{A}\rangle_\nu$ and

$\Gamma_0 = \|\tilde{A}\|_\nu^2$. Here $\tilde{A} \perp \mathbf{1}$; and $P \geq 0$ [II, Theorem 'Transfer operator, ground-state chain and mass gap' (`iv:thm:gap`)](ii), so $\|P|_{\mathbf{1}^\perp}\| = \sup\operatorname{spec}(P|_{\mathbf{1}^\perp}) = 1 - \operatorname{gap} P = e^{-m(\mu_\beta)}$ by [II, Theorem `iv:thm:gap`](iii). Hence $\Gamma_1 \leq e^{-m(\mu_\beta)}\Gamma_0$.

Finally, as $\beta \downarrow 0$: $\epsilon = 2\beta + O(\beta^2)$, $x, \theta, q_* = O(\beta)$, so $\mathcal{T}_Q, \mathcal{T}_p = O(\beta^5)$; $\bar{u}, a_* = O(\beta)$ and $\eta = 2\beta/\sqrt{\tau_0} + O(\beta^2)$. Hence $\log V = 2\beta + O(\beta^2)$ and $\log L = -2\eta + O(\beta^2)$. □

Lemma 5.7 (The constants on the whole range)**.** *Let $d \geq 3$ and $0 < \beta < \beta_0(d)$. Then $\epsilon < 8.06 \cdot 10^{-4}$, $\mathcal{T}_Q < 8.2 \cdot 10^{-4}$, $\mathcal{T}_p < 1.5 \cdot 10^{-5}$, $\eta < 1.14 \cdot 10^{-3}$, $V(\beta) < 1.00085$ and*

$$L(\beta) \;>\; 0.9969 - 8.2 \cdot 10^{-4}\, \frac{N^4}{\tau_0^2}.$$

Hence, if $N^4 \leq 1200\,\tau_0^2$ (i.e. $N \leq 4$, or $N = 5$ and $\rho \cong \bar{\rho}$), then $L(\beta) > 0.0129$ and $D(\beta) < \log(1.00085/(0.9969 - 8.2 \cdot 10^{-4} N^4/\tau_0^2))$; if $N \leq 3$ (e.g. $G = U(1)$, $SU(2)$, $SU(3)$), then $L(\beta) > 0.73$ and $D(\beta) < 0.32$. Moreover all the constants $\epsilon, x, \theta, q_*, \mathcal{T}_Q, \mathcal{T}_p, \bar{u}, a_*, \eta$ increase with β, so V increases, L decreases, and D increases with β.

Proof. $\beta_0(d)$ decreases with d, and $\beta < \beta_0(d)$ means $\epsilon < \epsilon_0(d) := \frac{1}{e^2 K(K+2)} \leq \frac{1}{168 e^2} < 8.056 \cdot 10^{-4}$. By [III, Lemma `al:lem:constants`], $x < \frac{1}{K+2}$ and $\theta < \frac{1}{K+1}$; so $q_* \leq x < \frac{1}{K+2}$ and $\frac{q_*^5}{1-q_*} < \frac{1}{(K+2)^4(K+1)}$. With $K = 8d - 12 \geq 12$: $(24d-42)\theta < \frac{24d-42}{8d-11} < 3$, $(8d-11)\theta < 1$, $(1+\frac{1}{K})^{24(d-2)} = (1+\frac{1}{K})^{3K-12} < e^3$ and $(1+\frac{1}{K})^K < e$. Hence $\mathcal{T}_Q < 1.00485\, e^6/(14^4 \cdot 13) < 8.12 \cdot 10^{-4}$ and $\mathcal{T}_p < 1.00081\, e^2/(14^4 \cdot 13) < 1.49 \cdot 10^{-5}$. Next $\beta < \beta_0(3) < 4.027 \cdot 10^{-4}$ gives $N\bar{u} < 4.03 \cdot 10^{-4}/N$, $a_* < 4.18 \cdot 10^{-4}$ (as $\sqrt{\tau_0} \leq 1$), and, with $\tau_0 \geq \frac{1}{2}$, $\eta < \sqrt{2} \cdot 8.056 \cdot 10^{-4} + 2 \cdot 4.18 \cdot 10^{-4} \cdot 4.03 \cdot 10^{-4} < 1.14 \cdot 10^{-3}$. So $(1-\eta)^2(1-\mathcal{T}_Q)/(1+\epsilon^6) > 0.99691$, $(1 + a_*^2/\tau_0) e^{16\beta} < 1.00647$, which gives the bound on L; and $V < 1.000806 \cdot 1.0000149 + 0.0000149 < 1.00085$. If $N^4 \leq 1200\tau_0^2$, then $L > 0.9969 - 0.984 = 0.0129$; since $\tau_0 \in \{\frac{1}{2}, 1\}$, this condition means $N^4 \leq 300$ ($\iota = 0$) or $N^4 \leq 1200$ ($\iota = 1$). For $N \leq 3$, $N^4/\tau_0^2 \leq 4 \cdot 81 = 324$, so $L > 0.9969 - 0.2657 > 0.73$ and $D < \log(1.00085/0.73) < 0.32$. Monotonicity is clear from the definitions. □

Corollary 5.8 (Two-sided bounds on the whole range)**.** *Let $d \geq 3$, G metrisable and ρ irreducible and non-trivial of dimension N, with $N^4 \leq 1200\,\tau_0^2$ (i.e. $N \leq 4$, or $N = 5$ and $\rho \cong \bar{\rho}$; e.g. $U(1)$, $SU(2)$, $SU(3)$, $SU(4)$). For every $0 < \beta < \beta_0(d)$,*

$$4 \log \frac{1}{q(\beta)} \;\leq\; m(\mu_\beta) \;\leq\; 4 \log \frac{1}{u(\beta)} + D(\beta), \qquad D(\beta) = O(\beta),$$

with $0 < D(\beta) < \log(1.00085/(0.9969 - 8.2 \cdot 10^{-4} N^4/\tau_0^2))$, and $D(\beta) < 0.32$ if $N \leq 3$; explicitly,

$$4 \log \frac{1}{\beta} - 4 \log(4e^3 K) - 4 \log \frac{e^{4\beta} - 1}{4\beta} \;\leq\; m(\mu_\beta) \;\leq\; 4 \log \frac{1}{\beta} + 4 \log \frac{2N^2}{1+\iota} + 4\beta + D(\beta).$$

The same bounds hold for $m_{\Lambda_s}(\beta)$ for every periodic spatial lattice with all sides ≥ 4.

If ρ is any irreducible non-trivial representation (no condition on N), the upper bounds hold for all sufficiently small β (Theorem 5.6); in particular $m(\mu_\beta) = 4 \log(1/\beta) + O(1)$ as $\beta \downarrow 0$, and the same holds uniformly for the finite-volume gaps on spatial lattices with all sides ≥ 4.

Proof. The lower bounds are Theorem 1.2(ii) and (i), with $q(\beta) = 4e^3 K\beta \cdot \frac{e^{4\beta}-1}{4\beta}$. The upper bounds are Theorem 5.6 and Lemma 5.7 (which also gives $\eta \leq 1$), with $\log \frac{1}{u} \leq \log \frac{2N^2}{(1+\iota)\beta} + \beta$ (Lemma 5.1(iii)). $D > 0$ because $V \geq 1 \geq L$. □

Remark 5.9 (Numbers, the first version, and checks). (i) Numerically (`python3` with `mpmath`), the supremum of $D(\beta)$ on $(0, \beta_0(d))$, its limit as $\beta \uparrow \beta_0(d)$ (D is increasing, Lemma 5.7), is

$4.8\cdot10^{-3}$, $4.9\cdot10^{-2}$ and $3.8\cdot10^{-3}$ for $SU(2)$, $SU(3)$ and $U(1)$ in $d=3$, and $1.4\cdot10^{-3}$, $1.0\cdot10^{-2}$ and $1.3\cdot10^{-3}$ in $d=4$; and $D(\beta)/\beta\to6$ for $SU(2)$, $\to2(1+2\sqrt{2})\approx7.66$ when $\iota=0$. So on the whole range the upper bound is $4\log(1/u)$ up to a few per cent. The two-sided window of Corollary 5.8 still has width about $4\log(4e^3K)+4\log\frac{2N^2}{1+\iota}$ (about 35.1 for $SU(2)$ in $d=4$). Since the strong-coupling expansion predicts $m=4\log(1/u)+O(u)$, it is expected (heuristic) to come entirely from the constant of the lower bound (Remark 10.3(ii)).

(ii) The first version of this section (9 November 2026; see the repository history and `journal/2026-11-09b.md`) obtained the upper bound from the duplicated expansion of Section 2: the four sides of Q form the unique minimal connecting set, with the exactly computable term $\Phi_F(X_0)=2\tau_0u^4$, and every other connecting set has at least five plaquettes. There the remainder, with sup-norm weights $\approx4\beta\cdot eK$ per plaquette, had to be compared with the leading term τ_0u^4, $u\approx(1+\iota)\beta/(2N^2)$, which restricted the result to $\beta\lesssim10^{-16}$. Lemma 5.2 removes this comparison: the remainder in Lemma 5.5 is relative to the tube itself, and the only price is the factor $(\int|\varphi_\beta|/u)^4\le e^{16\beta}N^4/\tau_0^2$.

(iii) Checks (a compass, not part of the proofs). Exact transfer-matrix computations for $\mathbb{Z}_2$ and $\mathbb{Z}_3$ in $d=3$ on a 3×3 spatial torus (physical spaces of dimension 2^{10} and 3^{10}): all sixteen terms of the decomposition in the proof of Lemma 5.2 are non-negative, the covariance dominates $\langle H_Q\rangle$, $\langle H_Q\rangle/(\tau_0u^4)\to1$, and for $\beta<\beta_0(3)$ the bounds of Lemma 5.5 hold with room (e.g. $\langle H_Q\rangle/(\tau_0u^4)=0.9999997\ge L=0.998$ at $\beta=4\cdot10^{-4}$ for $\mathbb{Z}_2$). On this lattice the true gap is $\approx3\log(1/u)$ (a torelon around a side of length 3), so Theorem 5.6(ii) is valid but not sharp there, as Remark 10.3(iii) expects. For S_3 with its two-dimensional representation, on a spatial graph consisting of one square, the decomposition was checked against brute-force summation over gauge-fixed configurations. Lemma 5.4 (with $\mathcal{L}=S_Q$ and $\mathcal{L}=S_A$) was checked by exhaustive enumeration on the tori $(3,3,3)$ and $(4,4,4)$ up to five plaquettes (the only five-plaquette sets are the caps: the five faces other than F of a unit cube $\neq Q$ sharing a face F with Q, resp. of a unit cube containing p_0) and on $(3,3,3,3)$ and $(4,3,3,3)$ up to four plaquettes.

6 A sharper lower bound: temporal plaquettes at their operator norm

In the proof of Theorem 4.1 every plaquette of a connecting set is bounded by the sup norm $\|g_p\|_\infty\le e^{4\beta}-1\approx4\beta$. For the temporal plaquettes this is far from sharp: viewed as the kernel of an operator between two time slices, $f_\beta(UV^{-1})$ has operator norm $\max_{r\neq0}u_r(\beta)$, which is $u(\beta)\approx(1+\iota)\beta/(2N^2)$ when $\dim\rho\le49$ (Corollary 6.5). The transfer-matrix representation of the terms of the duplicated loop expansion lets every temporal plaquette but those of one time layer carry this weight. The result is a lower bound of the form $4\log(1/u)-4\log(eK)-O(\beta)$ (when $u_*=u$), whose additive constant does not depend on G. Throughout this section G is metrisable when the infinite-volume theory of [II] is used (as in Theorem 1.2), u_r is as in Section 5, and

$$u_*(\beta):=\sup_{r\neq0}u_r(\beta)\ \le\ \epsilon(\beta),$$

the inequality because $u_r=d_r^{-1}\int f_\beta\overline{\chi_r}\,dU$ and $\int|\chi_r|\le1$. The ratios of partition functions are controlled by the following sharpening of [III, Proposition 'Removing plaquettes' (`al:prop:remove`)]: only closed polymers have non-zero activity.

Lemma 6.1 (Removing plaquettes: closed polymers)**.** *Let Λ be periodic with all sides ≥3, $0<\beta<\beta_0(d)$ and $\theta_5(\beta):=x(\beta)^5/(1-x(\beta))$, so that $\theta_5\le\theta$ and $\theta_5=O(\beta^5)$. Then for every $\mathcal{Q}\subseteq\mathcal{P}$*

$$e^{-\theta_5(\beta)|\mathcal{P}\setminus\mathcal{Q}|}\ \le\ \frac{\Xi(\mathcal{Q})}{\Xi(\mathcal{P})}\ \le\ e^{\theta_5(\beta)|\mathcal{P}\setminus\mathcal{Q}|}.$$

Proof. Let γ be a polymer and ℓ a link lying in exactly one plaquette p of γ. Then $U_p = AU_\ell^{\pm 1}B$ with A, B independent of U_ℓ, no other factor of $\prod_{p' \in \gamma} f_\beta(U_{p'})$ depends on U_ℓ, and $\int f_\beta(AU^{\pm 1}B)\, dU = 0$ ([III, Lemma `al:lem:facts`](c)); so $z(\gamma) = 0$. Hence $z(\gamma) \neq 0$ only if every link of $E(\gamma)$ lies in at least two plaquettes of γ, and then $|\gamma| \geq 5$ by Lemma 5.4 with $\mathcal{L} = \emptyset$. In the proof of [III, Lemma 'The Kotecký–Preiss condition for plaquette polymers' (`al:lem:KPsums`)] the sum over n may therefore start at $n = 5$, which gives $\sum_{\gamma \cap S \neq \emptyset} |z(\gamma)| e^{|\gamma|} \leq \theta_5 |S|$ for every $S \subseteq \mathcal{P}$; and [III, Lemma `al:lem:ratio`] applies with $a(\gamma) = |\gamma|$, because $(K+1)\theta_5 \leq (K+1)\theta < 1$. The proof of [III, Proposition `al:prop:remove`] then gives the claim. □

Lemma 6.2 (Layer operators)**.** *Let $\beta > 0$, let Π_0 be the projection onto the constants in $L^2(G)$, and let F_f be the integral operator on $L^2(G)$ with kernel $f_\beta(UV^{-1})$. Then $F_f = \sum_{r \neq 0} u_r \Pi_r$ (notation of [F, Lemma 'The one-link operator' (`lem:kop`)]), so $F_f \geq 0$, $\|F_f\| = u_*$ and $\operatorname{Tr} F_f = e^\beta/c_0 - 1$. The integral operator G_g on $L^2(G \times G)$ with kernel $g(UV^{-1}, U'V'^{-1}) := (1 + f_\beta(UV^{-1}))(1 + f_\beta(U'V'^{-1})) - 1$ is*

$$G_g = (\Pi_0 + F_f) \otimes (\Pi_0 + F_f) - \Pi_0 \otimes \Pi_0 = F_f \otimes \Pi_0 + \Pi_0 \otimes F_f + F_f \otimes F_f,$$

and $G_g \geq 0$, $\|G_g\| = u_*$, $\operatorname{Tr} G_g = (e^\beta/c_0)^2 - 1 \leq e^{2\beta} - 1$.

Proof. The kernel of $f_\beta = \sum_{r \neq 0} (c_r/c_0)\chi_r$ is the kernel of k_β/c_0 minus that of Π_0, so [F, Lemma `lem:kop`] gives $F_f = \sum_{r \neq 0} \frac{c_r}{d_r c_0} \Pi_r = \sum_{r \neq 0} u_r \Pi_r$, with $u_r > 0$; its norm is $\sup_{r \neq 0} u_r$ and its trace $\sum_{r \neq 0} u_r d_r^2 = (\operatorname{Tr} k_\beta - c_0)/c_0 = e^\beta/c_0 - 1$ by [F, eq. `eq:trace`]. The kernel of Π_0 is 1, which gives the formula for G_g. Its three summands are positive and act on the mutually orthogonal subspaces $\Pi_0^\perp \otimes \Pi_0$, $\Pi_0 \otimes \Pi_0^\perp$ and $\Pi_0^\perp \otimes \Pi_0^\perp$, so $\|G_g\| = \max(u_*, u_*, u_*^2) = u_*$, because $u_* \leq 1$: $|u_r| \leq \int |\chi_r| w_\beta\, dU/(d_r c_0) \leq 1$. Finally $\operatorname{Tr} G_g = (1 + \operatorname{Tr} F_f)^2 - 1$, and $c_0 \geq 1$. □

Lemma 6.3 (Transfer bound for duplicated terms)**.** *Let Λ be periodic, let A, B be bounded measurable functions of the spatial links of the time slices 0 and t, $0 < t < L_0$, and $F := (A - A')(B - B')$. For $X \subseteq \mathcal{P}$ let $T_s(X)$ be the set of temporal plaquettes of X in layer s, $T(X) := \bigcup_s T_s(X)$, $S(X) := X \setminus T(X)$ (its spatial plaquettes), $S_s(X)$ the spatial plaquettes of X at slice s, and $m(X) := \min_s |T_s(X)|$. Then*

$$|\Phi_F(X)| \;\leq\; 4\|A\|_\infty \|B\|_\infty\, \gamma_g^{|S(X)|}\, \big(e^{2\beta} - 1\big)^{m(X)}\, u_*^{|T(X)| - m(X)},$$

where $\gamma_g := \|g_p\|_\infty$ (the same for all p; $\gamma_g \leq e^{4\beta} - 1$ by Lemma 2.2(a)).

Proof. As in the proof of [F, Theorem 'Transfer matrix formalism' (`thm:tm`)], now for the duplicated system: label the spatial links of slice s in both copies by $(U^{(s)}, U'^{(s)}) \in \Omega_s \times \Omega_s$ and integrate out the temporal links of each layer. For layer s the integrand contains g_p for the temporal plaquettes $p \in T_s(X)$ and no factor for the others; the temporal links of the two copies are integrated independently, and every factor is a class function of the plaquette holonomy in each copy, invariant under inversion. So the computation in the proof of [F, Lemma `lem:K`](ii) shows that the integral over the temporal links of layer s is the kernel of

$$\mathcal{K}_s := \Big(\bigotimes_{\ell \in T_s(X)} G_g \otimes \bigotimes_{\ell \notin T_s(X)} (\Pi_0 \otimes \Pi_0) \Big)(P_{\mathcal{G}} \otimes P_{\mathcal{G}}),$$

where temporal plaquettes are identified with the spatial links on which they are based and $P_{\mathcal{G}}$ is the projection onto the gauge-invariant subspace (denoted P in [F]). Let $\mathbb{M}_s$ be multiplication by the product of the g_p, $p \in S(X)$ at slice s, times $A(U^{(0)}) - A(U'^{(0)})$ if $s = 0$ and times $B(U^{(t)}) - B(U'^{(t)})$ if $s = t$. Then, exactly as in [F, Theorem `thm:tm`], $\Phi_F(X) = \operatorname{Tr}\big(\mathbb{M}_0 \mathcal{K}_0 \mathbb{M}_1 \mathcal{K}_1 \cdots \mathbb{M}_{L_0 - 1} \mathcal{K}_{L_0 - 1}\big)$; every $\mathcal{K}_s$ is trace class. By Hölder's inequality for

traces, $|\Phi_F(X)| \le \|\mathcal{K}_{s_0}\|_1 \prod_{s\neq s_0}\|\mathcal{K}_s\| \prod_s \|\mathbb{M}_s\|$ for any s_0. By Lemma 6.2, $\|\mathcal{K}_s\| \le u_*^{|T_s(X)|}$ and $\|\mathcal{K}_s\|_1 \le (e^{2\beta}-1)^{|T_s(X)|}$ (and $\|\Pi_0\otimes\Pi_0\| = \mathrm{Tr}(\Pi_0\otimes\Pi_0) = 1$, $\|P_{\mathcal{G}}\otimes P_{\mathcal{G}}\| = 1$); by Lemma 2.2(a), $\|\mathbb{M}_s\| \le \gamma_g^{|S_s(X)|}$ times $2\|A\|_\infty$ or $2\|B\|_\infty$ at $s = 0, t$. Choose s_0 with $|T_{s_0}(X)| = m(X)$. □

Theorem 6.4 (Sharper covariance bound and gap)**.** *Let $d \ge 2$, $0 < \beta < \beta_0(d)$, let Λ be periodic with all sides ≥ 3 and girth g (Lemma 3.1), and let A, B be bounded measurable functions of finitely many spatial links of the time slices 0 and t, with $1 \le t \le L_0/2$; S is the set of these links. Put $\hat q(\beta) := eK\, e^{2(K+1)\theta_5(\beta)}(e^{4\beta}-1) \le q(\beta)$ and, with θ_5 as in Lemma 6.1,*

$$q_u(\beta) := eK\, e^{2(K+1)\theta_5(\beta)}\, u_*(\beta).$$

Then $\hat q < 1$ and, with $c = c(\beta)$ as in Theorem 1.2,

$$|\mathrm{Cov}_{\mu_{\Lambda,\beta}}(A,B)| \;\le\; \frac{2\|A\|_\infty\|B\|_\infty}{1-\hat q}\, c^{|S|}\, q_u^{gt}.$$

Consequently, for $d \ge 3$: (i) $m_{\Lambda_s}(\beta) \ge 4\log(1/q_u(\beta))$ for every periodic spatial lattice Λ_s with all sides ≥ 4; (ii) $m(\mu_\beta) \ge 4\log(1/q_u(\beta))$.

Proof. $\hat q \le q < 1$ because $(K+1)\theta < 1$ ([III, Lemma `al:lem:constants`], Lemma 4.2). As in the proof of Theorem 4.1, $2\,\mathrm{Cov}(A,B) = \sum_X \Phi_F(X) r_X^2$, $r_X := \Xi(\mathcal{P}\setminus\overline{X})/\Xi(\mathcal{P})$, where only sets $X \in \mathcal{X}_S$ contribute in which every link of $E(X)\setminus S$ lies in at least two plaquettes of X and some component meets both $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$. Fix such an X. Temporal links are never in S, so Lemma 3.2, applied to a component of X that meets layer s, shows that $T_s(X)$ is empty or has at least g elements, for every layer s. Put $n(X) := |T(X)| - m(X)$. If some layer contains no temporal plaquette of X, then $m(X) = 0$, and by Corollary 3.4 the connecting component has at least g temporal plaquettes in every inner layer or in every outer layer, so $n(X) \ge g\min(t, L_0 - t) = gt$. Otherwise every layer has at least $m(X) \ge g$ temporal plaquettes and $n(X) \ge (L_0-1)m(X) \ge (L_0-1)g \ge gt$. Let $\gamma := e^{4\beta}-1$. Since $u_* \le \epsilon \le \gamma$, $e^{2\beta}-1 \le \gamma$ and $\gamma_g \le \gamma$, Lemma 6.3 gives

$$|\Phi_F(X)| \le 4\|A\|_\infty\|B\|_\infty\, \gamma^{|X|-n(X)} u_*^{n(X)} \le 4\|A\|_\infty\|B\|_\infty\, \gamma^{|X|}\,(u_*/\gamma)^{gt},$$

and $|X| \ge n(X) \ge gt$. The rest is the computation in the proof of Theorem 4.1, with $r_X^2 \le e^{2\theta_5|\overline{X}|} \le e^{2\theta J_S} e^{2\theta_5(K+1)|X|}$ (Lemma 6.1) in place of $e^{2\theta|\overline{X}|}$, keeping the factor $e^{2\theta_5(K+1)|X|}$:

$$2|\,\mathrm{Cov}(A,B)| \le 4\|A\|_\infty\|B\|_\infty \Big(\frac{u_*}{\gamma}\Big)^{gt} c^{|S|} \sum_{n\ge gt} \big(eK\, e^{2\theta_5(K+1)}\gamma\big)^n = \frac{4\|A\|_\infty\|B\|_\infty\, c^{|S|}}{1-\hat q}\Big(\frac{u_*}{\gamma}\Big)^{gt}\hat q^{\,gt},$$

and $\hat q u_*/\gamma = q_u$.

(i) As in the proof of Theorem 1.2(i): for gauge-invariant bounded A on Ω_s, the bound with $g = 4$ (all spatial sides ≥ 4, Lemma 3.1) is uniform in L_0, and [F, Corollary `cor:gapfromcorr`] gives $m_{\Lambda_s}(\beta) \ge 4\log(1/q_u)$.

(ii) Let ψ be a real bounded measurable function of finitely many spatial links of the time-zero slice, and $\tilde\psi := \psi - \nu(\psi)$ with ν and P as in [II, Corollary 'The time-slice chain' (`iv:cor:chain`)]. By [II, Corollary `iv:cor:chain`](iv), $\langle\tilde\psi, P^t\tilde\psi\rangle_\nu = \mathrm{Cov}_{\mu_\beta}(\psi(U^{(0)}), \psi(U^{(t)}))$, and by [II, Theorem 'Uniqueness and convergence' (`iv:thm:unique`)](iii) this is the limit of the covariances of the periodised observables on periodic lattices with all sides tending to infinity (and spatial sides ≥ 4). The bound above, with $g = 4$, gives $\langle\tilde\psi, P^t\tilde\psi\rangle_\nu \le C_\psi q_u^{4t}$ for all $t \ge 1$. Since $P \ge 0$ [II, Theorem 'Transfer operator, ground-state chain and mass gap' (`iv:thm:gap`)](ii), the spectral measure σ_ψ of $\tilde\psi$ with respect to P satisfies $\int \lambda^t\, d\sigma_\psi \le C_\psi q_u^{4t}$, hence $\sigma_\psi((q_u^4, 1]) = 0$. Such ψ are dense in the real $L^2(\nu)$, by [II, Lemma `iv:lem:measures`](iv) with the link set $E^{(0)}$ and the unitary map $B \mapsto B(U^{(0)})$ (as in the proof of [II, Theorem

iv:thm:gap](i)), so the $\tilde{\psi}$ are dense in the real part of $\mathbf{1}^{\perp}$; since P maps real functions to real functions, the spectral projection of $P|_{\mathbf{1}^{\perp}}$ for $(q_u^4, 1]$ vanishes, $\sup\operatorname{spec}(P|_{\mathbf{1}^{\perp}}) \le q_u^4$, and $m(\mu_\beta) = -\log\sup\operatorname{spec}(P|_{\mathbf{1}^{\perp}}) \ge 4\log(1/q_u)$ by [II, Theorem iv:thm:gap](iii). □

Corollary 6.5 (The window on the whole range)**.** *Let $d \ge 3$, $0 < \beta < \beta_0(d)$, and let ρ be irreducible and non-trivial with $N \le 49$. Then $u_*(\beta) = u(\beta)$, so*

$$-4\log(eK) - 8(K+1)\theta_5(\beta) \;\le\; m(\mu_\beta) - 4\log\frac{1}{u(\beta)} \;\le\; D(\beta),$$

the upper inequality when $N^4 \le 1200\tau_0^2$ (Corollary 5.8); the lower one also holds for $m_{\Lambda_s}(\beta)$ on spatial lattices with all sides ≥ 4. As $\beta \downarrow 0$ the width of the window tends to $4\log(eK)$, which is 13.9 for $d = 3$ and 16.0 for $d = 4$, for every such G and ρ.

Proof. For irreducible $r \notin \{0, \rho, \bar{\rho}\}$, the multiplicity $m_r(n, m)$ of r in $\rho^{\otimes n} \otimes \bar{\rho}^{\otimes m}$ vanishes for $n+m \le 1$ and satisfies $d_r m_r(n, m) \le N^{n+m}$; the series of the proof of [F, Lemma lem:cr] gives, with $t = \beta/(2N)$, $c_r \le d_r^{-1} \sum_{k\ge 2}(2Nt)^k/k! = (e^\beta - 1 - \beta)/d_r$, so $u_r \le e^\beta - 1 - \beta \le \frac{1}{2}\beta^2 e^\beta$. By Lemma 5.1(iii), $u_\rho = u_{\bar{\rho}} = u \ge (1+\iota)\beta e^{-\beta}/(2N^2)$. Hence $u_* = u$ as soon as $\beta e^{2\beta} \le (1+\iota)/N^2$, which holds for $\beta < \beta_0(d) \le \beta_0(3) < 4.03\cdot 10^{-4}$ when $N \le 49$. The lower bound is Theorem 6.4 with $\log(1/q_u) = \log(1/u) - \log(eK) - 2(K+1)\theta_5$; the upper bound is Corollary 5.8. Finally $\theta_5(\beta) \to 0$ and $D(\beta) \to 0$ (Theorem 5.6). □

Remark 6.6 (Comparison and what remains). (i) Numerically (python3 with mpmath; u computed exactly), at $\beta = \beta_0(4) \approx 1.54 \cdot 10^{-4}$ for $SU(2)$ the lower bounds are $4\log(1/q_u) = 24.68$ (Theorem 6.4), $4\log(1/q) = 5.59$ (Theorem 1.2) and 5.89 (Part II), and the upper bound is 40.67; the window between the two bounds of Corollary 6.5 is $4\log(eK) + O(\beta)$, about 16.0 ($d = 4$; 13.9 for $d = 3$) on the whole range, against 35.1 for Theorem 1.2 and Corollary 5.8 ($SU(2)$, $d = 4$). (The first version of this section had θ in place of θ_5, and its window at $\beta_0(d)$ was 24.0 and 21.9; now $8(K+1)\theta_5 < 2.1 \cdot 10^{-4}$.) Section 7 raises the lower bound at $\beta_0(4)$ to 39.29 (Remark 7.10).

(ii) What remains is polymer entropy: the factor eK per required temporal plaquette comes from the rooted counting of all connected sets ([III, Lemma al:lem:rooted]), whereas a connecting surface with exactly four temporal plaquettes in each layer is essentially rigid: if T_s and T_{s+1} denote its temporal plaquettes in two consecutive layers, the symmetric difference of the corresponding sets of spatial links is contained in the edges of the spatial plaquettes of X at slice $s+1$ (a horizontal link lying in no such plaquette lies in 0 or 2 plaquettes of X). So a tube can change position only by paying for spatial plaquettes; Section 7 turns this into a proof that the loss tends to 0 with β (Theorem 7.8). The strong-coupling expansion predicts $m = 4\log(1/u) + O(u)$; the classical results [Sch83, Sch84a, Sch84b] (as reported in [FO25]) determine m to all orders.

(iii) The structural point is that the transfer-matrix view weighs the temporal plaquettes of all but one layer by the operator norm of their kernel, u_* ($= u \approx (1+\iota)\beta/(2N^2)$ for $\dim\rho \le 49$), instead of the sup norm $\approx 2\beta$ (or $\approx 4\beta$ in the duplicated system); only the spatial plaquettes, which a minimal tube does not need, are bounded by sup norms.

(iv) Checks (a compass): for S_3 with its two-dimensional representation on (time $\mathbb{Z}/2$) × (one square), the terms $\Phi_F(X)$ of the duplicated expansion for five sets X were computed by brute-force summation over gauge-fixed configurations and by the transfer representation of the proof of Lemma 6.3; they agree to all printed digits and satisfy the bound of Lemma 6.3 (theory/checks/lower-bound/).

7 The sharp constant: counting by fronts

The remaining loss $4\log(eK)$ in Theorem 6.4 is entropy: the rooted counting of all connected sets charges every temporal plaquette a factor eK, although a surface with exactly four temporal plaquettes in each layer is rigid. In this section the connecting sets are organised by their *fronts*, the temporal plaquettes of a single layer. A front can change only where a marker (a spatial plaquette or a support) touches it, so an unchanged front of k edges costs a^k per layer, with $a = e^{2(K+1)\theta_5}u_*$, and carries no entropy. The outcome is $m \geq 4\log(1/u_*) - O(\beta)$, the sharp constant: together with Section 5, $m(\mu_\beta) = 4\log(1/u(\beta)) + O(\beta)$ (here $u_* = u$, Corollary 6.5). Counting the fronts and clusters that meet a given front or cluster at a vertex, and using the exact numbers of fronts with four and five edges, makes the bound effective on the whole range: for $SU(2)$ the bound for $m(\mu_\beta)$ lies within 1.4 ($d = 4$) and 0.7 ($d = 3$) of $4\log(1/u)$ for all $\beta < \beta_0(d)$ (Remark 7.10).

Throughout this section $d \geq 3$, $0 < \beta < \beta_0(d)$, G is metrisable whenever the infinite-volume gap $m(\mu_\beta)$ is concerned, Λ is periodic with all sides ≥ 3 and all spatial sides ≥ 4 (so $g = 4$), A, B are bounded measurable functions of finitely many spatial links of the slices 0 and t, $1 \leq t \leq L_0/2$, with link sets S_A, S_B, $S := S_A \cup S_B$, and $F := (A - A')(B - B')$. Let Γ be the nearest-neighbour graph of Λ_s; we identify a temporal plaquette of layer s with the edge of Γ on which it is based, a spatial link at time s with an edge of Γ, and S_A, S_B with sets of edges of Γ. For a set E of edges, $V(E)$ is the set of their endpoints. A *front in* Γ is a non-empty set of edges, connected (any two of its edges are linked by a chain of edges of the set, consecutive ones sharing an endpoint), in which every vertex has degree ≥ 2. Put

$$P_v := 2(d-1)(d-2), \qquad K_S := 4(2d-5), \qquad N_k^v := \max_v \#\{\varphi : \ |\varphi| = k, \ v \in V(\varphi)\},$$

φ ranging over the fronts in Γ. P_v is the number of plaquettes of Λ_s containing a given vertex; a link of Λ_s lies in $2(d-2)$ plaquettes of Λ_s; and a plaquette of Λ_s shares a link with K_S others.

Definition 7.1 (Fronts, clusters, markers, segments)**.** Let $X \subseteq \mathcal{P}$ be such that every link of $E(X) \setminus S$ lies in at least two plaquettes of X, and such that some layer s_* contains no temporal plaquette of X; times are represented in $\{s_* + 1, \dots, s_* + L_0\}$.

(a) For a layer s, $T_s \subseteq E(\Gamma)$ is the set of temporal plaquettes of X in layer s. A *front* of layer s is a connected component of T_s.

(b) A *cluster* at slice s is a connected component (two plaquettes being connected when they share a link) of the set of spatial plaquettes of X at time s; $E(C) \subseteq E(\Gamma)$ is its set of links and $|C|$ its number of plaquettes. The *markers* at slice s are the clusters at slice s, together with S_A if $s \equiv 0$ and S_B if $s \equiv t$; $E(\mathfrak{m})$ is the edge set of a marker $\mathfrak{m}$.

(c) A marker $\mathfrak{m}$ at slice s *touches* a front φ of layer $s-1$ or s if $E(\mathfrak{m})$ and φ have a common vertex, i.e. some edge of $E(\mathfrak{m})$ equals or shares an endpoint with an edge of φ.

(d) A *segment* is a pair $\sigma = (\varphi, [s_1, s_2))$, $s_1 < s_2$, such that φ is a front of every layer $s \in [s_1, s_2)$, no marker at a slice $s \in (s_1, s_2)$ touches φ, and some marker at slice s_1 and some marker at slice s_2 touch φ. We write $\ell(\sigma) := s_2 - s_1$ and $|\sigma| := |\varphi|$.

Lemma 7.2 (Fronts and segments)**.** *Let X be as in Definition 7.1.*

(i) Every vertex of Γ has degree 0 or at least 2 in T_s; every front of a layer is a front in Γ and has at least 4 edges.

(ii) For every slice s, $T_{s-1} \triangle T_s \subseteq \bigcup_{\mathfrak{m}} E(\mathfrak{m})$, the union over the markers $\mathfrak{m}$ at slice s.

(iii) *If a front φ of layer $s-1$ is touched by no marker at slice s, then φ is a front of layer s; and if a front of layer s is touched by no marker at slice s, it is a front of layer $s-1$.*

(iv) *Every front of every layer belongs to exactly one segment, i.e. for every front φ of a layer s there is exactly one segment $(\varphi, [s_1, s_2))$ with $s \in [s_1, s_2)$.*

(v) *$|X| = \sum_\sigma |\sigma|\,\ell(\sigma) + \sum_C |C|$, the sums over the segments and the clusters of X.*

Proof. (i) This is the argument of Lemma 3.2 (temporal links are never in S), and a non-empty graph of minimum degree ≥ 2 contains a cycle, of length ≥ 4 (Lemma 3.1). (ii) A link at time s lies in the temporal plaquettes based on it in layers $s-1$ and s, and in the spatial plaquettes of time s containing it. If it lies in no marker at slice s, it is not in S and in no spatial plaquette of X, so it lies in 0 or 2 plaquettes of X: in both of T_{s-1}, T_s or in neither. (iii) Let φ be a front of layer $s-1$ touched by no marker at slice s. By (ii), $\varphi \subseteq T_s$. If an edge $e \in T_s \setminus \varphi$ shared an endpoint with an edge of φ, then $e \notin T_{s-1}$ (φ is a component of T_{s-1}), so $e \in E(\mathfrak{m})$ for a marker $\mathfrak{m}$ at slice s by (ii), and $\mathfrak{m}$ would touch φ. So φ is a component of T_s. The second statement is symmetric. (iv) Let φ be a front of layer s. Going up, let $s_2 > s$ be the first slice at which a marker touches φ; by (iii) and induction, φ is a front of every layer in $[s, s_2)$. Such a slice exists: otherwise φ would be a front of every later layer, including s_*. Likewise let $s_1 \leq s$ be the last slice $\leq s$ at which a marker touches φ. Then $(\varphi, [s_1, s_2))$ is a segment, and it is the only one containing (φ, s), because a segment cannot contain a slice touching φ in its interior. (v) Every plaquette of X lies in exactly one cluster or in exactly one pair (φ, s), hence by (iv) in exactly one segment. □

Definition 7.3 (Incidence graph)**.** For X as in Definition 7.1, $\mathcal{I}(X)$ is the graph whose vertices are the segments and the markers of X; a segment $(\varphi, [s_1, s_2))$ is joined to every marker at slice s_1 or s_2 that touches φ, and a cluster at slice 0 (resp. t) that contains an edge of S_A (resp. S_B) is joined to S_A (resp. S_B).

Lemma 7.4 (Connectivity and covering)**.** *Let X be as in Definition 7.1.*

(i) *If two plaquettes of X share a link, the vertices of $\mathcal{I}(X)$ containing them are equal or joined by a path of length at most two. A plaquette containing a link of S_A (resp. S_B) lies in a vertex equal or joined to S_A (resp. S_B).*

(ii) *If $X \in \mathcal{X}_S$, every vertex of $\mathcal{I}(X)$ is joined by a path to S_A or to S_B; if moreover a component of X contains a plaquette of $\mathcal{P}_{S_A}$ and one of $\mathcal{P}_{S_B}$, then $\mathcal{I}(X)$ contains a path from S_A to S_B.*

(iii) *If π is a path in $\mathcal{I}(X)$ from S_A to S_B, then $\sum_{\sigma\in\pi} \ell(\sigma) \geq \min(t, L_0 - t) = t$.*

Proof. (i) Two temporal plaquettes of the same layer sharing a (temporal) link lie in the same front, hence in the same segment. A temporal plaquette and a spatial plaquette of time s sharing a link e: the cluster C of the latter contains e, so it touches the front $\varphi \ni e$ (of layer $s-1$ or s), and the segment of φ has s as an end slice by Lemma 7.2(iv); so they are joined. Two spatial plaquettes sharing a link lie at the same time, in the same cluster. Two temporal plaquettes based on the same edge e in layers $s-1$ and s: let $\varphi \ni e$ and $\varphi' \ni e$ be their fronts. If $\varphi = \varphi'$ and no marker at slice s touches φ, they lie in the same segment; if some marker touches $\varphi = \varphi'$ at slice s, both segments end at s and are joined to it. Let $\varphi \neq \varphi'$. If $\varphi \subseteq T_s$, then $\varphi \subseteq \varphi'$ (it is connected and meets φ'), and if also $\varphi' \subseteq T_{s-1}$ then $\varphi = \varphi'$; so, by symmetry, we may assume $\varphi \not\subseteq T_s$. Follow a path of edges $e = e_0, e_1, \ldots, e_m$ in φ (consecutive edges sharing an endpoint) ending outside T_s, and let e_j be its first edge outside φ'. Then $e_j \notin T_s$ (otherwise it would lie in the component φ' of its neighbour e_{j-1}), so $e_j \in T_{s-1} \setminus T_s$, and by Lemma 7.2(ii) $e_j \in E(\mathfrak{m})$ for a marker $\mathfrak{m}$ at slice s. This $\mathfrak{m}$ touches φ ($e_j \in \varphi$) and

φ' (e_j shares an endpoint with $e_{j-1} \in \varphi'$), so both segments end at s and are joined to $\mathfrak{m}$. Finally, a temporal plaquette based on an edge of S_A lies in layer -1 or 0, its front is touched by S_A, and its segment ends at the slice 0; a spatial plaquette containing a link of S_A lies at time 0, in a cluster joined to S_A. (ii) Each component of $X \in \mathcal{X}_S$ contains a plaquette with a link in S_A or S_B; apply (i) along chains of adjacent plaquettes. (iii) Assign to a marker at slice s the point s and to a segment $(\varphi, [s_1, s_2])$ the interval $[s_1, s_2]$, in the representation of Definition 7.1. Joined vertices get intersecting sets, so the union of the sets along π is an interval containing the points assigned to S_A and S_B. These points are L_0 and $L_0 + t$ when s_* is an outer layer, and t and L_0 when s_* is an inner one; the layers between them are the t inner layers in the first case and the $L_0 - t$ outer layers in the second. Every such layer, a unit interval between consecutive integers, lies in $[s_1, s_2]$ for some segment $\sigma = (\varphi, [s_1, s_2))$ of π (markers are points), hence in its set of layers. □

Lemma 7.5 (Small fronts)**.** *Let φ be a front in Γ (all sides of Λ_s are ≥ 4).*

(i) $|V(\varphi)| \leq |\varphi|$.

(ii) If $|\varphi| = 4$, then φ is the boundary of a plaquette of Λ_s or a straight cycle $\{\{v + je_\lambda, v + (j+1)e_\lambda\} : 0 \leq j < 4\}$ around a side of length $L_\lambda = 4$.

(iii) If $|\varphi| = 5$, then φ is a straight cycle around a side of length 5.

Consequently, if n_j is the number of spatial directions in which Λ_s has side j,

$$N_4^v = P_v + n_4 \leq (d-1)(2d-3), \qquad N_5^v = n_5 \leq d-1, \qquad N_k^v \leq (d-1)\binom{(4d-6)k}{k-1} \quad (k \geq 4).$$

Proof. A closed walk of length m in Γ lifts to a walk in $\mathbb{Z}^{d-1}$ from 0 to a point w with $w_\lambda \in L_\lambda \mathbb{Z}$ for every λ. If m_λ is its number of steps in direction λ, then $m = \sum_\lambda m_\lambda$, $m_\lambda \geq |w_\lambda|$ and $m_\lambda \equiv w_\lambda \pmod 2$. So if $w = 0$, m is even; and if $w_\lambda \neq 0$, then $m_\lambda \geq L_\lambda \geq 4$. Hence for $m \leq 5$ either $w = 0$, or exactly one component w_λ is non-zero, $w_\lambda = \pm L_\lambda$, and then $m - m_\lambda$ is even. (a) Γ has no cycle of length 3: 3 is odd and < 4. (b) Let C be a cycle of length 4. If $w = 0$, every m_λ is even; the four steps are not all in one direction (such a walk visits at most three vertices), so there is one step in each sense in each of two directions λ, λ', and the four vertices are the corners of a unit square of the plane $\{\lambda, \lambda'\}$: C is the boundary of a plaquette. If $w \neq 0$, then $m_\lambda = 4 \geq L_\lambda = |w_\lambda| \geq 4$, so all four steps go in direction λ, in the same sense: C is a straight cycle and $L_\lambda = 4$. (c) A cycle of length 5 has $w \neq 0$; then $5 - m_\lambda$ is even, so $m_\lambda = 5$, and $4 \leq L_\lambda \leq m_\lambda$ with $L_\lambda \equiv m_\lambda \pmod 2$ gives $L_\lambda = 5 = |w_\lambda|$: a straight cycle.

(i) $2|V(\varphi)| \leq \sum_v \deg_\varphi(v) = 2|\varphi|$. (ii) Γ is a simple graph (all sides are ≥ 3), so three vertices span at most three edges; hence $|V(\varphi)| = 4$, every vertex has degree 2, and the connected graph φ is a cycle; apply (b). (iii) If $|V(\varphi)| = 5$, φ is a cycle, as in (ii); apply (c). If $|V(\varphi)| = 4$, φ is a simple graph with four vertices and five edges, the complete graph minus one edge, which contains a triangle, contrary to (a). Three vertices span at most three edges.

The plaquettes of Λ_s containing a vertex v are the four plaquettes of each coordinate plane that have v as a corner, P_v in all, and the straight cycles through v are one for each direction with side 4 (resp. 5); this gives N_4^v and N_5^v. For the last bound: a front containing v contains at least two of the $2(d-1)$ edges at v, and for each edge e the connected sets of k edges containing e number at most $\binom{(4d-6)k}{k-1}$, by [III, Lemma 'Rooted counting' (`al:lem:rooted`)] for the graph on $E(\Gamma)$ in which two edges are adjacent when they share an endpoint (root e, $J = K = 4d-6$, $n = k-1$); summing over e counts every front through v at least twice. □

Lemma 7.6 (Local counts)**.** *Let X be as in Definition 7.1.*

(i) Given a marker $\mathfrak{m}$ at slice s and integers $k \geq 4$, $\ell \geq 1$, at most $2|V(E(\mathfrak{m}))|N_k^v$ candidate segments $(\varphi, [s_1, s_2))$ (that is, φ a front in Γ with $|\varphi| = k$, $s_2 - s_1 = \ell$ and $s \in \{s_1, s_2\}$) have a vertex of φ in $V(E(\mathfrak{m}))$.

(ii) Given a segment σ and $n \geq 1$, at most $2|\sigma|P_v\binom{K_S n}{n-1}$ clusters C with $|C| = n$, at an end slice of σ, touch its front; at most $2(d-2)|S_A|\binom{K_S n}{n-1}$ clusters C with $|C| = n$ at slice 0 contain an edge of S_A, and likewise for S_B.

(iii) $|V(E(C))| \leq 4|C|$ for every cluster C, and $|V(S_A)| \leq 2|S_A|$, $|V(S_B)| \leq 2|S_B|$.

Proof. (i) The front contains one of the $|V(E(\mathfrak{m}))|$ vertices of $E(\mathfrak{m})$; given this vertex there are at most N_k^v choices of φ; and $s \in \{s_1, s_2\}$ (two choices) together with ℓ determines the interval. (ii) The cluster lies at one of the two end slices; it contains a vertex of the front φ of σ (at most $|V(\varphi)| \leq |\sigma|$ choices, Lemma 7.5(i)) and a plaquette of that slice containing this vertex (P_v choices), and it is a connected set of n plaquettes of the slice containing that plaquette, of which there are at most $\binom{K_S n}{n-1}$ by [III, Lemma 'Rooted counting' (`al:lem:rooted`)] (root the plaquette, $J = K = K_S$). For S_A: an edge of S_A, a plaquette of slice 0 containing it ($2(d-2)$ choices), and the same count. (iii) A plaquette has four vertices, a link two. □

Lemma 7.7 (Tree sums)**.** *Let $\mathcal{V}$ be a set, $w : \mathcal{V} \to [0, \infty)$, and $\mathcal{N}(v) \subseteq \mathcal{V}$ for $v \in \mathcal{V}$ (the possible children of v). For $v \in \mathcal{V}$ let $\Psi(v)$ be the sum, over all finite rooted trees whose root is labelled v and in which the children of a vertex labelled v' carry distinct labels in $\mathcal{N}(v')$, of the product of w over the non-root labels. If $h : \mathcal{V} \to [0, \infty)$ satisfies $\sum_{c \in \mathcal{N}(v)} w(c)e^{h(c)} \leq h(v)$ for all v, then $\Psi(v) \leq e^{h(v)}$.*

Proof. Let Ψ_n be the same sum restricted to trees of depth $\leq n$; $\Psi_0 = 1$. A tree of depth $\leq n$ is a set of distinct children of the root with subtrees of depth $\leq n-1$, so $\Psi_n(v) \leq \prod_{c \in \mathcal{N}(v)}(1 + w(c)\Psi_{n-1}(c)) \leq \exp \sum_c w(c)\Psi_{n-1}(c)$. By induction $\Psi_n \leq e^h$, and $\Psi = \lim_n \Psi_n$. □

Theorem 7.8 (The sharp constant)**.** *Let $\gamma_g := \|g_p\|_\infty$ (so $\gamma_g \leq e^{4\beta} - 1$, and $\gamma_g \leq \gamma_2(\beta)$ if ρ has no trivial summand, Lemma 2.2(a),(d)), let θ_5 be as in Lemma 6.1, and put $a := e^{2(K+1)\theta_5}u_*$ and $b := e^{2(K+1)\theta_5}\gamma_g$. For $\lambda_C, \lambda_T \geq 0$ put*

$$\eta_C := 2P_v \sum_{n \geq 1} \binom{K_S n}{n-1}(be^{\lambda_C})^n, \qquad \kappa_C := 8P_v \sum_{n \geq 1} n\binom{K_S n}{n-1}(be^{\lambda_C})^n,$$

$$\eta_T := 8 \sum_{k \geq 4} N_k^v \, e^{\lambda_T k} \frac{a^k}{1 - a^k}, \qquad \nu := 2 \sum_{k \geq 4} k N_k^v \, e^{\lambda_T k} a^{k-4}, \qquad \delta := \kappa_C \, \nu,$$

and suppose that the four series converge, $\eta_C \leq \lambda_T$, $\eta_T \leq \lambda_C$ and $a^4(1 + \delta) < 1$. Then for every $\delta' > \delta$ with $a^4(1 + \delta') \leq 1$ there is $C_{A,B}$, independent of t and L_0, with

$$|\mathrm{Cov}_{\mu_{\Lambda,\beta}}(A, B)| \leq C_{A,B}\big(a^4(1 + \delta')\big)^t + \frac{2\|A\|_\infty \|B\|_\infty c^{|S|}}{1 - \hat{q}} \, \hat{q}^{\,4L_0}.$$

Consequently $m_{\Lambda_s}(\beta) \geq 4\log(1/u_*) - 8(K+1)\theta_5 - \log(1 + \delta)$, with δ and the hypotheses taken for the spatial lattice Λ_s; and

$$m(\mu_\beta) \;\geq\; 4\log\frac{1}{u_*} - 8(K+1)\theta_5 - \log(1 + \delta_\infty),$$

where δ_∞, and the hypotheses, are taken with $N_4^v = P_v$, $N_5^v = 0$ and $N_k^v = (d-1)\binom{(4d-6)k}{k-1}$ for $k \geq 6$ (which bound the counts when all spatial sides are ≥ 6, Lemma 7.5). As $\beta \downarrow 0$ the hypotheses hold, with $\lambda_T = O(\beta)$ and $\lambda_C = O(\beta^4)$, and $\delta = 64P_v N_4^v \gamma_g(1 + O(\beta))$; if ρ has no trivial summand, $\delta_\infty \leq 128P_v^2\beta(1 + O(\beta))$.

Proof. As in the proof of Theorem 6.4, $2\operatorname{Cov}(A,B) = \sum_X \Phi_F(X) r_X^2$ over the sets $X \in \mathcal{X}_S$ without free links outside S that have a component meeting $\mathcal{P}_{S_A}$ and $\mathcal{P}_{S_B}$. Those X with a temporal plaquette in every layer have $|X| \geq 4L_0$ (Lemma 3.2), and the computation in the proof of Theorem 4.1, with $\hat{q}$ in place of q (as in the proof of Theorem 6.4), bounds their contribution by the second term. For the other X, Lemma 6.3 (with $m(X) = 0$), $r_X^2 \leq e^{2\theta_5 J_S} e^{2\theta_5 (K+1)|X|}$ (Lemma 6.1 and $|\overline{X}| \leq (K+1)|X| + J_S$) and Lemma 7.2(v) give

$$|\Phi_F(X)| r_X^2 \leq 4\|A\|_\infty \|B\|_\infty e^{2\theta_5 J_S}\, W(X), \qquad W(X) := \prod_\sigma a^{|\sigma|\ell(\sigma)} \prod_C b^{|C|}.$$

Fix δ' as in the statement and put $z := 1/(a^4(1+\delta')) \geq 1$. For each such X choose a path $\pi(X)$ in $\mathcal{I}(X)$ from S_A to S_B of minimal length (Lemma 7.4(ii)) and a spanning forest $\mathcal{F}(X)$ of $\mathcal{I}(X)$ each of whose trees contains exactly one vertex of $\pi(X)$ (possible by Lemma 7.4(ii)). By Lemma 7.4(iii), $z^t \leq z^{\sum_{\sigma\in\pi(X)} \ell(\sigma)}$, so

$$z^t W(X) \leq \prod_{\sigma\in\pi(X)} \left(z a^{|\sigma|}\right)^{\ell(\sigma)} \prod_{\sigma\notin\pi(X)} a^{|\sigma|\ell(\sigma)} \prod_C b^{|C|}.$$

The set X is the union of the plaquette sets of the vertices of $\mathcal{I}(X)$, so it is determined by the pair $(\pi(X), \mathcal{F}(X))$, the vertices being described geometrically (a segment by (φ, s_1, s_2), a cluster by its slice and its plaquettes). Hence $z^t \sum_X W(X)$ is at most the sum, over all sequences $\pi = (S_A, v_1, \dots, v_r, S_B)$ of segments and clusters in which consecutive terms are joined in the sense of Definition 7.3, of $\prod_{v\in\pi} w_\pi(v)\Psi(v)$. Here $w_\pi(\sigma) = (za^{|\sigma|})^{\ell(\sigma)}$, $w_\pi(C) = b^{|C|}$ and $w_\pi(S_A) = w_\pi(S_B) = 1$, and Ψ is the tree sum of Lemma 7.7 with $w(\sigma) = a^{|\sigma|\ell(\sigma)}$, $w(C) = b^{|C|}$ and children given by Definition 7.3 (supports excluded).

Decorations. Let $h(C) := \lambda_C|C|$, $h(\sigma) := \lambda_T|\sigma|$, and $h(S_A) := |S_A|(\lambda_T' + \eta_T/2)$ with $\lambda_T' := 2(d-2)\sum_n \binom{K_S n}{n-1}(be^{\lambda_C})^n$, and likewise for S_B. By Lemma 7.6: for a cluster C, $\sum_\sigma a^{|\sigma|\ell(\sigma)} e^{\lambda_T|\sigma|} \leq 2|V(E(C))| \sum_k N_k^v e^{\lambda_T k} \sum_{\ell\geq 1} a^{k\ell} \leq |C|\eta_T \leq \lambda_C|C|$; for a segment σ, $\sum_C b^{|C|} e^{\lambda_C|C|} \leq |\sigma|\eta_C \leq \lambda_T|\sigma|$; for S_A, the clusters contribute at most $|S_A|\lambda_T'$ and the segments at most $2|V(S_A)|\sum_k N_k^v e^{\lambda_T k} a^k/(1-a^k) \leq |S_A|\eta_T/2$. By Lemma 7.7, $\Psi \leq e^h$.

The path. Since $za^4 = 1/(1+\delta')$ and $a \leq 1$, $\sum_{\ell\geq 1}(za^k)^\ell = za^k/(1-za^k) \leq a^{k-4}/\delta'$ for $k \geq 4$. Hence, by Lemma 7.6(i), for a marker $\mathfrak{m}$ the segments σ joined to $\mathfrak{m}$, weighted by $|\sigma| w_\pi(\sigma) e^{h(\sigma)}$, sum to at most $|V(E(\mathfrak{m}))|\nu/\delta'$; and by Lemma 7.6(ii),(iii), for a segment σ the clusters C joined to σ, weighted by $|V(E(C))| b^{|C|} e^{h(C)}$, sum to at most $|\sigma|\kappa_C$. A minimal path has the form S_A, possibly a cluster C_0 at slice 0, then $\sigma_1, C_1, \sigma_2, \dots, C_{r-1}, \sigma_r$ (clusters are not joined to clusters), possibly a cluster C_r at slice t, and S_B. For a segment σ let $R(\sigma)$ be the sum, over the possible continuations of a path after σ (namely S_B; or C, S_B; or C, σ' followed by a continuation of σ'), of the product of $w_\pi(v)e^{h(v)}$ over the vertices v of the continuation. Then

$$R(\sigma) \leq e^{h(S_B)}(1 + |\sigma|\eta_C) + \sum_{C\sim\sigma} b^{|C|} e^{h(C)} \sum_{\sigma'\sim C} w_\pi(\sigma') e^{h(\sigma')} R(\sigma'),$$

and by induction on the length of the continuations $R(\sigma) \leq M|\sigma|$ with $M := e^{h(S_B)}(1+\eta_C)/(1-\delta/\delta')$: the double sum is at most $M\sum_{C\sim\sigma} b^{|C|} e^{h(C)} |V(E(C))|\nu/\delta' \leq M|\sigma|\kappa_C\nu/\delta' = (\delta/\delta')M|\sigma|$, and $1 + |\sigma|\eta_C \leq (1+\eta_C)|\sigma|$. The beginning of the path is treated in the same way:

$$z^t \sum_X W(X) \leq e^{h(S_A)} M \Big(|V(S_A)| + \sum_{C_0} b^{|C_0|} e^{h(C_0)} |V(E(C_0))| \Big) \frac{\nu}{\delta'} =: C_0 < \infty,$$

the sum over the clusters C_0 at slice 0 containing an edge of S_A, which is at most $8(d-2)|S_A|\sum_n n\binom{K_S n}{n-1}(be^{\lambda_C})^n$. Hence $\sum_X W(X) \leq C_0(a^4(1+\delta'))^t$, which is the covariance bound with $C_{A,B} = 2C_0\|A\|_\infty\|B\|_\infty e^{2\theta_5 J_S}$.

Gaps. Let A be bounded, measurable and gauge invariant on Ω_s. Applying the covariance bound to the real and imaginary parts of $\overline{A}(t)$ and $A(0)$, letting $L_0 \to \infty$ at fixed t (the second term tends to 0) and using [F, Theorem 'Ground-state expectations' (`thm:gs`)], we get $|\langle \Omega, \hat{A}^* e^{-tH} \hat{A}\Omega\rangle - |\langle A\rangle_\nu|^2| \le C_A(a^4(1+\delta'))^t$ for all $t \ge 0$, with C_A independent of t ($t = 0$ is trivial), hence $m_{\Lambda_s}(\beta) \ge -\log(a^4(1+\delta'))$ by [F, Theorem 'Gap $\Leftrightarrow$ clustering in time' (`thm:gapclust`)](ii). (The bound is not uniform in L_0 for t up to $L_0/2$, as [F, Corollary `cor:gapfromcorr`] would require, unless β is small: near $\beta_0(d)$ the term $\hat{q}^{4L_0}$ decays more slowly than $(a^4(1+\delta'))^{L_0/2}$.) For $m(\mu_\beta)$ argue as in the proof of Theorem 6.4(ii), with q_u^{4t} replaced by $(a^4(1+\delta'))^t$: the infinite-volume covariance is the limit over periodic lattices with all sides tending to infinity, on which (sides ≥ 6) the counts are bounded as in the definition of δ_∞, so the covariance bound holds with any $\delta' > \delta_\infty$ and the second term vanishes in the limit. Finally let $\delta' \downarrow \delta$ (resp. δ_∞), and note $4\log(1/a) = 4\log(1/u_*) - 8(K+1)\theta_5$.

Small β. Here $a, b = O(\beta)$. Take $\lambda_T := \eta_C|_{\lambda_C=1} = O(\beta)$ and then $\lambda_C := \max(\eta_T, \beta^4) \le 1$; since η_C increases with λ_C, both conditions hold, $\lambda_C = O(\beta^4)$ (as $\eta_T = O(a^4)$), $\kappa_C = 8P_v b(1+O(\beta))$, $\nu = 8N_4^v(1+O(\beta))$, and $\delta = 64P_vN_4^v\gamma_g(1+O(\beta))$, $a^4(1+\delta) \to 0$. If ρ has no trivial summand, $\gamma_g \le \gamma_2 = 2\beta + O(\beta^2)$, and $N_4^v = P_v$ in δ_∞. □

Lemma 7.9 (The hypotheses on the whole range)**.** *For every $d \ge 3$ and $0 < \beta < \beta_0(d)$, every G and ρ, and every spatial lattice with all sides ≥ 4, the hypotheses of Theorem 7.8 hold with $\lambda_C = \lambda_T = \frac{1}{10}$, both for δ (with the bounds of Lemma 7.5 for N_k^v) and for δ_∞; moreover $\eta_C < 0.023$ and $\eta_T < 10^{-9}$.*

Proof. The series increase with a, b and the N_k^v, and we use the largest bounds for the latter, $N_4^v \le (d-1)(2d-3)$, $N_5^v \le d-1$. Since $u_* \le \epsilon$ and $\gamma_g \le e^{4\beta} - 1 = \epsilon(2+\epsilon)$, $a \le \bar{a} := e^{2(K+1)\theta_5}\epsilon$ and $b \le \bar{b} := e^{2(K+1)\theta_5}\epsilon(2+\epsilon)$, which increase with β; so it suffices to bound the series at $\beta = \beta_0(d)$, where $\epsilon = \epsilon_0 := 1/(e^2K(K+2))$ and $x = 1/(K+2)$. Then $2(K+1)\theta_5 \le 2(K+1)x^5/(1-x) = 2(K+2)^{-4} < 6\cdot 10^{-5}$, so $\bar{a} \le 1.0001\,\epsilon_0$ and $y := \bar{b}e^{1/10} \le 2.212\,\epsilon_0 \le 0.3/(K(K+2))$. We write $K = 8d-12 \ge 12$, so that $K_S = K-8$, $4d-6 = K/2$, $2P_v = (K^2-16)/16$ and $(d-1)(2d-3) = K(K+4)/32$, and use $\binom{Jn}{n-1} \le e(eJ)^{n-1}$ for $n \ge 2$ (from $\binom{m}{j} \le (em/j)^j$). With $w := eK_Sy \le 0.82/(K+2) \le 0.06$,

$$\eta_C \le 2P_vy\Big(1 + \frac{ew}{1-w}\Big) \le \frac{K^2-16}{16}\cdot\frac{0.3}{K(K+2)}\cdot 1.18 < 0.023.$$

With $r := e^{1.1}(K/2)\bar{a} \le 0.21/(K+2) \le 0.015$, the terms $k \ge 6$ of η_T are at most $8(d-1)e^{1.1}\bar{a}\sum_{k\ge6} r^{k-1}(1-\bar{a}^6)^{-1} < 10^{-10}$, and the terms $k = 4, 5$ at most $8\frac{K(K+4)}{32}e^{0.4}\bar{a}^4(1+10^{-6}) + 8(d-1)e^{0.5}\bar{a}^5 < 10^{-10}$, using $\bar{a} \le 1.0001/(e^2K(K+2)) \le 8.1\cdot 10^{-4}$; so $\eta_T < 10^{-9}$. All series converge ($w, r < 1$). Finally $\kappa_C \le 8P_vy\big(1 + e((1-w)^{-2}-1)\big) < 0.11$ and, bounding every N_k^v by $(d-1)e(eK/2)^{k-1}$, $\bar{a}^4\nu \le 2(d-1)e^{1.1}\bar{a}\sum_{k\ge4} kr^{k-1} < 10^{-6}$; so $a^4(1+\delta) \le \bar{a}^4 + 0.11\,\bar{a}^4\nu < 10^{-6}$. □

Remark 7.10 (Numbers, the first version, and checks). (i) For $SU(2)$ (ρ has no trivial summand and $u_* = u$, computed exactly), with λ_C, λ_T the smallest solution of the two conditions (obtained by iterating them from $\lambda_C = 0$), the lower bound of Theorem 7.8 for $m(\mu_\beta)$ lies below $4\log(1/u)$ by 0.0018, 0.018, 0.17, 0.90 and 1.38 at $\beta = 10^{-7}$, 10^{-6}, 10^{-5}, $\beta_0(4)/2$ and $\beta_0(4)$ for $d = 4$, and by 0.0002, 0.0020, 0.020, 0.35 and 0.62 at $\beta = 10^{-7}$, 10^{-6}, 10^{-5}, $\beta_0(3)/2$ and $\beta_0(3)$ for $d = 3$. The loss increases with β: a and b do (as θ_5, $u = I_2/I_1$ and γ_2 do), $\eta_C, \eta_T, \kappa_C, \nu$ increase with a, b and the λ's, and the smallest solution, the limit of the iteration of an increasing map started at 0, increases with (a, b); so its supremum over the range is its limit at $\beta_0(d)$. For the finite-volume gap on spatial lattices with all sides ≥ 4 the loss is at most 1.56 ($d = 4$) and 0.83 ($d = 3$). (With the choice $\lambda_C = \lambda_T = \frac{1}{10}$ of Lemma 7.9 instead of

the smallest solution, the loss at $\beta_0(d)$ is 1.75 and 0.87.) With the upper bound (Corollary 5.8; $D < 1.4 \cdot 10^{-3}$ and $D < 4.9 \cdot 10^{-3}$, D increasing, Remark 5.9), for all $0 < \beta < \beta_0(d)$

$$-1.38 \le m(\mu_\beta) - 4\log\frac{1}{u(\beta)} \le 0.0014 \quad (d = 4),$$
$$-0.63 \le m(\mu_\beta) - 4\log\frac{1}{u(\beta)} \le 0.005 \quad (d = 3)$$

(Numerical: an evaluation of the explicit constants, `python3` with `mpmath`). Theorem 6.4 loses $4\log(eK)$, i.e. 16.0 and 13.9, throughout. As $\beta \downarrow 0$, $\delta_\infty/\beta \to 128P_v^2$, which is 18432 for $d = 4$ and 2048 for $d = 3$.

(ii) The first version of this section (commit `1766bb3`) counted the fronts and clusters meeting a marker or a front through an edge rather than through a vertex, bounded N_k by $e(e(4d-6))^{k-1}$ for every k, and used $e^{4\beta} - 1$ and θ in place of γ_g and θ_5; its δ/β tended to $3.4 \cdot 10^6$ ($d = 4$) and $4.1 \cdot 10^5$ ($d = 3$), and near $\beta_0(d)$ it was weaker than Theorem 6.4. That version said that its hypotheses fail at $\beta_0(4)$; an independent referee found this to be false (the iteration had been started at $\lambda_C = 1$ instead of 0): they hold, with a loss of 27.7.

(iii) The constants are still far from optimal. The dominant term of δ comes from clusters of one plaquette at an end of a segment with a four-edge front; most of these are impossible, since such a plaquette at slice s needs each of its links in $T_{s-1} \cup T_s$. For the family of tubes whose fronts are boundaries of plaquettes, with at most three spatial plaquettes per slice, an exact transfer-matrix sum (first done by the referee) gives the growth factor $a^4(1+b)$ per layer up to $O(b^2)$, i.e. an effective δ of about $b \approx 2\beta$ rather than $128P_v^2\beta$ (Numerical). Section 8 proves $\delta = b(1 + O(\beta))$.

(iv) Checks (a compass): the constants of (i), and those of Lemma 7.9 for $3 \le d \le 12$; an exhaustive enumeration of the fronts with at most six edges through a vertex on all spatial tori with sides in $\{4, \dots, 7\}$ ($d = 3, 4$), which confirms Lemma 7.5 and the value of P_v, and gives also $N_6^v = 12\binom{d-1}{2} + 96\binom{d-1}{3} + 18(d-2)n_4 + n_6$; the transfer-matrix sum of (iii); a test of Lemmas 7.2 and 7.4 on 400 random sets X on the periodic lattice $8 \times 5 \times 5$ (the referee's independent test covered 4513 pairs (X, s_*) on five lattices); and the enumeration of K_S and $2(d-2)$. The scripts are in `theory/checks/lower-bound/`.

Corollary 7.11 (The gap is $4\log(1/u) + O(\beta)$, on the whole range)**.** *Let G be metrisable and ρ irreducible and non-trivial with $N \le 49$. Then for every $0 < \beta < \beta_0(d)$*

$$-8(K+1)\theta_5(\beta) - \log(1 + \delta_\infty(\beta)) \;\le\; m(\mu_\beta) - 4\log\frac{1}{u(\beta)},$$

with δ_∞ as in Theorem 7.8 for any λ_C, λ_T satisfying its hypotheses (for instance $\frac{1}{10}$); the same holds for $m_{\Lambda_s}(\beta)$, with δ in place of δ_∞, on every periodic spatial lattice with all sides ≥ 4. If $N^4 \le 1200\tau_0^2$, then also $m(\mu_\beta) - 4\log(1/u(\beta)) \le D(\beta)$. As $\beta \downarrow 0$, $m(\mu_\beta) = 4\log(1/u(\beta)) + O(\beta)$, and the same holds for $m_{\Lambda_s}(\beta)$, uniformly in the spatial lattices with all sides ≥ 4.

Proof. Theorem 7.8 and Lemma 7.9, with $u_* = u$ (Corollary 6.5). The upper bound is Corollary 5.8 when $N^4 \le 1200\tau_0^2$, and for small β the last sentence of Theorem 5.6 (which has no condition on N; for the finite-volume gaps, Theorem 5.6(ii)). As $\beta \downarrow 0$, $\theta_5 = O(\beta^5)$, $\delta, \delta_\infty = O(\beta)$ (Theorem 7.8) and $D(\beta) = O(\beta)$. □

8 The first-order constant: junctions

In Theorem 7.8 the constant δ is about $2 \cdot 10^4\beta$ ($d = 4$). Its dominant term counts a cluster of one plaquette at an end of a segment with a four-edge front, together with any front that touches it, although almost all of these configurations violate the condition that no link of

X is free (Remark 7.10(iii)). In this section all markers and fronts that meet at a slice are grouped into a *junction*; the path of Section 7 becomes a path of junctions and segments, and the weight of each segment is split between its two end junctions. The local form of the no-free-link condition then leaves exactly one junction of order b through which a path can pass: the *cap*, one plaquette closing a four-edge front at a slice where the front continues. Every other junction is of order b^2, or carries a further front and with it a factor a^2, or lets the path continue with a front of at least six edges, which the next segment suppresses by a factor a^2. The outcome is $\delta = b\,(1 + O(\beta))$ (Theorem 8.11), the first-order constant of the method; $b \approx 2\beta$ when ρ has no trivial summand. For $SU(2)$ the lower bound for $m(\mu_\beta)$ lies within $5 \cdot 10^{-4}$ ($d = 4$) and $1.1 \cdot 10^{-3}$ ($d = 3$) of $4\log(1/u)$ on the whole range (Remark 8.12).

The setting is that of Section 7; in addition we put $T_v := 4(P_v - 1)$, an upper bound for the number of plaquettes of Λ_s other than a given one that share a vertex with it.

Definition 8.1 (Junctions)**.** Let X be as in Definition 7.1 and s a slice. A *front-pair* at slice s is a pair $p = (\varphi, \lambda)$ with $\lambda \in \{s-1, s\}$ and φ a front of layer λ touched by a marker at slice s; we write $|p| := |\varphi|$. Front-pairs at different slices are different objects, even when the pairs coincide (as for a segment of length 1, whose front φ of layer s gives (φ, s) at the slices s and $s+1$). Let H_s be the graph whose vertices are the markers and the front-pairs at slice s, two of them being adjacent when their edge sets ($E(\mathfrak{m})$, resp. φ) have a common vertex. A *junction* at slice s is a connected component J of H_s. We write $M(J)$ and $\Phi(J)$ for its markers and front-pairs, $E(J)$ for the union of their edge sets, $E_\mu(J)$ for the union of the fronts of its front-pairs of layer μ, $E_M(J) := \bigcup_{\mathfrak{m}\in M(J)} E(\mathfrak{m})$, and $|J|_C$ for the number of plaquettes of the clusters in $M(J)$.

Lemma 8.2 (Junctions)**.** *Let X be as in Definition 7.1.*

(i) Every junction contains a marker, and every front-pair is adjacent to a marker of its junction. For every segment $\sigma = (\varphi, [s_1, s_2))$, (φ, s_1) is a front-pair at slice s_1 and $(\varphi, s_2 - 1)$ one at slice s_2, and every front-pair arises in this way from exactly one segment. The junctions containing these two front-pairs are the end junctions *of σ; they lie at different slices.*

(ii) If J is a junction at slice s and $e \in E(J)$, then every marker at slice s whose edge set contains e belongs to J, and for every front ψ of layer $\mu \in \{s-1, s\}$ with $e \in \psi$, (ψ, μ) is a front-pair and belongs to J.

(iii) If J contains no support, then for every $e \in E(J)$

$$n_J(e) := [e \in E_{s-1}(J)] + [e \in E_s(J)] + \pi_J(e) \;\geq\; 2,$$

where $\pi_J(e)$ is the number of plaquettes of the clusters of J that contain e.

(iv) $|X| = \sum_\sigma |\sigma|\,\ell(\sigma) + \sum_J |J|_C$, the sums over the segments and the junctions of X.

Proof. (i) A front-pair is touched by a marker at its slice, which is adjacent to it. A front φ of layer s touched by a marker at slice s lies in a segment $(\varphi, [s_1, s_2))$ with $s \in [s_1, s_2)$ (Lemma 7.2(iv)); no marker at a slice in (s_1, s_2) touches φ, so $s = s_1$. Likewise a front-pair $(\varphi, s-1)$ is the end of its segment, $s = s_2$. Uniqueness is Lemma 7.2(iv), and $s_1 < s_2$. (ii) A marker whose edge set contains e shares the endpoints of e with the vertex of J whose edge set contains e, so it is adjacent to it. Let $\psi \ni e$ be a front of layer μ. If ψ is touched by a marker at slice s, (ψ, μ) is a front-pair adjacent to that vertex of J. Otherwise no marker at slice s contains e, so $e \in \varphi$ for some $(\varphi, \nu) \in \Phi(J)$. If $\nu = \mu$, then $\psi = \varphi$, the component of T_μ containing e; if $\nu \neq \mu$, then ψ is a front of layer ν as well (Lemma 7.2(iii)), so again $\psi = \varphi$. In both cases ψ is touched by a marker at slice s after all, a contradiction. (iii) If the

link e of time s were in S, then $s \equiv 0$ and $e \in S_A$, or $s \equiv t$ and $e \in S_B$, and that support would belong to J by (ii). So the link e of time s is not in S, and it lies in a plaquette of X (a plaquette of a cluster, or the temporal plaquette based on e in the layer of a front-pair). Hence it lies in at least two plaquettes of X. These are the temporal plaquettes based on e in layers $s-1$ and s, which by (ii) are counted by the first two terms, and the spatial plaquettes of time s containing e, which lie in clusters containing e, hence by (ii) in clusters of J. (iv) Lemma 7.2(v); every cluster lies in exactly one junction. □

Definition 8.3 (Junction graph)**.** For X as in Definition 7.1, $\mathcal{J}(X)$ is the bipartite graph whose vertices are the junctions and the segments of X, every segment being joined to its two end junctions; J_A and J_B are the junctions containing S_A and S_B.

Lemma 8.4 (Connectivity and covering for junctions)**.** *Let $X \in \mathcal{X}_S$ be as in Definition 7.1, with a component that contains a plaquette of $\mathcal{P}_{S_A}$ and one of $\mathcal{P}_{S_B}$. Then every vertex of $\mathcal{J}(X)$ is joined by a path to J_A or to J_B, there is a path from J_A to J_B, and every such path $J_A = J_0, \sigma_1, J_1, \dots, \sigma_r, J_r = J_B$ satisfies $\sum_i \ell(\sigma_i) \geq t$.*

Proof. Send a segment of $\mathcal{I}(X)$ to itself and a marker to its junction. If a segment and a marker are joined in $\mathcal{I}(X)$, the marker touches the front of the segment at an end slice s, i.e. it is adjacent in H_s to the corresponding front-pair, so the segment is joined to the image of the marker; a cluster joined to S_A shares an edge with it, hence lies in J_A. So paths of $\mathcal{I}(X)$ become walks of $\mathcal{J}(X)$, and Lemma 7.4(ii) gives the first two statements. The last is proved as Lemma 7.4(iii), a junction at slice s being assigned the point s. □

Weights. For a junction J and a segment σ put

$$w(J) := b^{|J|_C} \prod_{p \in \Phi(J)} a^{|p|/2}, \qquad w(\sigma) := a^{|\sigma|(\ell(\sigma)-1)}.$$

Every segment has exactly two end front-pairs (Lemma 8.2(i)), so by Lemma 8.2(iv) the weight $W(X)$ of the proof of Theorem 7.8 factorises as $W(X) = \prod_J w(J) \prod_\sigma w(\sigma)$.

Lemma 8.5 (Fronts with six edges)**.** *If all sides of Λ_s are ≥ 4, every front with six edges is a cycle, and $N_6^v \leq (2(d-1))^5/2 = 16(d-1)^5$.*

Proof. Let φ be a front with six edges. Γ has no triangles (proof of Lemma 7.5), and a graph on four vertices with at least five edges contains one, so $|V(\varphi)| \in \{5, 6\}$. If $|V(\varphi)| = 6$, all degrees are 2 and φ is a cycle. If $|V(\varphi)| = 5$, the degrees are $(4,2,2,2,2)$ or $(3,3,2,2,2)$. Following the edges at the vertices of degree ≥ 3 through the vertices of degree 2, in the first case φ is the union of two cycles through the vertex of degree 4, of lengths ≥ 4 each: too many edges. In the second case it is either two cycles joined by a path (at least $4+4+1$ edges) or two vertices x, y joined by three internally disjoint paths of lengths $l_1 + l_2 + l_3 = 6$, with $l_i + l_j \geq 4$ (no cycles of length ≤ 3) and at most one $l_i = 1$ (Γ is simple); so $l_1 = l_2 = l_3 = 2$ and x, y have three common neighbours. But a common neighbour z of $x \neq y$ gives $y - x \equiv e + e'$ with unit vectors $e = z - x$, $e' = y - z$, and as all sides are ≥ 4 every class $y - x \not\equiv 0$ is represented by at most two ordered pairs (e, e'): by (e, e') and (e', e) if $y - x \equiv e + e'$ with $e \neq \pm e'$, and by (e, e) and $(-e, -e)$ if $y - x \equiv 2e$ (the second only when the side in that direction is 4). Hence φ is a 6-cycle. A 6-cycle through v is traversed by two closed walks of length 6 from v, and there are at most $(2(d-1))^5$ such walks. □

Lemma 8.6 (Junctions with two front-pairs)**.** *Let J be a junction without support at slice s whose front-pairs are exactly $p = (\varphi, \lambda)$ and $p' = (\varphi', \lambda')$, with $|\varphi| = 4$, and let $n := |J|_C$.*

(i) *If $\lambda \neq \lambda'$, then $\varphi \triangle \varphi' \subseteq E_M(J)$ and $\varphi' \subseteq E_M(J) \cup \varphi$; if $\lambda = \lambda'$, then $\varphi \cup \varphi' \subseteq E_M(J)$. Every plaquette of J has at least $5 - n$ edges in $\varphi \cup \varphi'$. Every plaquette of J is joined to $V(\varphi)$ by a chain of plaquettes of J, consecutive ones sharing a vertex and the first having a vertex in $V(\varphi)$.*

(ii) *If $|\varphi'| = 4$ and $n = 1$, then $\lambda \neq \lambda'$, $\varphi = \varphi'$, and the plaquette of J is the one bounded by φ (a* cap*).*

(iii) *If $|\varphi'| = 4$ and $n = 2$, then $\lambda \neq \lambda'$, φ and φ' are the boundaries of two distinct plaquettes q, q' that share a vertex, and q, q' are the plaquettes of J (a* shift *of the tube if q, q' share a link, a* jump *if they share only a vertex).*

(iv) *If $|\varphi'| = 4$ and $n = 3$, then $\varphi' \neq \varphi$, every plaquette of J is the plaquette bounded by φ, the one bounded by φ', or the unique plaquette containing two given distinct edges $e \in \varphi$, $e' \in \varphi'$, and at least one plaquette of J is of the last kind.*

(v) *If $|\varphi'| = 5$, then $n \geq 4$.*

Proof. (i) By Lemma 8.2(iii), an edge of $E(J)$ outside $E_M(J)$ has $n_J(e) \geq 2$ from the two layers alone; this gives the first statements. An edge of a plaquette of J outside $\varphi \cup \varphi'$ lies in no front of J, so it lies in two plaquettes of J; two plaquettes share at most one link (proof of Lemma 5.4), so a plaquette has at most $n - 1$ such edges. For the chains: let $\mathcal{A}$ be the set of plaquettes of J joined to $V(\varphi)$ by such chains, $\mathcal{B}$ the others, and suppose $\mathcal{B} \neq \emptyset$. Plaquettes sharing a link share a vertex, so each cluster lies in $\mathcal{A}$ or in $\mathcal{B}$, and a cluster in $\mathcal{B}$ shares no vertex with φ or with a cluster in $\mathcal{A}$. By the first statement every edge of φ' lies in $E(\mathcal{A}) \cup \varphi$ or in $E(\mathcal{B})$. If every edge of φ' lies in $E(\mathcal{A}) \cup \varphi$, the clusters in $\mathcal{B}$ are adjacent in H_s only to each other (a common vertex with φ' would be one with $\mathcal{A}$ or φ), and J is disconnected. If every edge lies in $E(\mathcal{B})$, then p' and the clusters in $\mathcal{B}$ are adjacent only to each other (p and the clusters in $\mathcal{A}$ share no vertex with $E(\mathcal{B})$), and again J is disconnected. Otherwise, φ' being connected, some vertex is an endpoint of an edge of each kind, and then lies in $V(E(\mathcal{B})) \cap V(E(\mathcal{A}) \cup \varphi)$, a contradiction. (ii) If $\lambda = \lambda'$, φ and φ' are distinct components of T_λ, so vertex-disjoint, and (i) gives $8 \leq 4n$. So $\lambda \neq \lambda'$, the four edges of the plaquette lie in $\varphi \cup \varphi'$, and $\varphi \triangle \varphi'$ lies in its boundary. Two distinct cycles of length 4 in Γ share at most one edge (two plaquette boundaries by the proof of Lemma 5.4; a plaquette contains no two collinear edges, so a straight cycle shares at most one edge with a plaquette boundary; distinct straight cycles are edge-disjoint), so $\varphi \neq \varphi'$ would give $|\varphi \triangle \varphi'| \geq 6$. Hence $\varphi = \varphi'$ contains the boundary of the plaquette, and they are equal. (iii) If $\lambda = \lambda'$, (i) shows that the two plaquettes have all their edges in $\varphi \cup \varphi'$, so they are the plaquettes bounded by φ and φ' (a cycle of length 4 contained in the union of two vertex-disjoint ones is one of them); but these have disjoint vertex sets, and J would be disconnected. So $\lambda \neq \lambda'$. Every edge of $E_M(J)$ outside $\varphi \cup \varphi'$ lies in both plaquettes, so there is at most one such edge e_0. If $\varphi = \varphi'$, each plaquette has three edges in φ and is bounded by φ, which is impossible for two distinct plaquettes. So $\varphi \neq \varphi'$; a plaquette whose boundary is neither φ nor φ' has at most one edge in each and at most one further edge e_0, three in all, which is impossible. So the boundaries of the two plaquettes are φ and φ', and J is connected only if they share a vertex. (iv) By (i) every plaquette has two edges in $\varphi \cup \varphi'$. A plaquette with two edges in one cycle of length 4 is the plaquette that cycle bounds (two edges of a plaquette at a corner, or opposite ones, determine it, and a straight cycle contains at most one edge of a plaquette). If $\varphi' = \varphi$, all three plaquettes would be the one bounded by φ. Otherwise a plaquette not bounded by φ or φ' has exactly one edge $e \in \varphi$ and one edge $e' \in \varphi'$, $e \neq e'$, and is the only plaquette containing both. As there are three plaquettes, at least one is of this kind. (v) A front with five edges is a straight cycle (Lemma 7.5), with five collinear edges, around a side of length 5. By (i) these

lie in $E_M(J) \cup \varphi$. A plaquette has at most one edge on a given line; φ has at most one if it bounds a plaquette, and none if it is straight (its direction has side 4). So $5 \le n+1$. □

Lemma 8.7 (Counting junctions with two front-pairs)**.** *Let $p = (\varphi, \lambda)$ be a front-pair candidate at slice s with $|\varphi| = 4$ (a front in Γ and a layer). For $n \ge 1$ and $k \ge 4$ let $Z_{n,k}$ be the number of pairs (J, p') with J a junction without support at slice s of some set X as in Definition 7.1, with $\Phi(J) = \{p, p'\}$, $|J|_C = n$ and $|p'| = k$. Then $Z_{1,4} \le 1$, $Z_{2,4} \le 4(P_v - 1) - K_S$, $Z_{3,4} \le 48\binom{18}{2}(d-2)(2d-3)$, and for all n, k*

$$Z_{n,k} \le 2\binom{4P_v + T_v n}{n} X_{n,k},$$

where $X_{n,4} := (n+1)(2d-3)$, $X_{n,5} := 0$ if $n \le 3$ or if no side of Λ_s equals 5 and $X_{n,5} := \frac{4}{5}(n+1)$ otherwise, and $X_{n,k} := \min\{\binom{4n+4}{k}, (4n+4)N_k^v\}$ for $k \ge 6$.

Proof. $Z_{1,4}$ and $Z_{2,4}$: Lemma 8.6(ii),(iii); for (iii), q' is a plaquette other than q with a vertex in $V(q)$, and counting the pairs (v, q') with $v \in V(q) \cap V(q')$, at most $4(P_v - 1)$, counts the K_S plaquettes that share a link with q twice. $Z_{3,4}$: by Lemma 8.6(iv), choose $e \in \varphi$ (4 ways), a plaquette $P \ni e$ ($2(d-2)$ ways), an edge $e' \ne e$ of P (3 ways), the cycle $\varphi' \ni e'$ of length 4 (at most $2d-3$ ways: the plaquettes containing e' and the straight cycle through it), the layer λ' (2 ways), and two further plaquettes among the at most $2+16$ allowed by Lemma 8.6(iv). For the general bound: by Lemma 8.6(i) the set of the n plaquettes of J, together with φ, is connected in the graph in which φ is joined to the at most $4P_v$ plaquettes with a vertex in $V(\varphi)$ and plaquettes sharing a vertex are joined; by [III, Lemma 'Rooted counting' (`al:lem:rooted`)] there are at most $\binom{4P_v + T_v n}{n}$ such sets, and they determine the clusters. Given them, φ' is a front contained in $F := E_M(J) \cup \varphi$, with $|F| \le 4n+4$ and $|V(F)| \le 4n+4$, at one of two layers. Fronts with $k \ge 6$ edges in F number at most $\binom{|F|}{k}$ and at most $|V(F)|N_k^v$; fronts with four edges in F are plaquette boundaries (at most $2(d-2)|F|/4$, each edge of F lying in $2(d-2)$ plaquettes of the slice) or straight cycles, which are edge-disjoint and so at most $|F|/4$; straight cycles with five edges are edge-disjoint, at most $|F|/5$, and occur only for $n \ge 4$, and only if a side of Λ_s equals 5 (Lemma 8.6(v)). □

Lemma 8.8 (Marked tree sums)**.** *In the setting of Lemma 7.7, let h satisfy its hypothesis, and let $\omega_1, \omega_2 : \mathcal{V} \to [0, \infty)$. Let $\Psi_i(v)$ be the same sum as $\Psi(v)$ over trees with one marked non-root vertex u, with $w(u)$ replaced by $\omega_i(u)$, and $\Psi_{12}(v)$ the sum over trees with two distinct marked non-root vertices u_1, u_2, with $w(u_j)$ replaced by $\omega_j(u_j)$. If $g_1, g_2, g_{12} : \mathcal{V} \to [0, \infty)$ satisfy*

$$G_i(v) := \sum_{c \in \mathcal{N}(v)} (\omega_i(c) + w(c) g_i(c)) e^{h(c)} \le g_i(v) \qquad (i = 1, 2),$$
$$G_1(v) G_2(v) + \sum_{c \in \mathcal{N}(v)} (\omega_1(c) g_2(c) + \omega_2(c) g_1(c) + w(c) g_{12}(c)) e^{h(c)} \le g_{12}(v)$$

for all v, then $\Psi_i \le g_i e^h$ and $\Psi_{12} \le g_{12} e^h$.

Proof. By induction on the depth n, as in Lemma 7.7, with $\Psi_{\cdot,n}$ the sums over trees of depth $\le n$. A tree with one mark is a set of children of the root of which exactly one carries the mark or has it in its subtree, so

$$\Psi_{i,n}(v) \le \Psi_n(v) \sum_c (\omega_i(c) \Psi_{n-1}(c) + w(c) \Psi_{i,n-1}(c)) \le e^{h(v)} G_i(v).$$

With two marks, either one child carries or contains both, or two distinct children carry or contain one each:

$$\frac{\Psi_{12,n}(v)}{\Psi_n(v)} \le \sum_c \big(\omega_1(c)\Psi_{2,n-1}(c) + \omega_2(c)\Psi_{1,n-1}(c) + w(c)\Psi_{12,n-1}(c)\big) \\ + \prod_{i=1,2}\sum_c \big(\omega_i(c)\Psi_{n-1}(c) + w(c)\Psi_{i,n-1}(c)\big),$$

which is at most $g_{12}(v)$ by the induction hypothesis. □

Definition 8.9 (Constants of the junction expansion)**.** Let a, b be as in Theorem 7.8, and take N_4^v, N_5^v from Lemma 7.5, $N_6^v \le 16(d-1)^5$ (Lemma 8.5) and $N_k^v \le (d-1)\binom{(4d-6)k}{k-1}$ for $k \ge 7$. For $\xi_F, \xi_C, h_J, F \ge 0$ put $c_J := e^{h_J}/(1-a^4)$ and, for $j = 0, 1, 2$,

$$Y_j := \sum_{n\ge1} n^j \binom{K_S n}{n-1} (be^{\xi_C})^n, \qquad A_j := e^{c_J}\sum_{k\ge4} k^j N_k^v a^{k/2} e^{\xi_F k}, \qquad B_j := \sum_{k\ge4} k^j N_k^v f(k) e^{\xi_F k},$$
$$f(4) := 1, \qquad f(k) := Fk^2 e^{\xi_F k} a^{k-4} \quad (k \ge 5),$$

and let $g_F, g_C, g_F', g_C', \kappa_F, \kappa_C \ge 0$ satisfy

$$P_v g_C Y_1 + B_0 + g_F A_1 \le g_F, \qquad 4P_v g_C Y_1 + 8B_0 + 8g_F A_1 \le g_C,$$
$$P_v g_C' Y_1 + A_0 + g_F' A_1 \le g_F', \qquad 4P_v g_C' Y_1 + 8A_0 + 8g_F' A_1 \le g_C',$$
$$g_F g_F' + P_v \kappa_C Y_2 + g_F' B_1 + g_F A_1 + \kappa_F A_2 \le \kappa_F,$$
$$g_C g_C' + 4P_v \kappa_C Y_2 + 8(g_F' B_1 + g_F A_1 + \kappa_F A_2) \le \kappa_C.$$

Finally let

$$\Sigma_4 := b + (4P_v - 4 - K_S)b^2 + 48\binom{18}{2}(d-2)(2d-3)\,b^3 + 2\sum_{n\ge4}(n+1)(2d-3)\binom{4P_v + T_v n}{n} b^n$$
$$+ 2\sum_{n\ge1}\binom{4P_v + T_v n}{n} b^n \sum_{k\ge5} X_{n,k} f(k) + 4P_v e^{4\xi_F}\big(\kappa_C Y_2 + 4(g_C g_F' + g_C' g_F)Y_1 + 16\kappa_F Y_0\big),$$

with $X_{n,k}$ as in Lemma 8.7.

Lemma 8.10 (Junction sums)**.** *In the setting of Definition 8.9 suppose that the series converge and that*

$$P_v Y_0 + A_0 \le \xi_F, \qquad 4P_v Y_0 + 8A_0 \le \xi_C, \qquad kP_v Y_0 a^{k/2} e^{\xi_F k} \le h_J \quad (k \ge 4).$$

Let p be a front-pair candidate at slice s, $k := |p|$, and let J range over the junctions without support at slice s containing p, of sets X as in Definition 7.1. Then:

(i) $\displaystyle\sum_J b^{|J|_C} \prod_{\psi\in\Phi(J)\setminus\{p\}} (a^{|\psi|/2} e^{c_J}) \le kP_v Y_0 e^{\xi_F k}$;

(ii) with $\mathcal{S}(p) := \sum_J \sum_{p'\in\Phi(J)\setminus\{p\}} b^{|J|_C} \prod_{\psi\in\Phi(J)\setminus\{p,p'\}} (a^{|\psi|/2} e^{c_J}) f(|p'|)$*:* $\mathcal{S}(p) \le \Sigma_4$ *if* $k = 4$*, and* $\mathcal{S}(p) \le kP_v e^{\xi_F k}(g_C Y_1 + g_F k Y_0)$ *for all* k*.*

Proof. Apply Lemmas 7.7 and 8.8 with $\mathcal{V}$ the set of candidate clusters at slice s (connected sets of plaquettes of the slice) and candidate front-pairs, $w(C) := b^{|C|}$, $w(\psi) := a^{|\psi|/2} e^{c_J}$, the children of a cluster being the clusters and the front-pairs (of either layer) with which it shares a vertex, and the children of a front-pair being the clusters and the front-pairs of

the other layer with which it shares a vertex; $h(C) := \xi_C|C|$, $h(\psi) := \xi_F|\psi|$. Counting as in Lemma 7.6 (a cluster through a vertex: a plaquette at the vertex and a connected set through it; a front-pair of a given layer through a vertex: $N^v_{k'}$ fronts of each size k'), the children of a front-pair ψ contribute at most $|\psi|(P_vY_0 + A_0)$ to the condition of Lemma 7.7 and those of a cluster C at most $4|C|P_vY_0 + 8|C|A_0$, since $|V(\psi)| \le |\psi|$ and $|V(E(C))| \le 4|C|$; so $\Psi \le e^h$. In the same way the six inequalities of Definition 8.9 are the hypotheses of Lemma 8.8 with $\omega_1(\psi) := f(|\psi|)$, $\omega_2(\psi) := w(\psi)$, $\omega_i(C) := 0$, $g_1 = g_F|\cdot|$, $g_2 = g'_F|\cdot|$, $g_{12} = \kappa_F|\cdot|^2$ on front-pairs and $g_1 = g_C|\cdot|$, $g_2 = g'_C|\cdot|$, $g_{12} = \kappa_C|\cdot|^2$ on clusters (using $|v| \le |v|^2$).

A junction $J \ni p$ without support is a connected subgraph of this graph, and p is adjacent to a cluster C_0 of J (Lemma 8.2(i)); so J is the label set of a tree rooted at p in which C_0 is a child of p, and a pair (J, p') is such a tree with a marked vertex. The trees in which C_0 is a child of the root contribute at most $w(C_0)\Psi(C_0)\cdot\Psi(p)$, and those with marks at most the corresponding sums of the decompositions in the proof of Lemma 8.8. Summing over the at most $kP_v\binom{K_S n}{n-1}$ clusters C_0 with n plaquettes touching p gives (i) and the second bound in (ii). For the first bound in (ii), split the pairs (J, p'): if $\Phi(J) = \{p, p'\}$ and $|p'| = 4$, they are counted by $Z_{n,4}$ (Lemma 8.7) with weight b^n; if $\Phi(J) = \{p, p'\}$ and $|p'| \ge 5$, by $Z_{n,k}$ with weight $b^n f(k)$; if $|\Phi(J)| \ge 3$, choose a further front-pair p'' and mark it with $\omega_2 = w$: these terms are bounded by the doubly marked sum, at most $\sum_{C_0} w(C_0) e^{\xi_C|C_0|} e^{4\xi_F} (\kappa_C|C_0|^2 + g_C|C_0|\,4g'_F + g'_C|C_0|\,4g_F + 16\kappa_F)$. □

Theorem 8.11 (The first-order constant)**.** *In the setting of Definition 8.9 and Lemma 8.10, let $\delta' > 0$ and suppose that*

$$\Sigma_4 < \delta', \qquad P_v(g_C Y_1/5 + g_F Y_0) < F\delta', \qquad a^4(1+\delta') \le 1.$$

Then there is $C_{A,B}$, independent of t and L_0, with

$$\left|\mathrm{Cov}_{\mu_{\Lambda,\beta}}(A,B)\right| \le C_{A,B}\big(a^4(1+\delta')\big)^t + \frac{2\|A\|_\infty\|B\|_\infty c^{|S|}}{1-\hat q}\,\hat q^{\,4L_0},$$

and $m_{\Lambda_s}(\beta) \ge 4\log(1/u_*) - 8(K+1)\theta_5 - \log(1+\delta')$. With the values $N^v_4 = P_v$, $N^v_5 = 0$, valid when all spatial sides are ≥ 6, the same holds for $m(\mu_\beta)$. Let δ_J be the infimum of the admissible δ' (for the respective values of N^v_4, N^v_5). As $\beta \downarrow 0$ the hypotheses hold for every $\delta' > b(1+c\beta)$ with a constant c, so $\delta_J = b\,(1+O(\beta))$; if ρ has no trivial summand, $\delta_J \le 2\beta(1+O(\beta))$ and

$$m(\mu_\beta) \;\ge\; 4\log\frac{1}{u_*(\beta)} - 2\beta + O(\beta^2).$$

Proof. As in the proof of Theorem 7.8, the sets X with a temporal plaquette in every layer give the second term, and for the others $|\Phi_F(X)|r_X^2 \le 4\|A\|_\infty\|B\|_\infty e^{2\theta_5 J_S} W(X)$ with $W(X) = \prod_J w(J)\prod_\sigma w(\sigma)$. Put $z := 1/(a^4(1+\delta')) \ge 1$. For each such X choose a path $\pi(X)$ of minimal length from J_A to J_B in $\mathcal{J}(X)$ and a spanning forest $\mathcal{F}(X)$ of $\mathcal{J}(X)$ each of whose trees contains exactly one vertex of $\pi(X)$ (Lemma 8.4); then $z^t \le z^{\sum_{\sigma\in\pi(X)}\ell(\sigma)}$, and X is determined by $(\pi(X), \mathcal{F}(X))$. On $\pi(X)$ we combine each segment with the factors $a^{|\sigma|/2}$ of its two end junctions: a path segment carries $(za^{|\sigma|})^{\ell(\sigma)}$, and a path junction J the weight $b^{|J|_C}\prod a^{|\psi|/2}$ over its front-pairs ψ that are not ends of path segments. Every intermediate junction of the path contains no support, since $\pi(X)$ is simple and J_A, J_B are its ends; and junctions off the path contain no support, since every support lies in J_A or J_B.

Decorations. The children of a junction in $\mathcal{F}(X)$ are segments with an end in it, those of a segment the junction at its other end. A path segment has no children. In Lemma 7.7 we label a junction off the path by the pair (J, p_0), p_0 the front-pair at which it is reached from its parent, with possible children the segments with an end at the other front-pairs of J, and a segment by the pair (σ, end) of the end at which it is reached, with possible child the junction at its other

end; so the possible children depend only on the label. Give $(\sigma,\cdot)$ the weight $w(\sigma)$ and $h := h_J$, and (J,p_0) the weight $w(J)$ and $h := c_J(|\Phi(J)|-1)$. The segments with an end at a given front-pair ψ of a junction contribute at most $\sum_{\ell\ge1} a^{|\psi|(\ell-1)}e^{h_J} \le c_J$, which gives the condition of Lemma 7.7 at junctions; at a segment ending at a front-pair p^* with $|p^*|=k$, the junctions $J' \ni p^*$ contribute $\sum w(J')e^{h(J')} = a^{k/2}\sum b^{|J'|_C}\prod_{\psi\neq p^*}(a^{|\psi|/2}e^{c_J}) \le a^{k/2}kP_vY_0e^{\xi_F k} \le h_J$ by Lemma 8.10(i). Hence the decorations of a path junction J are bounded by e^{c_J} for each of its front-pairs that is not an end of a path segment.

The path. For a front-pair p at a path junction that is the start of the next path segment, let $\mathcal{V}(p)$ be the sum over the continuations of the path (the segment, of any length ℓ, the junction J' at its far end with its decorations, and then either $J'=J_B$ or a further start $p'\in\Phi(J')$ and its continuation) of the product of the weights just described. Since $\sum_{\ell\ge1}(za^k)^\ell \le a^{k-4}/\delta'$ for $k\ge4$,

$$\mathcal{V}(p) \le \frac{a^{|p|-4}}{\delta'}\Big(E_B(|p|) + \sup_{p^*}\sum_{J'\ni p^*}\sum_{p'} b^{|J'|_C}\prod_{\psi\neq p^*,p'}(a^{|\psi|/2}e^{c_J})\,\mathcal{V}(p')\Big),$$

where p^* ranges over the front-pairs with the front of p at the far end and $E_B(k)$ bounds the contribution of the junctions $J'=J_B$. A junction $J'\ni p^*$ containing S_B is the label set of a tree rooted at p^* in which S_B occurs; so $E_B(k)$ is at most the singly marked sum of Lemma 8.8 with $\omega(S_B) := e^{h(S_B)}$, $h(S_B) := |V(S_B)|(P_vY_0+2A_0)$ (which bounds the subtrees of S_B, whose children are the clusters and front-pairs touching it), and $\omega := 0$ on all other vertices, S_B being excluded from the unmarked sums. The function $g := G|\cdot|$ satisfies the hypothesis of that lemma for a constant G depending on $|S_B|$ (the term $\omega(S_B)e^{h(S_B)}$ appears only at the neighbours of S_B, and $|v|\ge1$), so $E_B(k) \le Gke^{\xi_F k} \le E_Bk^2e^{\xi_F k}$ with $E_B := G$. We show by induction on the length of the continuations that $\mathcal{V}(p) \le Mf(|p|)$ for a constant M. If $|p|=4$, Lemma 8.10(ii) bounds the right-hand side by $(E_B(4)+M\Sigma_4)/\delta' \le M$ when $M \ge E_B(4)/(\delta'-\Sigma_4)$. If $|p|=k\ge5$, it is at most $a^{k-4}(E_Bk^2e^{\xi_F k} + MkP_ve^{\xi_F k}(g_CY_1+g_FkY_0))/\delta' \le Mf(k)$ when $M \ge E_B/(F\delta' - P_v(g_CY_1/5+g_FY_0))$, since $(g_CY_1+g_FkY_0)/k \le g_CY_1/5+g_FY_0$. The beginning of the path is treated in the same way: the junctions $J_A\ni S_A$ with an exit p_1 are trees rooted at S_A with a marked front-pair p_1 (weight $f(|p_1|)$), and the singly marked bound gives a factor $C_A<\infty$ depending on $|S_A|$. So $z^t\sum_X W(X) \le MC_A$, which is the covariance bound.

Gaps: as in the proof of Theorem 7.8.

Small β. Here $a,b = O(\beta)$. Take $h_J := a^2$, $\xi_F := 2P_vb$, $\xi_C := 8P_vb$: for small β, $Y_0 \le b(1+O(\beta))$, $A_j = O(\beta^2)$, so the three conditions of Lemma 8.10 hold. Take $F := 3P_vN_4^v$. Then $B_0 = N_4^v+O(\beta)$, $B_1 = 4N_4^v+O(\beta)$, and the six inequalities hold with $g_F = N_4^v(1+O(\beta))$, $g_C = 8N_4^v(1+O(\beta))$, $g_F', g_C' = O(\beta^2)$ and $\kappa_F,\kappa_C = O(\beta^2)$. Then $\Sigma_4 = b+O(\beta^2)$ (the terms after the first are $O(b^2)$, $O(b\,a^2F)$ and $O(\beta^3)$), and $P_v(g_CY_1/5+g_FY_0) = \frac{13}{5}P_vN_4^vb(1+O(\beta)) < F\delta'$ for δ' near b. Finally $b = e^{2(K+1)\theta_5}\gamma_g$, $\theta_5 = O(\beta^5)$, and $\gamma_g \le \gamma_2 = 2\beta+O(\beta^2)$ if ρ has no trivial summand. □

Remark 8.12 (Numbers and checks). (i) For $SU(2)$, with the parameters of Definition 8.9 found by iterating the conditions, the lower bound of Theorem 8.11 for $m(\mu_\beta)$ lies below $4\log(1/u)$ by $2.0\cdot10^{-7}$, $2.0\cdot10^{-6}$, $2.00\cdot10^{-5}$, $1.7\cdot10^{-4}$ and $4.9\cdot10^{-4}$ at $\beta=10^{-7}$, 10^{-6}, 10^{-5}, $\beta_0(4)/2$ and $\beta_0(4)$ ($d=4$; $\delta_J/b = 1.00001$, 1.0001, 1.0012, 1.08, 1.49), and by $2.0\cdot10^{-7}$, $2.0\cdot10^{-6}$, $2.00\cdot10^{-5}$, $4.2\cdot10^{-4}$ and $1.09\cdot10^{-3}$ for $d=3$; on spatial lattices with all sides ≥4 the loss is at most $5.3\cdot10^{-4}$ ($d=4$) and $1.12\cdot10^{-3}$ ($d=3$). These values extend to the whole range: every left side in the hypotheses of Lemma 8.10, Definition 8.9 and Theorem 8.11 is non-decreasing in (a,b), and $a(\beta)$, $b(\beta)$ increase with β (as θ_5, $u=I_2/I_1$ and γ_2 do), so parameters admissible at $\beta_0(d)$ remain admissible for every smaller β. With the upper bound (Corollary 5.8), for all

$0 < \beta < \beta_0(d)$

$$-5.0 \cdot 10^{-4} \le m(\mu_\beta) - 4 \log \frac{1}{u(\beta)} \le 1.4 \cdot 10^{-3} \quad (d = 4),$$
$$-1.1 \cdot 10^{-3} \le m(\mu_\beta) - 4 \log \frac{1}{u(\beta)} \le 4.9 \cdot 10^{-3} \quad (d = 3)$$

(Numerical: an evaluation of the explicit constants, `python3` with `mpmath`). As $\beta \downarrow 0$ the two bounds are $-2\beta(1 + O(\beta))$ and $6\beta(1 + O(\beta))$ (Remark 5.9); Section 9 improves both to $-\kappa(\beta) + O(\beta^2)$. Near $\beta_0(4)$ the largest corrections to $\Sigma_4 \approx b$ come from junctions with two front-pairs and at least four plaquettes, from exits with at least six edges, and from junctions with three or more front-pairs (each about $0.15\, b$ at $\beta_0(4)$); all are crude, and the first-order term b is the cap. The largest single source of the last two is the crude bound $N_7^v \le (d-1)\binom{7(4d-6)}{6}$ (about $3.9 \cdot 10^8$ for $d = 4$), whereas on large tori the fronts with seven edges are the 108 ($d = 4$) and 12 ($d = 3$) pairs of plaquettes sharing a link through a given vertex (an independent referee's enumeration); with these values δ_J/b at $\beta_0(4)$ would be about 1.40. Without the special structure of $SU(2)$, i.e. with $u_* \le \epsilon$ and $\|g_p\|_\infty \le e^{4\beta} - 1$, the hypotheses still hold at $\beta_0(d)$ for $3 \le d \le 6$, with losses 0.051, 0.21, 0.44, 0.70 (and 0.023, 0.094, 0.21, 0.36 when ρ has no trivial summand), so for every G and ρ on the whole range (Numerical, values for spatial sides ≥ 6).

(ii) The cap is a plaquette q at a slice where a tube with front ∂q continues, and its weight b is the sup norm of g_q. Its true contribution is an integral of characters; Section 9 evaluates it, together with the corresponding terms of the upper bound.

(iii) Checks (`theory/checks/lower-bound/`, a compass): the constants of (i); an exhaustive enumeration of the junctions with two front-pairs, an entry with four edges and at most two plaquettes (and three for $d = 3$) on spatial tori 6×6, 7×7, 4×6, $6 \times 6 \times 6$ and $4 \times 6 \times 6$, which finds exactly the cap, the 8 ($d = 3$) and 32 ($d = 4$) junctions of Lemma 8.6(iii) with four-edge exits ($= 4(P_v - 1) - K_S$), no junction with three plaquettes and a four-edge exit in $d = 3$, and none with a straight entry and a four-edge exit; and a test of Lemmas 8.2 and 8.4 on 400 random sets X on the periodic lattice $8 \times 5 \times 5$ (2228 junctions). An independent referee's checks, with separate code, confirmed the numbers of (i), Lemma 8.6 on nine tori in $d = 3, 4$ (with up to four plaquettes in $d = 3$, three in $d = 4$; also $Z_{3,4} = 0$ in $d = 4$, so the b^3 term of Σ_4 is an overestimate), the junction lemmas on about 23600 junctions of random connecting sets in $d = 3, 4$, and Lemma 8.8 on random weighted systems.

Corollary 8.13 (The gap to first order, on the whole range)**.** *Let G be metrisable and ρ irreducible and non-trivial with $N \le 49$. For every $0 < \beta < \beta_0(d)$ for which the hypotheses of Theorem 8.11 hold (by Remark 8.12, a numerical evaluation, all $\beta < \beta_0(d)$ when $3 \le d \le 6$),*

$$-8(K+1)\theta_5(\beta) - \log\bigl(1 + \delta_J(\beta)\bigr) \;\le\; m(\mu_\beta) - 4 \log \frac{1}{u(\beta)} \;\le\; D(\beta),$$

the upper bound when $N^4 \le 1200\tau_0^2$; and $-2\beta + O(\beta^2) \le m(\mu_\beta) - 4\log(1/u(\beta)) \le D(\beta) = O(\beta)$ as $\beta \downarrow 0$.

Proof. Theorem 8.11 with $u_* = u$ (Corollary 6.5) and Corollary 5.8, or Theorem 5.6 for small β. □

9 The cap by characters: the first-order term

In Theorem 8.11 the first-order constant b of the lower bound is the cap, a plaquette q at a slice where a tube with front ∂q continues, weighed by the sup norm of g_q (Remark 8.12(ii)); and the first-order term $2(1 + 2/\sqrt{\tau_0})\beta$ of $D(\beta)$ in Theorem 5.6 comes from two sup-norm

bounds in the proof of Lemma 5.5. In this section both are evaluated by characters. In the transfer-matrix representation the cap is an operator on the class functions of the holonomy of q in the two copies; on the plaquette states $\chi_\rho(U_q)$, $\chi_{\bar\rho}(U_q)$ its norm, relative to the tube, is

$$\kappa(\beta) := \frac{1}{\tau_0}\int (\operatorname{Re}\chi_\rho)^2 f_\beta \, dU = \frac{\beta}{N\tau_0}\int (\operatorname{Re}\chi_\rho)^3 \, dU + O(\beta^2),$$

and the same number appears in the upper bound. The outcome is $m(\mu_\beta) = 4\log(1/u(\beta)) - \kappa(\beta) + O(\beta^2)$ (Corollary 9.6). The first-order term vanishes when the centre of G acts on ρ by a scalar that is neither 1 nor a cube root of unity: for $U(1)$, and for $SU(N)$ with $N \neq 3$ and the fundamental representation, $m(\mu_\beta) = 4\log(1/u(\beta)) + O(\beta^2)$. For $SU(3)$ it does not vanish: $\rho\otimes\rho\otimes\rho$ contains the trivial representation, so three sheets of fundamental flux can meet along the edges of the cap, and the cap can reverse the orientation of the flux through the tube; here $\int(\operatorname{Re}\chi_\rho)^3\,dU = \frac14$ and $m(\mu_\beta) = 4\log(1/u(\beta)) - \beta/6 + O(\beta^2)$. Two further changes make the remainder of order β^2: the transfer bound is applied junction by junction, with the gauge projections at the vertices of each junction (Lemma 9.2), and path segments whose fronts have at least five edges are summed without the factor $1/\delta'$ of the proof of Theorem 8.11.

Throughout this section ρ is irreducible and non-trivial, with N, ι, τ_0, u_r, u as in Section 5; $L^2_{\mathrm{cl}}(G)$ is the space of square-integrable class functions on G, with orthonormal basis $(\chi_r)_r$; and

$$u_2(\beta) := \sup_{r\notin\{0,\rho,\bar\rho\}} u_r(\beta), \qquad \sigma_6(\beta) := \sum_{r\neq 0} d_r^2 u_r(\beta)^6.$$

Lemma 9.1 (The cap matrix element)**.** *For every $\beta > 0$:*

(i) For $a, b \in \{\rho, \bar\rho\}$ the numbers $\varphi_{ab} := \int \overline{\chi_a}\, f_\beta\, \chi_b\, dU$ are real and non-negative, with $\varphi_{\bar\rho\bar\rho} = \varphi_{\rho\rho}$ and $\varphi_{\bar\rho\rho} = \varphi_{\rho\bar\rho}$. If P_ρ is the orthogonal projection of $L^2_{\mathrm{cl}}(G)$ onto the span of χ_ρ and $\chi_{\bar\rho}$, and M_f is multiplication by f_β, then $\|P_\rho M_f P_\rho\| = \kappa(\beta)$; and $\int \operatorname{Re}\chi_\rho\, f_\beta\, \overline{\chi_\rho}\, dU = \tau_0\kappa(\beta)$.

(ii) $0 \le \kappa(\beta) \le \epsilon(\beta)$; $\int(\operatorname{Re}\chi_\rho)^3\,dU \ge 0$; and $\bigl|\kappa(\beta) - \frac{\beta}{N\tau_0}\int(\operatorname{Re}\chi_\rho)^3\,dU\bigr| \le \frac12\beta^2 e^\beta(1+\beta)$.

(iii) If $\rho(z) = \zeta\mathbf{1}$ for a central element z of G with $\zeta \neq 1$ and $\zeta^3 \neq 1$, then $\int(\operatorname{Re}\chi_\rho)^3\,dU = 0$, so $\kappa(\beta) = O(\beta^2)$. This applies to $U(1)$ with $\rho(e^{i\theta}) = e^{i\theta}$, to $\mathbb{Z}_n$ ($n \neq 1, 3$) with a faithful character, and to $SU(N)$ with the fundamental representation for $N = 2$ and $N \ge 4$. For $SU(3)$ with the fundamental representation, $\int(\operatorname{Re}\chi_\rho)^3\,dU = \frac14$, so $\kappa(\beta) = \frac{\beta}{6} + O(\beta^2)$.

(iv) $u_2(\beta) \le e^\beta - 1 - \beta$ and $\sigma_6(\beta) \le (2-\iota)N^2 u(\beta)^6 + u_2(\beta)^5(e^\beta - 1)$.

Proof. (i) Inserting the absolutely and uniformly convergent expansion $f_\beta = \sum_{r\neq0} d_r u_r \chi_r$ gives $\varphi_{ab} = \sum_r d_r u_r \int \overline{\chi_a}\chi_r\chi_b\,dU$, where $\int\overline{\chi_a}\chi_r\chi_b\,dU$ is the multiplicity of a in $r\otimes b$; as $u_r > 0$, $\varphi_{ab} \ge 0$. Since f_β is real and $\overline{\chi_r} = \chi_{\bar r}$, $\varphi_{\bar a\bar b} = \overline{\varphi_{ab}} = \varphi_{ab}$, which gives the two equalities. If $\iota = 1$, P_ρ is the projection onto the real function χ_ρ ($\|\chi_\rho\| = 1$), so $\|P_\rho M_f P_\rho\| = \varphi_{\rho\rho}$, which is $\int(\operatorname{Re}\chi_\rho)^2 f_\beta = \kappa$ as $\tau_0 = 1$. If $\iota = 0$, $\chi_\rho \perp \chi_{\bar\rho}$, and in this basis $P_\rho M_f P_\rho$ has diagonal entries $\alpha := \varphi_{\rho\rho}$ and off-diagonal entries $\gamma := \varphi_{\rho\bar\rho}$, so its eigenvalues are $\alpha \pm \gamma$, with eigenvectors $\chi_\rho \pm \chi_{\bar\rho}$; its norm is $\alpha + \gamma$, attained at $\chi_\rho + \chi_{\bar\rho} = 2\operatorname{Re}\chi_\rho$, and therefore equals the Rayleigh quotient of $\operatorname{Re}\chi_\rho$, which is κ because $\|\operatorname{Re}\chi_\rho\|^2 = \tau_0$ (Lemma 5.1(i)). Finally $\int \operatorname{Re}\chi_\rho f_\beta\overline{\chi_\rho}$ equals $\varphi_{\rho\rho} = \kappa$ if $\iota = 1$, and $\frac12(\varphi_{\rho\rho} + \varphi_{\rho\bar\rho}) = \frac12\kappa$ if $\iota = 0$.

(ii) $\kappa = \|P_\rho M_f P_\rho\| \ge 0$, and $\kappa \le \|M_f\| = \|f_\beta\|_\infty \le \epsilon$, because $1 + f_\beta \in [e^{-2\beta}, e^\beta]$ (proof of Lemma 2.2(a)). Expanding $(\operatorname{Re}\chi_\rho)^3 = \frac18(\chi_\rho + \overline{\chi_\rho})^3$ writes $\int(\operatorname{Re}\chi_\rho)^3$ as a combination with positive coefficients of the integrals $\int \chi_\rho^j \overline{\chi_\rho}^{3-j}$, $0 \le j \le 3$, which are multiplicities of the trivial representation, hence non-negative. Put $y := \frac{\beta}{N}\operatorname{Re}\chi_\rho \in [-\beta, \beta]$ and $R := e^y - 1 - y \in [0, \frac12\beta^2 e^\beta]$. As $\int y = 0$, $c_0 = 1 + \int R$, and $f_\beta = e^y/c_0 - 1$ gives $f_\beta - y = (R - \int R - y\int R)/c_0$;

so $|f_\beta - y| \le \frac{1}{2}\beta^2 e^\beta(1+\beta)$, since $c_0 \ge 1$ and $|R - \int R| \le \frac{1}{2}\beta^2 e^\beta$. Hence $|\kappa - \tau_0^{-1}\int(\operatorname{Re}\chi_\rho)^2 y| \le \frac{1}{2}\beta^2 e^\beta(1+\beta)$, and $\tau_0^{-1}\int(\operatorname{Re}\chi_\rho)^2 y = \frac{\beta}{N\tau_0}\int(\operatorname{Re}\chi_\rho)^3$.

(iii) Haar measure is invariant under $U \mapsto zU$, and $\chi_\rho(zU) = \zeta\chi_\rho(U)$, $|\zeta| = 1$; so $\int \chi_\rho^j \overline{\chi_\rho}^{3-j} = \zeta^{2j-3}\int \chi_\rho^j \overline{\chi_\rho}^{3-j}$, and $\zeta^{2j-3} \in \{\zeta^{-3}, \zeta^{-1}, \zeta, \zeta^3\}$ is not 1. Take $z = -1$ for $U(1)$, a generator of $\mathbb{Z}_n$, and $z = e^{2\pi i/N}\mathbf{1}$ for $SU(N)$, for which $\zeta = e^{2\pi i/N}$ and $\zeta^3 = 1$ only if $N \in \{1, 3\}$. For $SU(3)$ the same argument removes the terms with $j = 1, 2$, and $\int \chi_\rho^3 = \int \overline{\chi_\rho}^3$ is the dimension of the space of $SU(3)$-invariant tensors $T = \sum T_{ijk} e_i \otimes e_j \otimes e_k \in (\mathbb{C}^3)^{\otimes 3}$ (the trace of the projection $\int U^{\otimes 3}\, dU$). The diagonal matrices $\operatorname{diag}(z_1, z_2, z_3)$, $z_1 z_2 z_3 = 1$, multiply T_{ijk} by $z_i z_j z_k$, so $T_{ijk} = 0$ unless $\{i, j, k\} = \{1, 2, 3\}$. Let $H \cong SU(2)$ act on the span of e_1, e_2 and fix e_3. Then $T = T^{(1)} + T^{(2)} + T^{(3)}$, where $T^{(m)}$ collects the terms with e_3 in the m-th factor; H preserves each of these three subspaces and acts on each as on $\mathbb{C}^2 \otimes \mathbb{C}^2$, whose H-invariants are the multiples of $e_1 \otimes e_2 - e_2 \otimes e_1$ (their dimension is $\int \chi_{1/2}^2 = 1$, and the determinant of $\mathbb{C}^2$ is invariant). So T is antisymmetric under the exchange of the values 1 and 2, and likewise for the other pairs: $T_{\sigma(1)\sigma(2)\sigma(3)} = \operatorname{sgn}(\sigma) T_{123}$, and T is a multiple of the determinant tensor, which is invariant. So $\int \chi_\rho^3 = 1$, $\int(\operatorname{Re}\chi_\rho)^3 = \frac{1}{8}(1+1) = \frac{1}{4}$, and (ii) with $N = 3$, $\tau_0 = \frac{1}{2}$ gives $\kappa = \frac{\beta}{6} + O(\beta^2)$.

(iv) The first bound is in the proof of Corollary 6.5. In σ_6 the terms $r \in \{\rho, \bar\rho\}$ contribute $(2-\iota)N^2 u^6$, and the others at most $u_2^5 \sum_{r \ne 0} d_r^2 u_r = u_2^5(e^\beta/c_0 - 1) \le u_2^5(e^\beta - 1)$ (Lemma 6.2). □

Lemma 9.2 (Transfer bound by pieces)**.** *Let X be as in Definition 7.1, and s a slice. The* pieces *at slice s are the junctions at slice s and the fronts of layer $s-1$ touched by no marker at slice s (which are fronts of layer s as well, Lemma 7.2(iii)). For a junction J at slice s, let M_J be multiplication by the product of the g_p over the plaquettes p of the clusters of J, times $A(U^{(0)}) - A(U'^{(0)})$ if $S_A \in M(J)$ and times $B(U^{(t)}) - B(U'^{(t)})$ if $S_B \in M(J)$; for $\lambda \in \{s-1, s\}$ let $K_{J,\lambda}$ be the tensor product over $e \in E(J)$ of $G_g^{1/2}$ if $e \in E_\lambda(J)$ and of $\Pi_0 \otimes \Pi_0$ otherwise (notation of Lemma 6.2); and let $O_J := K_{J,s-1} M_J K_{J,s}$, an operator on $L^2(G^{E(J)} \times G^{E(J)})$. For an untouched front φ let $E(\varphi) := \varphi$ and $O_\varphi := G_g^{\otimes\varphi}$. For a piece π let P^π be the orthogonal projection onto the functions of $(U_e, U'_e)_{e \in E(\pi)}$ that are invariant, in each copy, under the gauge transformations at the vertices of $V(E(\pi))$ (acting on the links of $E(\pi)$). Then*

$$|\Phi_F(X)| \;\le\; \prod_s \prod_{\pi \text{ at } s} \|P^\pi O_\pi P^\pi\|,$$

and $\|P^\varphi O_\varphi P^\varphi\| \le u_*^{|\varphi|}$, $\|P^J O_J P^J\| \le \|M_J\| \prod_{p \in \Phi(J)} u_*^{|p|/2}$.

Proof. In the proof of Lemma 6.3, $\Phi_F(X) = \operatorname{Tr}(\mathbb{M}_0 \mathcal{K}_0 \cdots \mathbb{M}_{L_0-1}\mathcal{K}_{L_0-1})$ and $\mathcal{K}_s = \tilde{\mathcal{K}}_s \mathbf{P}$, where $\mathbf{P} := P_{\mathcal{G}} \otimes P_{\mathcal{G}}$ and $\tilde{\mathcal{K}}_s \ge 0$ is the tensor product over the spatial links of G_g (on $T_s(X)$) and $\Pi_0 \otimes \Pi_0$ (elsewhere). It commutes with $\mathbf{P}$, so $\mathcal{K}_s = \tilde{\mathcal{K}}_s^{1/2} \mathbf{P} \tilde{\mathcal{K}}_s^{1/2}$, where $\tilde{\mathcal{K}}_s^{1/2}$ is the tensor product of the operators $G_g^{1/2}$ and $\Pi_0 \otimes \Pi_0$. By cyclicity of the trace and $\mathbf{P}^2 = \mathbf{P}$,

$$\Phi_F(X) = \operatorname{Tr}\Big(\prod_s \mathbf{P}\, \mathcal{O}_s \mathbf{P}\Big), \qquad \mathcal{O}_s := \tilde{\mathcal{K}}_{s-1}^{1/2}\, \mathbb{M}_s\, \tilde{\mathcal{K}}_s^{1/2}$$

(indices modulo L_0). For a layer s_* without temporal plaquettes, $\tilde{\mathcal{K}}_{s_*}^{1/2}$ is the rank-one projection Π onto the constants, and $\mathbf{P}\Pi = \Pi$; so $\mathbf{P}\mathcal{O}_{s_*+1}\mathbf{P} = \Pi(\mathbf{P}\mathcal{O}_{s_*+1}\mathbf{P})$ has trace norm at most its operator norm, and Hölder's inequality for traces gives $|\Phi_F(X)| \le \prod_s \|\mathbf{P}\mathcal{O}_s\mathbf{P}\|$.

Fix s. A spatial link of slice s that lies in T_{s-1}, in T_s or in the edge set of a marker at slice s belongs to exactly one piece (by Lemma 8.2(ii), and Lemma 7.2(iii)), and on the links of a junction J the factor $\tilde{\mathcal{K}}_\lambda^{1/2}$ is $K_{J,\lambda}$ (an edge of $E(J)$ lies in T_λ exactly when it lies in $E_\lambda(J)$, by Lemma 8.2(ii)), while on an untouched front φ the product $\tilde{\mathcal{K}}_{s-1}^{1/2}\tilde{\mathcal{K}}_s^{1/2}$ is $G_g^{\otimes\varphi}$. Moreover no link outside $E(\pi)$ at a vertex v of a piece π lies in T_{s-1}, in T_s or in a marker. Indeed, a marker

at slice s with the vertex v touches π if π is a front, and is adjacent in H_s to the vertex of J whose edge set has the endpoint v if $\pi = J$. A front ψ of layer $\lambda \in \{s-1, s\}$ through v is π itself if π is a front (fronts of one layer are vertex-disjoint). If $\pi = J$ and ψ is touched at slice s, then (ψ, λ) is a front-pair adjacent to the vertex of J with the endpoint v, hence in J; if ψ is not touched, it is a front of both layers, so it is the front of every front-pair of J through v, which is touched, and no front-pair of J passes through v; then a marker of J does, and touches ψ: a contradiction. Hence $\mathcal{O}_s = \bigotimes_\pi O_\pi \otimes (\Pi_0 \otimes \Pi_0)^{\otimes \mathcal{R}}$, with $\mathcal{R}$ the remaining links. Now $\mathbf{P} = \prod_v P_v \otimes P_v$, where $P_v = \int R_h^{(v)}\, dh$ is the projection for the gauge transformations $R_h^{(v)}$ at the vertex v of the slice; as Π_0 is invariant under left and right translations, $R_h^{(v)}$ acts on $\mathcal{O}_s$, from either side, only through the links of the piece containing v, and trivially if v lies in no piece. So $\mathbf{P}\mathcal{O}_s\mathbf{P} = \bigotimes_\pi P^\pi O_\pi P^\pi \otimes (\Pi_0 \otimes \Pi_0)^{\otimes \mathcal{R}}$, whose norm is $\prod_\pi \|P^\pi O_\pi P^\pi\|$. The last two bounds follow from $\|G_g^{1/2}\| = u_*^{1/2}$ (Lemma 6.2), $\|\Pi_0 \otimes \Pi_0\| = 1$ and $|E_\lambda(J)| = \sum |p|$ over the front-pairs p of J of layer λ. □

With $\|M_J\| \le \gamma_g^{|J|_C}$ (times $2\|A\|_\infty$, $2\|B\|_\infty$ for J_A, J_B) this is the bound of Lemma 6.3 for sets with $m(X) = 0$, split into the factors $w(J)$ and $w(\sigma)$ of Section 8 (before the factors $e^{2(K+1)\theta_5}$). The point of the splitting is that a single factor can now be computed exactly.

Lemma 9.3 (The cap by characters)**.** *Let $\beta > 0$ with $u_*(\beta) = u(\beta)$, and let J be a cap at slice s (Lemma 8.6(ii)): $M(J) = \{\{q\}\}$ and $\Phi(J) = \{(\partial q, s-1), (\partial q, s)\}$. Then $\|P^J O_J P^J\| \le u^4 \bar\kappa_{\rm cap}$, where*

$$\bar\kappa_{\rm cap} := \kappa + \frac{u_2^2}{u^2}\Big(1 + \frac{u_2^2}{u^2}\Big)\epsilon + \frac{\sigma_6}{u^4} + \sqrt{2}\, e^\beta \sigma_6^{1/2} + u^4\gamma_g,$$

all at β. As $\beta \downarrow 0$, $\bar\kappa_{\rm cap} = \kappa(\beta) + (2-\iota)N^2 u(\beta)^2 + O(\beta^3)$.

Proof. Here $E(J) = \partial q = \{e_1, \dots, e_4\}$, $O_J = (G_g^{1/2})^{\otimes 4} M_{g_q} (G_g^{1/2})^{\otimes 4}$, and $g_q = (1 + f_\beta(U_q))(1 + f_\beta(U_q')) - 1$. A function of the four links of the square ∂q that is invariant under the gauge transformations at its corners is a class function of the holonomy U_q (fix the gauge on three links), and U_q is Haar distributed when the U_{e_i} are; so $(U, U') \mapsto (U_q, U_q')$ identifies the range of P^J with $\mathcal{H} := L^2_{\rm cl}(G) \otimes L^2_{\rm cl}(G)$, with orthonormal basis $e_{aa'} := \chi_a(U_q)\chi_{a'}(U_q')$. Both factors of O_J commute with P^J (gauge transformations act by translations, which commute with G_g, and g_q is gauge invariant). As a function of U_{e_i}, $\chi_a(U_q)$ is a matrix coefficient of a or of $\bar a$, and $u_{\bar a} = u_a$; so by Lemma 6.2, $(G_g^{1/2})^{\otimes 4}$ multiplies $e_{aa'}$ by $\lambda^2_{aa'}$, where $\lambda_{a0} = u_a$, $\lambda_{0a'} = u_{a'}$, $\lambda_{aa'} = u_a u_{a'}$ $(a, a' \ne 0)$ and $\lambda_{00} = 0$. Hence $P^J O_J P^J$ is $\Lambda M_g \Lambda$ on $\mathcal{H}$, with $\Lambda e_{aa'} = \lambda^2_{aa'} e_{aa'}$ and $\langle e_{aa'}, M_g e_{bb'}\rangle = m_{ab} m_{a'b'} - \delta_{ab}\delta_{a'b'}$, where $m_{ab} := \int \overline{\chi_a}(1 + f_\beta)\chi_b\, dU = \delta_{ab} + \varphi_{ab}$ (the notation of Lemma 9.1, for all a, b). Since $\Lambda e_{00} = 0$, only $\mathcal{H}_1 \oplus \mathcal{H}_2 \oplus \mathcal{H}_{12}$ matters, the spans of the e_{a0}, the $e_{0a'}$ and the $e_{aa'}$ with $a, a' \ne 0$. With $m_{00} = 1 + \int f_\beta = 1$, $\varphi_{a0} = d_a u_a$ and $\varphi_{0b} = d_b u_b$ $(a, b \ne 0)$, $v_a := d_a u_a^3$, $D := \mathrm{diag}(u_a^2)_{a \ne 0}$ (of norm u^2, as $u_* = u$), $\Phi := (\varphi_{ab})_{a,b \ne 0}$ and $\mathfrak{m} := (m_{ab})_{a,b \ne 0}$ (compressions of M_f and M_{1+f}, so $\|\Phi\| \le \|f_\beta\|_\infty \le \epsilon$ and $\|\mathfrak{m}\| \le e^\beta$), the blocks of $\Lambda M_g \Lambda$ are: $D\Phi D$ from $\mathcal{H}_1$ to $\mathcal{H}_1$ and from $\mathcal{H}_2$ to $\mathcal{H}_2$; the rank-one operator $e_{0b'} \mapsto v_{b'} \sum_a v_a e_{a0}$, of norm $\sum_a v_a^2 = \sigma_6$, from $\mathcal{H}_2$ to $\mathcal{H}_1$, and its adjoint; $e_{bb'} \mapsto v_{b'} \sum_a (D\mathfrak{m}D)_{ab} e_{a0}$, of norm at most $u^4 e^\beta \sigma_6^{1/2}$, from $\mathcal{H}_{12}$ to $\mathcal{H}_1$, likewise to $\mathcal{H}_2$, and the adjoints; and $(D \otimes D)(\mathfrak{m} \otimes \mathfrak{m} - \mathbf{1})(D \otimes D)$, of norm at most $u^8\gamma_g$, on $\mathcal{H}_{12}$, because $\mathfrak{m} \otimes \mathfrak{m} - \mathbf{1}$ is a compression of M_g. For $D\Phi D$ write $D = u^2 P_\rho + D'$ with $\|D'\| \le u_2^2$; since P_ρ and D' act on orthogonal subspaces, $\|D\Phi D\| \le \max(u^4\kappa, u_2^4\epsilon) + u^2 u_2^2 \epsilon \le n_1 := u^4\kappa + u^2 u_2^2 \epsilon + u_2^4 \epsilon$ (Lemma 9.1(i)). The norm of a block operator is at most the largest eigenvalue of the matrix of the norms of its blocks, here

$$\begin{pmatrix} n_1 & \sigma_6 & n_y \\ \sigma_6 & n_1 & n_y \\ n_y & n_y & n_z \end{pmatrix}, \qquad n_y := u^4 e^\beta \sigma_6^{1/2}, \quad n_z := u^8 \gamma_g.$$

Its eigenvectors are symmetric or antisymmetric in the first two coordinates; the antisymmetric one has the eigenvalue $n_1 - \sigma_6$, and on the symmetric ones it acts as $\left(\begin{smallmatrix} n_1+\sigma_6 & \sqrt{2}n_y \\ \sqrt{2}n_y & n_z \end{smallmatrix}\right)$, with eigenvalues at most $n_1 + \sigma_6 + n_z + \sqrt{2}n_y = u^4\bar\kappa_{\rm cap}$. As $\beta \downarrow 0$: $u \ge (1+\iota)\beta e^{-\beta}/(2N^2)$ (Lemma 5.1(iii)) and $u_2 = O(\beta^2)$, so $(u_2/u)^2\epsilon = O(\beta^3)$, $\sigma_6/u^4 = (2-\iota)N^2u^2 + O(\beta^7)$ and $\sqrt{2}e^{\beta}\sigma_6^{1/2} + u^4\gamma_g = O(\beta^3)$ (Lemma 9.1(iv)). □

Theorem 9.4 (The lower bound to first order)**.** *Suppose that $u_* = u$. In the setting of Definition 8.9 and Lemma 8.10 put $b_{\rm cap} := e^{2(K+1)\theta_5}\min(\bar\kappa_{\rm cap}, \gamma_g)$ and $\Sigma_4^\chi := \Sigma_4 - b + b_{\rm cap}$. Let $\delta' > 0$ satisfy*

$$\Sigma_4^\chi < \delta', \qquad P_v(g_C Y_1/5 + g_F Y_0) < F(1+\delta'-a), \qquad a^4(1+\delta') \le 1.$$

Then the conclusions of Theorem 8.11 hold for this δ'. Let δ_χ be the infimum of the admissible δ' (with $N_4^v = P_v$, $N_5^v = 0$ for $m(\mu_\beta)$). As $\beta \downarrow 0$ the hypotheses hold for every $\delta' > b_{\rm cap} + (4P_v - 4 - K_S)b^2 + c\beta^3$, with a constant c; so

$$\delta_\chi \le \kappa(\beta) + (2-\iota)N^2u(\beta)^2 + (4P_v - 4 - K_S)\gamma_g^2 + O(\beta^3), \qquad m(\mu_\beta) \ge 4\log\frac{1}{u(\beta)} - \kappa(\beta) - O(\beta^2).$$

Proof. Weights. By Lemmas 9.2 and 9.3, and $r_X^2 \le e^{2\theta_5 J_S}e^{2(K+1)\theta_5|X|}$, every set X with an empty layer satisfies

$$|\Phi_F(X)|\, r_X^2 \le 4\|A\|_\infty\|B\|_\infty e^{2\theta_5 J_S}\, W^\chi(X), \qquad W^\chi(X) := \prod_J w^\chi(J)\prod_\sigma w(\sigma),$$

where $w^\chi(J) := w(J)$ for junctions other than caps and $w^\chi(J) := a^4 b_{\rm cap} \le w(J)$ for caps; the factor $e^{2(K+1)\theta_5}$ of each plaquette is distributed as in Section 8, one for the plaquette of the cap and the factors of the half layers with its two front-pairs. (A cap contains no support, so the bound applies to every cap of X.)

The path. The proof of Theorem 8.11 applies to W^χ, with two changes. First, the decorations are bounded as there, as $w^\chi \le w$, but a cap on the path now carries $b_{\rm cap}$: in the bound of Lemma 8.10(ii) for $k = 4$ the contribution $Z_{1,4}b$ of the cap becomes $b_{\rm cap}$, so Σ_4 becomes Σ_4^χ. Second, for a front-pair p with $k := |p| \ge 5$ we have $za^k = a^{k-4}/(1+\delta') \le a/(1+\delta')$, so $\sum_{\ell\ge1}(za^k)^\ell \le a^{k-4}/(1+\delta'-a)$ instead of a^{k-4}/δ', and the induction step for $|p| \ge 5$ holds when $M \ge E_B/(F(1+\delta'-a) - P_v(g_C Y_1/5 + g_F Y_0))$.

Small β. Take h_J, ξ_F, ξ_C as in the proof of Theorem 8.11, but $F := 3P_vN_4^v b$. Then $f(k) = O(\beta a^{k-4})$ for $k \ge 5$, so $B_0 = N_4^v(1+O(\beta))$ and $B_1 = 4N_4^v(1+O(\beta))$, and the six inequalities of Definition 8.9 hold with $g_F = N_4^v(1+O(\beta))$, $g_C = 8N_4^v(1+O(\beta))$ and $g_F', g_C', \kappa_F, \kappa_C = O(\beta^2)$; so $P_v(g_C Y_1/5 + g_F Y_0) = \frac{13}{5}P_vN_4^v b(1+O(\beta)) < F(1+\delta'-a)$. In Σ_4^χ the term of $Z_{2,4}$ is $(4P_v - 4 - K_S)b^2$; the terms of three plaquettes and of $n \ge 4$ plaquettes are $O(b^3)$; those with exits of $k \ge 5$ edges are $O(b\,Fa) = O(\beta^3)$ (the five-edge exits occur only for $n \ge 4$); and the last term is $O((\kappa_C + \kappa_F + g_F' + g_C')b) = O(\beta^3)$. So $\Sigma_4^\chi = b_{\rm cap} + (4P_v - 4 - K_S)b^2 + O(\beta^3)$, and $b_{\rm cap} \le \bar\kappa_{\rm cap}(1+O(\beta^5))$, $b = \gamma_g(1+O(\beta^5))$ give the bound on δ_χ by Lemma 9.3. Finally $m(\mu_\beta) \ge 4\log(1/u) - 8(K+1)\theta_5 - \log(1+\delta')$ with $\theta_5 = O(\beta^5)$, and $\gamma_g \le \gamma_2 = O(\beta)$, $u = O(\beta)$. □

Theorem 9.5 (The upper bound by characters)**.** *In the setting of Lemma 5.5 put $\eta_2 := a_* N\bar u/\tau_0$, $\kappa := \kappa(\beta)$, and*

$$V^\chi := (1+\kappa)(1+\mathcal{T}_p) + \mathcal{T}_p, \qquad L^\chi := \frac{(1+\kappa-\eta_2)^2(1-\mathcal{T}_Q)}{1+\epsilon^6} - \Big(1 + \frac{a_*^2}{\tau_0}\Big)e^{16\beta}\,\frac{N^4}{\tau_0^2}\,\mathcal{T}_Q.$$

Then $V^\chi \le V$ and, if $\eta \le 1$, $L^\chi \ge L$. If $\eta_2 \le 1$ and $L^\chi > 0$, then (i) and (ii) of Theorem 5.6 hold with $D^\chi(\beta) := \log(V^\chi/L^\chi)$ in place of $D(\beta)$. As $\beta \downarrow 0$, $D^\chi = -\log(1+\kappa) + 2\eta_2 + O(\beta^3) = -\kappa(\beta) + O(\beta^2)$; for $SU(2)$, $D^\chi(\beta) = \frac{3}{8}\beta^2 + O(\beta^3)$. Moreover $D^\chi \le -\log(1+\kappa) + D_0$, where $D_0 := \log\big((1+2\mathcal{T}_p)/L_0\big)$ and L_0 is L^χ with κ replaced by 0; D_0 increases with β.

Proof. In the proof of Lemma 5.5 two estimates are replaced by identities. First, $M_{S_A}(A^2) = \int A^2(1+f_\beta)\,dU = \tau_0 + \int(\operatorname{Re}\chi_\rho)^2 f_\beta\,dU = \tau_0(1+\kappa)$ (Lemma 5.1(i) and the definition of κ), instead of $M_{S_A}(A^2) \le (1+\epsilon)\tau_0$; so $\operatorname{Var}(A) \le \langle A^2\rangle \le (1+\mathcal{T}_p)\tau_0(1+\kappa) + \tau_0\mathcal{T}_p = \tau_0 V^\chi$. Second, by Lemma 9.1(i), $\hat a_\rho = \tau_0 + \int \operatorname{Re}\chi_\rho f_\beta \overline{\chi_\rho}\,dU - aNu = \tau_0(1+\kappa) - aNu \ge \tau_0(1+\kappa-\eta_2) \ge 0$, as $|a| \le a_*$ and $u \le \bar u$; so $M_Q \ge \frac{2}{1+\iota}\hat a_\rho^2 u^4 \ge \tau_0(1+\kappa-\eta_2)^2 u^4$. The rest of the proofs of Lemma 5.5 and Theorem 5.6 is unchanged and gives $\operatorname{Cov}(A, A_1) \ge \tau_0 u^4 L^\chi$, $\operatorname{Var}(A) \le \tau_0 V^\chi$ and the theorem. As $0 \le \kappa \le \epsilon$ (Lemma 9.1(ii)) and $\eta_2 \le \eta$, $V^\chi \le V$ and $L^\chi \ge L$. As $\beta \downarrow 0$: $\mathcal{T}_Q, \mathcal{T}_p = O(\beta^5)$, $a_* = O(\beta)$, $\eta_2 = O(\beta^2)$, so $\log V^\chi = \log(1+\kappa) + O(\beta^5)$ and $\log L^\chi = 2\log(1+\kappa) - 2\eta_2 + O(\beta^3)$. For $SU(2)$, $\kappa = O(\beta^2)$ by Lemma 9.1(iii); more precisely $\kappa = d_1 u_1 = 3I_3(\beta)/I_1(\beta) = \frac{1}{8}\beta^2 + O(\beta^4)$ ($u_j = I_{2j+1}/I_1$ for the spin-j representation, as in Remark 7.10), and $\eta_2 = \frac{1}{4}\beta^2 + O(\beta^3)$, so $D^\chi = \frac{3}{8}\beta^2 + O(\beta^3)$. Finally, $V^\chi \le (1+\kappa)(1+2\mathcal{T}_p)$ and $L^\chi \ge (1+\kappa)^2 L_0$, because $(1+\kappa-\eta_2)/(1+\kappa) \ge 1-\eta_2$; and $\mathcal{T}_p$, $\mathcal{T}_Q$, ϵ, a_*, η_2 increase with β (Lemma 5.7). □

Corollary 9.6 (The first-order term)**.** *Let G be metrisable and ρ irreducible and non-trivial. As $\beta \downarrow 0$,*

$$m(\mu_\beta) = 4\log\frac{1}{u(\beta)} - \kappa(\beta) + O(\beta^2) = 4\log\frac{1}{u(\beta)} - \frac{\beta}{N\tau_0}\int(\operatorname{Re}\chi_\rho)^3\,dU + O(\beta^2),$$

and the same holds for $m_{\Lambda_s}(\beta)$, uniformly in the periodic spatial lattices with all sides ≥ 4. In particular $m(\mu_\beta) = 4\log(1/u(\beta)) + O(\beta^2)$ for $U(1)$, and for $SU(N)$ with $N \ne 3$ and the fundamental representation; and for $SU(3)$ with the fundamental representation

$$m(\mu_\beta) = 4\log\frac{1}{u(\beta)} - \frac{\beta}{6} + O(\beta^2).$$

Proof. For small β, $u_* = u$ (proof of Corollary 6.5: as soon as $\beta e^{2\beta} \le (1+\iota)/N^2$). The lower bound is Theorem 9.4, with $\log(1+\delta_\chi) \le \delta_\chi$, and in finite volume the same theorem with the values of N_4^v, N_5^v of the lattice (bounded by Lemma 7.5, uniformly). The upper bound is Theorem 9.5, since $L^\chi \to 1$ and $\eta_2 \to 0$. The rest is Lemma 9.1(ii),(iii). □

Remark 9.7 (Numbers, meaning, and checks). (i) For $SU(2)$ ($\kappa = 3I_3/I_1$, $u = I_2/I_1$; u_2 and σ_6 bounded by Lemma 9.1(iv)), with the parameters found by iterating the conditions, the lower bound of Theorem 9.4 for $m(\mu_\beta)$ lies below $4\log(1/u)$ by $1.3\cdot10^{-12}$, $1.3\cdot10^{-10}$, $1.5\cdot10^{-8}$, $6.8\cdot10^{-6}$ and $1.28\cdot10^{-4}$ at $\beta = 10^{-7}$, 10^{-6}, 10^{-5}, $\beta_0(4)/2$ and $\beta_0(4)$ ($d=4$; $\delta_\chi/\beta^2 = 128.5$, 129.8, 151.7, 979, 3971), and by $3.2\cdot10^{-13}$, $3.3\cdot10^{-11}$, $3.4\cdot10^{-9}$, $1.0\cdot10^{-5}$ and $2.44\cdot10^{-4}$ for $d=3$ ($\delta_\chi/\beta^2 = 32.4$, 32.6, 34.5, 95, 219; at $\beta_0(3)$ most of the loss is $8(K+1)\theta_5$). On spatial lattices with all sides ≥ 4 the loss is at most $1.4\cdot10^{-4}$ ($d=4$) and $2.51\cdot10^{-4}$ ($d=3$). As $\beta \downarrow 0$, $\delta_\chi/\beta^2 \to \frac{1}{8}+\frac{1}{4}+4(4P_v - 4 - K_S)$, i.e. 128.375 ($d=4$) and 32.375 ($d=3$): the shifts and jumps of the tube, still at their sup norm, are now the leading term. The upper bound of Theorem 9.5 has $D^\chi = \frac{3}{8}\beta^2(1+O(\beta))$, and $D_0(\beta_0(d))$ is $4.8\cdot10^{-4}$ ($d=4$) and $2.4\cdot10^{-3}$ ($d=3$), where $\mathcal{T}_Q$, the five-plaquette remainder, dominates. As before (Remark 8.12(i)) the hypotheses of Theorem 9.4 are monotone in (a, b, b_{cap}), the right side of the second one decreases with a, and a, b, b_{cap} increase with β (checked on a grid for b_{cap}), so for all $0 < \beta < \beta_0(d)$

$$-1.3\cdot10^{-4} \le m(\mu_\beta) - 4\log\frac{1}{u(\beta)} \le 4.8\cdot10^{-4} \quad (d=4),$$
$$-2.5\cdot10^{-4} \le m(\mu_\beta) - 4\log\frac{1}{u(\beta)} \le 2.4\cdot10^{-3} \quad (d=3),$$

and $-128.4\beta^2 \le m(\mu_\beta) - 4\log(1/u(\beta)) \le 0.375\beta^2$ up to factors $1+O(\beta)$ ($d=4$; $-32.4\beta^2$ for $d=3$) (Numerical: an evaluation of the explicit constants, `python3` with `mpmath`). For

$SU(3)$ (κ and u by Weyl integration) $\kappa/\beta = 0.1667$ on the whole range, and the two bounds are $-\kappa(1+O(\beta))$ for small β (at $\beta = 10^{-7}$: $\delta_\chi = 1.6668 \cdot 10^{-8}$ and $D^\chi = -1.6667 \cdot 10^{-8}$, $\kappa = 1.6667 \cdot 10^{-8}$); near $\beta_0(4)$ the lower bound loses $1.1 \cdot 10^{-4}$ and $D^\chi \approx 9.0 \cdot 10^{-3}$, dominated by the factor $N^4/\tau_0^2 = 324$ in front of $\mathcal{T}_Q$.

(ii) Without the second change in the proof of Theorem 9.4 the method would not reach the second order: with $\sum_\ell (za^k)^\ell \le a^{k-4}/\delta'$ for $k \ge 5$ the constant F must be of order b/δ', and with $\delta' = O(\beta^2)$ the exits with six edges contribute $O(bFa^2) = O(\beta^2)$ with a large constant; numerically $\delta/\beta^2 \to 4.2 \cdot 10^3$ instead of 128.4 ($SU(2)$, $d = 4$).

(iii) Meaning. In the transfer-matrix picture the cap acts on the plaquette states $\chi_\rho(U_q)$ and $\chi_{\bar\rho}(U_q)$ by the matrix of Lemma 9.1(i); its eigenvectors are $\operatorname{Re}\chi_\rho$ and $\operatorname{Im}\chi_\rho$, which are even and odd under charge conjugation $U \mapsto \bar U$, with eigenvalues $\kappa = \alpha + \gamma$ and $\alpha - \gamma$. So to first order the C-even plaquette state is lowered by κ and, if $\iota = 0$, the C-odd one is shifted by $-(\alpha - \gamma)$ (Heuristic; only the lowest mass is bounded here). For $SU(3)$, $\alpha = O(\beta^2)$ and $\gamma = 3u + O(\beta^2)$: the two plaquette glueballs split at first order, $m_\pm \approx 4\log(1/u) \mp 3u$. The title and the first lines of the abstract of [OBS85] (complex characters, the subspace generated by the plaquette function and its complex conjugate, "mass splitting") suggest that this splitting was found there, in $2+1$ dimensions; we have not seen the paper.

(iv) Checks (`theory/checks/first-order/`, a compass): an exact computation for $\mathbb{Z}_3$ in $d = 3$ on the 3×3 spatial torus, in the sector of winding zero (dimension 3^8; there $\kappa = u$ exactly), gives $(m - 4\log(1/u))/u = -0.99875$ for the lowest C-even state and $+1.00025$ for the lowest C-odd state at $\beta = 10^{-3}$, tending to ∓ 1; for $\mathbb{Z}_2$ ($\kappa = 0$) on the 4×4 torus, winding zero, $(m - 4\log(1/u))/u^2 \to 1$, so there is no first-order term. A brute-force computation of $\|P^J O_J P^J\|$ on the full two-copy space of the four links of a cap ($\mathbb{Z}_3$: dimension 3^8; S_3 with its two-dimensional representation, for which $\int \chi_\rho^3 = 1$ and $\kappa \approx \beta/2$: dimension 6^8) agrees with the class-function formula of the proof of Lemma 9.3 to all printed digits, lies below $u^4\bar\kappa_{\mathrm{cap}}$, and is $u^4(\kappa + (2-\iota)N^2u^2)$ to six digits. Weyl integration gives $\int(\operatorname{Re}\chi)^3 = 0$ for $SU(2)$, $SU(4)$, $SU(5)$ and $\frac14$ for $SU(3)$, and $\kappa_{SU(3)}(\beta)/(\beta/6) = 1 + O(\beta)$.

(v) What remains at second order: the shifts and jumps of Lemma 8.6(iii), $4(P_v - 1) - K_S$ junctions of two plaquettes at their sup norm $b^2 \approx 4\beta^2$, whose character values should be of order N^2u^2, about $\beta^2/4$ for $SU(2)$ (Heuristic); the copy switch σ_6/u^4 and the term η_2, which come from the duplicated system and from the mean a; and, near $\beta_0(d)$, the crude counts of Remark 8.12(i) and $\mathcal{T}_Q$.

10 Two dimensions; discussion

Proposition 10.1 (Two dimensions)**.** *Let $d = 2$ and $0 < \beta < \beta_0(2)$, so that $K = 4$ and $q(\beta) < 0.909$ (Lemma 4.2).*

(i) For every $L_1 \ge 3$, the spatial lattice $\Lambda_s = \mathbb{Z}/L_1\mathbb{Z}$ has $m_{\Lambda_s}(\beta) \ge L_1 \log(1/q(\beta))$.

(ii) In infinite volume, $\operatorname{Cov}_{\mu_\beta}(A, B) = 0$ for all A, B as in Theorem 1.2(iii) (time ranges with $a_2 < b_1$); hence $m(\mu_\beta) = \infty$.

Proof. By Lemma 3.1, $g = L_1$, and Theorem 4.1 gives the bound with $q^{L_1\tau}$. (i) follows as in the proof of Theorem 1.2(i), with L_1 in place of 4. For (ii), the bound for the periodised observables on tori with $L_0, L_1 \to \infty$ tends to 0, since $\tau \ge 1$ and $g = L_1 \to \infty$, and the torus covariances converge to $\operatorname{Cov}_{\mu_\beta}(A, B)$ as in the proof of Theorem 1.2(iii) (note $\beta_0(2) < \beta_*(2)$, Lemma 4.2); then [II, Theorem `iv:thm:OS`](iii) applies with every $m > 0$. □

Remark 10.2 (Consistency with the exact solution). In $d = 2$ the transfer matrix on a spatial circle of length L_1 is diagonal in the characters of the holonomy, with eigenvalue ratios $(c_r/(d_r c_0))^{L_1}$ (`lattice/ymlattice/transfer2d.py`, tested against the Monte Carlo); so the

exact finite-volume gap is $L_1 \log(1/u(\beta))$ for $SU(2)$, linear in L_1, and the infinite-volume theory has no propagating states. Proposition 10.1 reproduces both features (with q in place of u). It is also consistent with [II, Proposition 'Gauge invariance of the ground-state chain' (`iv:prop:phys`)]: in $d = 2$ the spatial links of a time slice form a line, gauge-invariant functions of finitely many of them are constant, and P acts trivially on the physical sector. The difference between $d = 2$ and $d \geq 3$ is the girth of the spatial graph (L_1 versus 4): in $d \geq 3$ a closed surface can turn a corner in every time layer with four plaquettes, the minimal glueball.

Remark 10.3 (The strong-coupling test, sharpness, and the $U(1)$ test). (i) Principle 3 of the research program requires the correct asymptotics $m(\beta) \sim 4\log(1/u(\beta))$. Theorem 1.2 gives the lower half, $m \geq 4\log(1/\beta) - O(1)$ for all $\beta < \beta_0(d)$, and Theorem 5.6 the upper half, $m \leq 4\log(1/u(\beta)) + O(\beta)$, on the same range when $N^4 \leq 1200\tau_0^2$ and for sufficiently small β in general; Corollary 5.8 combines them. The upper bound uses a single correlator at time separations 0 and 1; by the positivity of P ([II, Remark 'Consequences for the numerics' (`iv:rem:numerics`)]) any larger pair of separations could be used instead. The additive constants of the two halves differ ($-4\log(4e^3K)$ versus $+4\log\frac{2N^2}{1+\iota}$); the strong-coupling expansion predicts $m = 4\log(1/u) + O(u)$, i.e. the upper one. (ii) The constant $-4\log(4e^3K)$ of Theorem 1.2 reflects the crude counting of [III] and the sup-norm bound $\|g_p\|_\infty \leq e^{4\beta} - 1$ on every plaquette. Section 6 removes the second loss for the temporal plaquettes, which gives the constant $-4\log(eK)$ relative to $4\log(1/u)$ as $\beta \downarrow 0$ (Corollary 6.5); counting by fronts (Section 7) removes the counting loss as well, so that the lower bound is $4\log(1/u_*) - O(\beta)$, i.e. $4\log(1/u) - O(\beta)$ when $u_* = u$, and grouping by junctions (Section 8) brings the $O(\beta)$ to $2\beta(1 + O(\beta))$ when ρ has no trivial summand. Evaluating the cap by characters (Section 9) brings it to $\kappa(\beta) + O(\beta^2)$, which is also the first-order term of the upper bound: $m = 4\log(1/u) - \kappa + O(\beta^2)$. (iii) The hypothesis $L_i \geq 4$ cannot be dropped from the sharper finite-volume bounds: if a spatial side equals 3, Theorem 4.1 holds only with $g = 3$, and closed strips of three plaquettes per layer around that side (torelons, carrying the flux of a Polyakov loop) give states of energy $\approx 3\log(1/u(\beta))$. For $\mathbb{Z}_2$ in $d = 3$ on a 3×3 spatial torus the transfer matrix preserves the winding class, and a high-precision computation of the top eigenvalues of the sectors of winding $(0,0)$ and $(1,0)$ gives a torelon energy $43.526 = 3\log(1/u)$ at $\beta = 5{\cdot}10^{-7}$, below the bound $4\log(1/q_u) = 44.086$ of Theorem 6.4(i) (Numerical; also $46.27 < 47.76$ at $\beta = 2\cdot 10^{-7}$; `theory/checks/lower-bound/z2_torelon_mp.py`). So Theorem 6.4(i) fails on such lattices. For the weaker bound $4\log(1/q)$ of Theorem 1.2(i) the crossing is expected only near $\beta \approx 10^{-12}$. (iv) Nothing here is non-abelian: the argument applies verbatim to $G = U(1)$, whose $d = 4$ theory is expected to be massless at large β (proved in [FS82] for the Villain action and asserted there for Wilson's; [III, Remark 'The $U(1)$ test' (`al:rem:u1test`)]). The result therefore concerns the strong-coupling endpoint only; it shows that the endpoint is controlled with the right leading behaviour (Corollary 5.8), which is what a renormalisation-group argument would need to reach.

Summary of results

Statement	Where	Status
Duplicated system; loop expansion with Ξ^2; cancellation of non-connecting terms	Lem. 2.2, 2.4, 2.5	Theorem
Surfaces without free edges have $\geq g$ temporal plaquettes per crossed layer ($g = 4$ for $d \geq 3$, spatial sides ≥ 4)	Lem. 3.1, 3.2, 3.3, Cor. 3.4	Theorem
Covariance bound $q^{g\tau}$ in finite and infinite volume	Thm. 4.1, 1.2(iii)	Theorem
$m(\mu_\beta) \geq 4\log(1/q(\beta)) = 4\log(1/\beta) - O(1)$, and $m_{\Lambda_s}(\beta) \geq 4\log(1/q(\beta))$ for spatial sides ≥ 4; $d \geq 3$, $\beta < \beta_0(d)$	Thm. 1.2	Theorem
$d = 2$: gap $\geq L_1 \log(1/q)$; infinite gap in infinite volume	Prop. 10.1	Theorem
Positivity: $\mathrm{Cov}(A(0), A(1))$ dominates the expectation in which the four sides of the cube carry φ_β (every β)	Lem. 5.2	Theorem
Cube identity $\sum_{r\neq 0} \hat{a}_{0,r}\hat{a}_{1,r}u_r^4$; five-plaquette lemma	Lem. 5.1, 5.4	Theorem
Matching upper bound $m \leq 4\log(1/u(\beta)) + D(\beta)$, $D = O(\beta)$ (infinite volume; finite volume, sides ≥ 3): for all $\beta < \beta_0(d)$ if $N^4 \leq 1200\tau_0^2$ ($D < 0.32$ if $N \leq 3$), for small β in general	Thm. 5.6, Lem. 5.7	Theorem
$4\log(1/q) \leq m(\mu_\beta) \leq 4\log(1/u) + D$ for all $\beta < \beta_0(d)$ ($N^4 \leq 1200\tau_0^2$; also finite volume, sides ≥ 4); $m = 4\log(1/\beta) + O(1)$	Cor. 5.8	Theorem
Ratio bound with $\theta_5 = x^5/(1-x)$ (only closed polymers count); $\|g_p\|_\infty \leq e^{2\beta} - 1 + \beta^2 e^\beta$ if ρ has no trivial summand	Lem. 6.1, 2.2(d)	Theorem
Transfer bound: the temporal plaquettes of all but one layer weigh $u_* = \max_{r\neq 0} u_r$, not $e^{4\beta} - 1$	Lem. 6.2, 6.3	Theorem
$m \geq 4\log(1/q_u)$ for all $\beta < \beta_0(d)$ (infinite volume; finite volume, sides ≥ 4), $= 4\log(1/u) - 4\log(eK) - O(\beta)$ for $\dim \rho \leq 49$; window $4\log(eK) + O(\beta)$ when also $N^4 \leq 1200\tau_0^2$	Thm. 6.4, Cor. 6.5	Theorem
Fronts and segments; incidence graph; fronts with 4 and 5 edges; local counts at vertices; tree sums	Lem. 7.2–7.7	Theorem
Sharp constant on the whole range: $m \geq 4\log(1/u_*) - 8(K+1)\theta_5 - \log(1+\delta)$, $\delta = O(\beta)$, hypotheses verified for all $\beta < \beta_0(d)$; $m(\mu_\beta) = 4\log(1/u(\beta)) + O(\beta)$ (also finite volume, sides ≥ 4)	Thm. 7.8, Lem. 7.9, Cor. 7.11	Theorem

Statement (continued)	Where	Status
$SU(2)$, all $\beta < \beta_0(d)$: $-1.38 \le m - 4\log(1/u) \le 0.0014$ ($d = 4$), $-0.63 \le m - 4\log(1/u) \le 0.005$ ($d = 3$)	Rem. 7.10	Numerical
Junctions: locality and local admissibility; junction graph; fronts with six edges; junctions with two front-pairs (cap, shift, jump); marked tree sums	Lem. 8.2–8.10	Theorem
First-order constant: $m \ge 4\log(1/u_*) - 8(K+1)\theta_5 - \log(1+\delta_J)$, $\delta_J = b(1+O(\beta))$, so $m(\mu_\beta) \ge 4\log(1/u(\beta)) - 2\beta + O(\beta^2)$ (ρ without trivial summand)	Thm. 8.11, Cor. 8.13	Theorem
$SU(2)$, all $\beta < \beta_0(d)$: $-5.0\cdot 10^{-4} \le m - 4\log(1/u) \le 1.4\cdot 10^{-3}$ ($d = 4$), $-1.1\cdot 10^{-3} \le m - 4\log(1/u) \le 4.9\cdot 10^{-3}$ ($d = 3$)	Rem. 8.12	Numerical
Cap matrix element: $\kappa = \|P_\rho M_f P_\rho\| = \frac{\beta}{N\tau_0}\int (\mathrm{Re}\,\chi_\rho)^3 + O(\beta^2)$; $O(\beta^2)$ if the centre acts on ρ by a scalar $\neq 1$ that is not a cube root of unity ($U(1)$, $SU(N)$, $N \neq 3$); $\frac{\beta}{6} + O(\beta^2)$ for $SU(3)$	Lem. 9.1	Theorem
Transfer bound by pieces (junctions with their gauge projections); the cap by characters, $\|P^J O_J P^J\| \le u^4(\kappa + (2-\iota)N^2u^2 + O(\beta^3))$	Lem. 9.2, 9.3	Theorem
First-order term: $m(\mu_\beta) = 4\log(1/u(\beta)) - \kappa(\beta) + O(\beta^2)$ (also finite volume, sides ≥ 4): $4\log(1/u) + O(\beta^2)$ for $U(1)$ and $SU(N)$, $N \neq 3$; $4\log(1/u) - \beta/6 + O(\beta^2)$ for $SU(3)$	Thm. 9.4, 9.5, Cor. 9.6	Theorem
$SU(2)$, all $\beta < \beta_0(d)$: $-1.3\cdot 10^{-4} \le m - 4\log(1/u) \le 4.8\cdot 10^{-4}$ ($d = 4$), $-2.5\cdot 10^{-4} \le m - 4\log(1/u) \le 2.4\cdot 10^{-3}$ ($d = 3$); as $\beta \downarrow 0$, between $-128.4\beta^2$ and $0.375\beta^2$ ($d = 4$)	Rem. 9.7	Numerical
$\mathbb{Z}_3$ (3×3 torus, winding zero): C-even and C-odd plaquette states at $4\log(1/u) \mp u$ ($\kappa = u$); $\mathbb{Z}_2$: no first-order term	Rem. 9.7(iv)	Numerical
Sides ≥ 4 needed for Thm. 6.4(i): torelon below the bound on a 3×3 torus ($\mathbb{Z}_2$)	Rem. 10.3(iii)	Numerical
Analyticity of $m(\beta) + 4\log u(\beta)$, as in the classical results; the second-order coefficient (shifts and jumps by characters); the C-odd mass	Rem. 9.7(iii),(v)	open

References

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